

Phys 310 Black Holes & Relativity

Numbered Transcript

PHYS 310 • Dr. Brian Keating

Contents

Lecture 1. Unveiling Cosmic Mysteries

Lecture 2. Seeing the Invisible

Lecture 3. When Gravity Bends Time

Lecture 4. Anatomy of a Black Hole

Lecture 5. Observing Black Holes

Lecture 6. Wormholes & Quantum Mechanics

Lecture 7. Entropy & Thermodynamics

Lecture 8. Journeys Through Space-time

Lecture 1: Unveiling Cosmic Mysteries

At a glance — Keating unveils the cosmic mysteries of black holes.

- 1.1 Welcome, everybody, to my third course at Peterson Academy. This one is the one I've been most excited for. Don't tell the previous students about that. But this class is actually related to an event which may have just saved my life. I'll explain that in just a bit.
- 1.2 But we're going to be talking about the most magical and mysterious objects that exist in the known universe. For a physicist like me who studies the origin of the universe, the properties of materials at low temperatures, how electrical conductivity works, you never, ever get to see things like what I do for my real research and my teaching. You never see that depicted in Hollywood, in blockbusters, and some would say the greatest film ever made, *Interstellar*. You get to see the objects that we're talking about today throughout all of media, and they captivate the audiences. They captivate the public imagination, and for good reason. They are the most interesting and exciting.
- 1.3 Perhaps the reason that we're all here, because black holes, wormholes, and Einstein's most outrageous legacy is perhaps the highest pinnacle of what human intellect has ever achieved. And there are cutting-edge discoveries taking place to this very day.
- 1.4 Today, I want to communicate to you why this is so exciting, why it's so important, why it's so meaningful. And it's an easy job to do because people are fascinated with these objects in a way that they're not when it comes to studying the properties of electrons at low temperatures or studying how fluids flow in different types of scenarios.
- 1.5 It's a treat for me to teach this course. It is my favorite course, and it ties into the previous courses that I've done in a very interesting and unique way, and will also tie into future courses that will be involved in Peterson Academy, including a very deep dive into the math and physics behind general relativity, behind Einstein's theories of special relativity, of time travel, of space warps, and all sorts of cool things that will be coming up in future courses at Peterson Academy.
- 1.6 But also into quantum mechanics. The physics of black holes actually relates to the microscopic physics of quantum mechanics, of particles, of forces and fields, again, in a way that no other branch of science does. I'm really glad that you're here today, and I'm really glad to be teaching my favorite of all the courses, at least so far. Every one of my kids is my favorite, and every one of the courses I teach here is my favorite.
- 1.7 We're going to take a journey really, truly to the edge of time. What is the meaning of where it is said that God is dividing by zero, black holes, singularities? You've heard of them, you've seen them. They're everywhere, from movies to the logo of the Grok AI system that's used in Elon Musk's Twitter and other ventures. Why are these things so important? Why do they captivate the imagination?

- 1.8** Well, I'm going to start with the end. Sometimes it's best, I found, to start with where we're going. We're going to the center of this object. You've seen them. You might have recognized them. As I said, they're logos for different things. They were featured prominently in blockbuster movies. Again, my favorite movie, which, as I said briefly, and I'll explain later, how *Interstellar* may have saved my life. It may be overdone.
- 1.9** But these black holes are the most fascinating objects because they embody all aspects of science, of knowledge, of time, of space, of what we say, relativity. What does that mean? They even impact philosophy and certainly into entertainment, because they're so powerful, they're so enigmatic, they contain all the pinnacle of what we've learned over the past 400 years or so of studying science as a scientific process using the scientific method. Again, it's a real treat for me to teach it.
- 1.10** Here are the major landmarks on the roadmap, if you will, to where we're going. And where we're going, we might not come back from because a black hole is a place where, as I said, God divides by zero, where the laws of physics break down completely, where time becomes space and space becomes time. What can that possibly mean? We're going to explore all of that—things like the causality of events. Can one event precede another event? Could you be older than your grandfather? These things can take place in the extremes of physics, only present in black holes. That's why they're so fascinating.
- 1.11** Here are the major landmarks. Here's the map of the territory we're going to go explore in these next eight lectures. You've heard perhaps of the event horizon. It's a movie also. A lot of these things are movies or based on people that have done tremendous work, and there'll be many, many more movies to come.
- 1.12** A black hole, simplest way to say it, it's a region of space and time, which is the fabric of the cosmos that we all exist in right now—we exist in space-time—where gravity is so extreme that nothing can escape from it, not even light.
- 1.13** Yet it has properties that are mundane, ordinary. For example, you've heard of the event horizon. We will explore what is the event horizon. What happens if you go beyond the event horizon? Would you notice it? Would you care? When do you start to care? Well, when a process called "spaghettification" occurs, you will definitely care when your atoms in your body get ripped apart. But don't worry, that won't come until tomorrow. So you have plenty of time to make out your will.
- 1.14** But we're going to talk about these different features. How is it possible we could even see a black hole? If nothing, not even light, certainly not sound, certainly not TikTok videos, nothing can escape from a black hole. How do we know they exist? It's bizarre. And how have we even thought that they might exist for nearly 300 years? That's what makes them so fascinating.
- 1.15** Along the way, we'll encounter some of the most fascinating people, events, human stories that have contributed to this subject matter. And ironically, it's a story just

because science is done by scientists, and scientists are human, despite the stereotypes that were just these walking automatons like Wikipedia on legs. No, it's a story of jealousy, of racism, of sexism, of all sorts of crazy things that have happened that humans have overcome in their quest to understand intellectually the most fascinating topic imaginable.

- 1.16 That will bring us things and names like Schwarzschild down here: the photon sphere, the accretion disk, and the singularity. We'll talk about all these different attributes and how they reveal the unseen reality of the black hole. We'll understand how black holes connect to other black holes, perhaps speculatively in the far future, allowing for the time travel that Hollywood loves to exploit. We'll explain so that you will be able to tell your friends and family what's real, what's fantasy, what's science, and what's science fiction.
- 1.17 This is going to be such a fun journey. I would pay to teach this course if I had to. But there is a human story, and we want to understand how the people came to make these enormous intellectual leaps and what we still do not yet know, because that's for you, that's for the audience, that's for the students at Peterson Academy and my students in San Diego to perhaps make a contribution to.
- 1.18 So black holes are not fiction. They are reality. We have observed them, and I'll explain how we've observed them. But what's so fascinating about them is that they were predicted before they were observed.
- 1.19 That's what a good scientist should do. A good scientist should not, like my kids, say, "Well, a Yeti came in last night and ate that last piece of cake, Dad," because that's predicting something after the fact. That's called a "retrodiction." It may be true. It may not be true.
- 1.20 But a better thing is to predict something that has not yet ever been seen, explore the consequences of if it did exist, what would be the signature of it, and then, clever scientists, maybe a year later, a decade later, or in the case of the experiments we'll talk about in this course, two and a half centuries later, proved the prediction was true and allowed us to not only see but hear and feel black holes' reality.
- 1.21 And this goes back—the first person to conjecture was a man by the name of John Michell. He was an English scientist who had a very strong grasp of the physics of the day. But the job of a scientist is to make him or herself obsolete and to have their theories, their conjectures, their observations overturned by better, more precise, more accurate, and more broadly and boldly predictive features, theories, and experiments in the future.
- 1.22 John Michell predicted the existence of black holes, but not in the way we currently understand it. He conjectured the existence of what was called a dark star. Again, it seems like an oxymoron, a dark star. What does that mean? A giant shrimp. I mean, you can get a jumbo shrimp, right? But a dark star was an object, a star, so luminous

that it was so massive. And its mass created a gravitational field so strong that to escape its gravitational field required almost infinite speed. So therefore, nothing could escape its gravitational pull. It's broadly the way that we think about a black hole, even as I described it, but lacks all the features, the physics, the math behind it. It was a conjecture. Based on the primitive math for its time, it was outrageous cutting-edge math, akin to the most sophisticated tools and technologies of the 1700s. But nowadays, we know it to be outdated.

1.23 About 100 and so years later, a man by the name of Karl Schwarzschild, working in the trenches of World War I, about to die, fell prey to the spell of black holes and worked out a solution to Einstein's most outrageous theory called general relativity, which we'll explore in this course at a high level, but I'll explore in much greater detail when we have a dedicated course on special and general relativity for those that are more mathematically inclined. Not everyone will need mathematics for this course. For those courses, my aim is to be a little bit more, as I say, white tofu or red meat or however you want to adhere to it for those that are interested in getting into the nuts and bolts of what scientists approach relativity by. But we'll get into that.

1.24 Karl Schwarzschild had the first notion of what's called an event horizon. Events being every point, not in space, but in space-time. Right now, we're at certain latitude, longitude, altitude. Those are three spatial coordinates, but we also exist at a moment in time. Every moment is traveling through time. We're all traveling through time. It turns out we're traveling through time at the speed of light. The maximum speed we could travel is the speed of light. You could either choose to spend some of that speed towards moving in space or moving in time, and we're basically stationary now, which means we're just going through time. Things we can influence in the future are possible. We can influence what's going to happen in the future. We can't change what's happening in the past. There are situations where past and future get intertwined, and they occur primarily near black holes. We'll talk about that, and that will bring up the context of the event horizon.

1.25 I said that we face strange forces that aren't scientific, and those are the most destructive in my mind because it takes smart people away from the scientific process. And we need as many smart people as possible, regardless of their race, gender, creed, whatever. It turns out this man, he looks old here, but he was a young man once, obviously, Subrahmanyan Chandrasekhar. He calculated the first ingredient of what could possibly turn into a black hole or not turn into a black hole. And he was discriminated against by a British physicist who tried to destroy his life because he disagreed with him, because he was, in part, racially biased against Indians. Chandrasekhar was Indian.

1.26 His name was Arthur Eddington. And he, Arthur Eddington, to make the story even more complex, did some of the greatest service to astronomy and physics and even help substantiate the existence of Einstein as the monumental genius that we know

today. And he was rumored to be homosexual. So we have all these weird things coming into play, discrimination and recrimination, biases. And we'll see later in just a second about influence against women who played a huge role in this field.

- 1.27** Science is done by scientists. We are humans, but we don't want to exclude anyone from the scientific process because we need as much help as possible. This is some of the most difficult things the human mind can ever accomplish. On a par with the greatest art ever made, the greatest music ever made, science is a part of culture. And the physical sciences, in particular, physics and relativity, some say, is the highest form of not only science, but perhaps civilization as it involves culture, communication, dissemination, testing, proofs, experiments, tests. All these things are relevant only in things like relativity.
- 1.28** You don't have a Picasso, and it wins a Nobel Prize because there's no objective standard for art. You might think it's beautiful. I might not. And it turns out Picasso was deeply influenced by the work of Albert Einstein. Art influences science. Science can influence art. But we have to think about how that comes into play. And we'll see in just a moment how art and science came together in this movie, *Interstellar*, which, again, I'm going to keep building this up, but maybe I shouldn't, but it's a cliffhanger that they use in movies, that I faced great danger, but *Interstellar* helped me get out of it.
- 1.29** So I said John Michell was the first person to really think quantitatively, not just qualitatively, that all this thing could exist. He had an idea that if you had a star that was so bright, it would actually be dark, counterintuitively, because the star would be massive in order to have that luminosity. Beyond a certain mass, Newton's laws predicted that gravity would take over, preventing the motion of objects unless they were traveling at the speed of light. And then you dial up the mass a little bit, you put in one more tablespoon of sugar into that massive object, that star, all of a sudden it's more massive than the speed of light would allow an escape from. So that's called the escape velocity of an object.
- 1.30** We'll explore what that is for the Earth. There's a notion of escape velocity, where, if you could fire a projectile, a cannonball, at such a speed, it could actually leave the Earth's gravitational field and go way beyond the orbit of the Earth, for example. It's much faster than any SpaceX rocket or anything we've ever devised before. In fact, we can't actually make something that goes that fast. But theoretically, it could take place.
- 1.31** This man, back in the 1700s, had this brilliant idea that a dark star could be massive enough to entrap and entomb the light that it was producing. So it would appear completely dark. Now, he was an incredible mathematical genius, but he wasn't as great at explicating and teaching other people why his idea should be taken seriously. Sometimes that's a good thing.

- 1.32** Ironically, he was so far ahead of his time that there would be nothing to do about it. There was no way to test, predict an image, simulate, and then observe these objects the way we have now. In fact, it took almost 300 years before the tests that he proposed, more or less, were verified. He was ahead of his time by a lot, so he never, ironically, escaped this velocity that was necessary for him to achieve scientific immortality, the way even Isaac Newton or a Galileo we will talk about in this class as well. He also invented things like seismology, the reverberations of the Earth's crust, can teach us about the dynamics of plate tectonics, of earthquakes. So he's considered that and also invented many other tools called the torsion balance.
- 1.33** So we fast forward about 200 years to the trenches of World War I, and we meet a person who's got the most ironic name you could imagine. So there's a term—sociologists or some people that study these things—there's a concept called the "apronym." It means your name is very appropriate for your job. So, for example, Usain Bolt is the fastest runner on Earth. It's a great name for a runner. Now, did his parents know he'd be a great runner? No, I don't know about that. But Schwarzschild means "black shield," which is very apt for the man who invented the concept of the event horizon around a black hole.
- 1.34** So what was occurring at this time, he was actually 41 years old, which is not a young man. He was very patriotic and volunteered; he was Jewish, and he volunteered to serve in the German army. At that time, it was long before the rise of the Third Reich. Most Jews in Germany were trying to show their patriotism and nationalism. They volunteered, many of them did, a very famous one, Fritz Haber, who invented the fertilization process called the Haber-Bosch process that makes nitrogen part of fertilizer, he was a German Jew as well, and he went on to invent chemical weapons. And tragically, the chemical weapons that the Nazis would later use were used to kill members of his own family.
- 1.35** So scientists are not above the petty and foolish actions of any other person. Never put a scientist up on a pedestal until you understand what he or she is famous for and understand the fact that, as I always say, knowledge and wisdom are not interchangeable. Knowledge, they say, is knowing that a tomato is a fruit, but wisdom is knowing not to put it in a fruit salad. Those are very big differences and many scientists don't know the difference.
- 1.36** Einstein got a letter from this man, Schwarzschild, from the trenches of World War I, and it said, "I have come up with a particular solution to your theory of general relativity." It's called now the Schwarzschild solution. Einstein immediately—and he had a huge ego. Einstein had a huge ego. He would say things like, "If this experiment doesn't work, then God is wrong because my theory is right." This is how egoistic, but he was also humble. He balanced humility and arrogance.
- 1.37** He mailed the solution from the trenches to Einstein. Einstein said, "You're a genius. This has got to be true. Too bad we'll never be able to see it." Einstein was amazing

because most of his things that he said were wrong turned out to be right, even though he invented them. So he predicted things we'll talk about: gravitational waves, gravitational lensing. He said, "They'll never be observed." Nowadays, it's done every day, literally every day.

1.38 And the most famous one is, he said, "The universe didn't begin with a Big Bang because of this thing called dark energy." But he said, "We'll never observe it. It doesn't really exist. I was wrong about it." And then 80 years later, the Nobel Prize was won for the discovery of dark energy. So I say, I aspire such that my blunders are me saying that I made a mistake. I try that with my wife. It doesn't usually work. And don't take that advice.

1.39 He predicted this feature of not dark stars, but of regions of space-time so dense, so highly curved, as we'll explain in a few slides, that gravity couldn't escape because space and time were flowing faster towards the singularity than the speed of light, effectively trapping information, not leading to spaghettification just yet, not leading to destruction. We'll see what happens as an observer moves through the event horizon, what actually happens.

1.40 Tragically, he died. He died three years before the verification of this phenomenon that he had predicted. He didn't die from—I mean, it would have been amazing if he died from some war injury or something. I think he died of a tragic disease, maybe cancer or something like that. But he was incredibly proficient. And to imagine doing this as your side hustle when you're in a battle, it's incredible. I guess it says here, he died from pemphigus. I don't know what that is, but it was a painful disease.

1.41 And then I said in the 1920s, 1930s, there was a young Indian astronomer, physicist working in isolation. India was not yet the technological superpower that it is now, and scientific superpower that it is now. And it's really thanks to this man, Subrahmanyan Chandrasekhar, who later emigrated to the United States, that he made this prediction that you could have a certain type of star, not a dark star, but a very white star called the white dwarf, and that these white dwarfs could only get to a certain size. And beyond that size, physics had to be radically altered.

1.42 And that size was effectively compressing the mass of the Earth into something the size of the city that we're in right now. So imagine the entire mass of the Earth compacted into a small enough region. It's not quite a black hole. We'll talk about that in a bit. It's called a white dwarf. And this is one of the key enabling intellectual triumphs that allowed us to start thinking about the quantitative, not just qualitative, like Michell or even Schwarzschild, but the quantitative predictions and observations that we could see, literally with telescopes, even at that time, called white dwarfs. And this discovery pushed the limits and the boundaries of physics and mathematics.

1.43 He presented it in the Royal Society, and the Royal Society was dominated by this man, Arthur Eddington. And Arthur Eddington said that he basically— it wasn't clear—he set

it up such that Chandrasekhar would give a talk. And then he was like the opening act for Arthur Eddington because he was the esteemed scientist. Chandrasekhar was a young nobody from nowhere. And people in that time in that part of England were not positively predisposed, shall we say, to people from India. There was a lot of tension between India. It was a colony of Great Britain.

- 1.44 So he said, "I think there should be a law of nature preventing a star from behaving in this absurd way." I mean, this is like a rap battle of the time. It sounds very conservative right now. This was like outrageous to say this in the middle of a lecture to all these other barristers and Nobel Prize winners and esteemed scientists.
- 1.45 It was an ambush because he was jealous. He knew that Chandrasekhar would probably be right. His math was perfect. He's an incredibly revered genius to this very day. And the vindication—as I say at the bottom—is that Chandra won the Nobel Prize, and Eddington never did. And Eddington deeply wanted to win the Nobel Prize. He was swooning for it. It was incredible how much he desired it.
- 1.46 So nowadays, let's fast forward to some modern heroes stories and the human side of science that's going on today, many of whom these people here I've either talked to, I have a podcast called Into the Impossible, a YouTube channel. But we'll look at a couple of these videos right now of modern-day heroes of mine that are working incredibly hard and sharing their knowledge and wisdom, not only with other scientists, as you have to do, but with the public.
- 1.47 I always say scientists work for the public. As much as we might be at Harvard, I happen to teach at a public university in California, where my boss or the regents of the state of California, but even at Harvard or MIT or Caltech, where I used to work or Stanford, where I used to work, their scientists are supported by the public, by taxpayers. I assume you students pay the taxes that you're owed to pay. So in reality, scientists work for the public. And so not many of us choose to communicate, disseminate the findings that you pay for.
- 1.48 I always find that negligent because if you worked at a real estate agency and your boss, the owner of the real estate firm, came in and she said, "Well, what are you working on, Brian?" I said, "You can't understand what I'm doing. It's very, very technical. And I use very specialized tools." I'd be gone the next day, right? You can't do that in a normal job, in a real job.
- 1.49 We scientists, most of us, would do what we do for free. I mean, that's the truth. We love it. We're curious. We're passionate. We've been doing it since we were kids, and now we get millions of dollars from taxpayers and from government and private foundations, in my case, to do research that we would pay to do, volunteer to do, and have volunteered and have paid to do in the past. So, I say take advantage of the great communicators that we have today.

- 1.50** Roger Penrose won the Nobel Prize in 2020 as a personal friend. He predicted, along with Stephen Hawking, aspects of how the singularity of black holes behave, which then influence aspects of how the Big Bang behaves, which is my other course, Cosmology at Peterson Academy. You can see more about that.
- 1.51** Stephen Hawking was his close collaborator. I met Stephen Hawking once in England in 1995. He gave a "speech" when he could still not really speak, but he could communicate using his speech synthesizer. And someone asked him a question, "Why did you write the book called A Brief History of Time?" It's rumored that 99% of people who have bought it never read it, and it took him five minutes to answer in his synthesized voice. And he said, "I needed to pay for my daughter's private school." He was very funny. I give him that.
- 1.52** Andrea Ghez, she was an incredible superstar. We'll find out more about her in just a second. She led one of the first teams to discover that our Milky Way harbors at its heart a supermassive black hole, millions of times more massive than the Sun.
- 1.53** Shep Doeleman took an image with his team of the event horizon of a black hole, or more correctly, we will distinguish between the singularity of a black hole, the event horizon of a black hole, the photon sphere of a black hole, the innermost stable circular orbit, ISCO, of a black hole, the Schwarzschild radius of a black hole, the rotation dynamics, the ergosphere of a black hole. Black holes are so cool. They have so many cool names, terms, and technology associated with them. He's a friend. He's been on my podcast.
- 1.54** Kip Thorne has been on my podcast. Kip Thorne was the scientific consultant for the movie Interstellar, which not only produced the most mesmerizing images ever made on film, in my opinion. If you watch Interstellar and you think it's okay, go back and watch Star Wars. Star Wars' big thing was like, stars move as you go. They make straight lines. They made computer simulations that took something like 150 hours per frame of video because Kip demanded that they be scientifically accurate, so much so that the producers and him and the videographers ended up writing a scientific paper on scientific visualization. We're going to take a look at that in a minute, along with Chris Nolan. Chris Nolan is not a scientist, but he loves looking at the scientific details, getting everything right.
- 1.55** I mentioned that scientists are human, and a lot of humanity has displayed awful types of discrimination. It's just the truth. For a long time, Einstein couldn't win the Nobel Prize because he was Jewish. Most Nobel Prizes were given out by Germans, Austrians, and members of the Aryan Scientific Society, and they claimed that Einstein was just doing Jewish science. Can you believe that? He wouldn't win a Nobel Prize for almost 20 years because he was a Jew.
- 1.56** Similarly, up until 2020, there had only been four women to ever win a Nobel Prize [in Physics], one of whom was at my institution, Maria Goeppert-Mayer, who discovered

how the nuclei of atoms behave. She was only the second to win it. When she won the Nobel Prize in 1963, the headline in our hometown newspaper in San Diego read, "La Jolla Housewife Wins Nobel Prize." One of my friends won the Nobel Prize in chemistry in 2018, and it said, "Frances Arnold Wins Chemistry Nobel Prize. Her son works at NASA's JPL." It's crazy. It still exists to this very day.

- 1.57** Now, these women, and I've interviewed many of them on my podcast, including Donna Strickland. If you've ever had LASIK eye surgery as I have, you owe her a great debt of thanks because she invented, along with her colleagues, the laser that makes these incredible pulses that if you told somebody, "Hey, I'm going to shoot this high-powered laser into your eye and you're going to give me \$2,200," you'd be like, "You're crazy. I'm not going to do that. Laser is bad." Now, she developed a way to do it completely safely. I've had it done, like I said. She's an incredible woman. She was the [third] woman to ever win it. It's incredible. Andrea's high school counselor—Andrea Ghez, who's at UCLA, our sister university to the north in Los Angeles—said she couldn't do it, she's never going to make it.
- 1.58** And she did it. She invented this tool, this technology, using telescopes in Hawaii that are owned in part by the University of California, where I work. She discovered that this supermassive black hole could be seen by the invisible force that it exerts on other objects. Brilliant, brilliant discovery. She did this. I talked to her colleague, who also won the Nobel Prize, Reinhard Genzel, also at UC at the time. These are incredible discoveries, and we're going to take a look at some of that in just a second.
- 1.59** Then, of course, there's the Hollywood connection between black holes. Maybe the reason some of you are here. And that is owing to the fact that these objects can almost do anything. So they're very convenient, if nothing else, plot devices to shore up all sorts of different loopholes and make things happen. Otherwise, it wouldn't be possible.
- 1.60** So *Interstellar* came out in 2014, and it was incredibly detailed. At the time, the computer technology doubles every 18 months or something like that. So this is more than something like 2 to the fourth or sixth power doubling cycles ago. It was much more primitive, even though it was incredibly advanced. Like I said, every frame of the visualization that you can find on YouTube at home if you're not able to watch it here, and so you'll be able to watch the animation. But each frame took hundreds of hours of the most powerful computer of the age. And they discover things that were not really well understood, at least in terms of how to visualize this.
- 1.61** By the end of this course, I will have failed if you cannot understand why these black hole images look the way they do. Why did Gargantua do that? Such that when you watch the movie for the first time or the second time or the 10th time like I finally have.
- 1.62** I better just bring up my story here because in 2015, I had to get my passport and my Global Entry card renewed, and I was sent to the San Diego–Tijuana border, where I

was a little bit nervous. I had never gotten this Global Entry. It's like a TSA PreCheck, and you have to get your mother's maiden name, your elementary school teacher's dog's endorsement. Anyway, you need a whole bunch of information, your fingerprint, your dental record. I don't know. But they asked all these questions.

- 1.63** I'm sitting in line, and I'm waiting. There are all these guards, huge, bulky guys. And I come up to the desk, and I am one person behind in front of me. He's going up, and they ask him, "So what do you do?" And he's like, "I'm a college professor." And he's like, "What do you teach?" And he's like, "I'm a scientist. And I do this and that." I couldn't quite hear. I was trying to eavesdrop, but I didn't want to get in trouble. So they were like, "Step back, sir." And so they made me step back to give him his privacy.
- 1.64** Then they called me up, and they're like, "What do you do?" "I'm a college professor." "Wait a second. You guys are part of some team? You guys are working together, trying to pull one over on us? What do you teach?" And I hadn't heard what he said. I was like, "Please, God, don't say astrophysicist." I said, "I teach astrophysics." And the guard says—he reaches for his gun—he's like, "Really? What do you think of *Interstellar*?" And I hadn't seen it at the time.
- 1.65** So I said, "Well, the black hole in Gargantua is rendered incorrectly in one frame because it has the general relativistic time dilation." He's like, "Oh, shut up, shut up. I believe you. You think you're smarter than me?" And I said, "No, you got the bigger gun." So anyway, it may have saved my life just knowing about black holes. But this movie captivated everyone from border guards to astrophysicists to Hollywood elite.
- 1.66** That is my goal. When you guys leave this course, you'll be able to understand why these black holes are so interesting, why they play the role, what's going on in the movie. It's confusing. It's advanced. I didn't fully grasp everything that happened from the very beginning, and so it took me seeing it twice. So now, what is it now? Twelve years later, I can finally pass my next border control exam.
- 1.67** Of course, it wasn't the first one to feature a black hole. Two movies in 1997, one of which I have a slight connection to because the movie *Contact* was based on a book co-written by Carl Sagan, who's one of my heroes in science, and his wife, Ann Druyan, who is more of an artist, an artistic type. She and her daughter, Sasha Sagan, were both guests on my podcast. So, I had a mother-daughter team talking about their work, their father's work and Ann's work on the movie *Contact*, which actually not only depicts black holes and wormholes—which are related, interrelated with each other, as we'll see—but also depicts radio astronomers, which is the actual sub-branch of astrophysics and cosmology that I do.
- 1.68** And it depicts a real-life person named Jill Tarter, the role played by Jodie Foster. That's a real-life person named Jill Tarter, who's a very good friend of mine. It's modeled on her real life. She was one of the first women to explore the search for extraterrestrial

intelligence, looking for aliens using telescopes for the first time. She's coming to visit us in just a couple of weeks in San Diego.

- 1.69** I never saw Event Horizon, but I'm told I'm not missing much. But if you have seen it and you want to learn about how a black hole and a wormhole can open an artificial gate to hell, you'll be in luck.
- 1.70** Now, other than that, there's also human drama associated with black holes, including Stephen Hawking's life and The Theory of Everything, which also came out in—I mean, these black hole movies seem to come out every 20 years, 21 years or something like that. But there are a lot of classic movies, some not so classic movies, but The Theory of Everything is based on Stephen Hawking's life. Can you imagine another—the only other movie really based on a scientist—I mean, Einstein really hasn't had a movie based on his life. Oppenheimer is really based on the life of a physicist during the atomic weapons construction phase in World War II. It's not really about his life as a human being, but it has some elements of that, I suppose.
- 1.71** And then in popular culture, one of my favorite songs, "Black Hole Sun" by Soundgarden. You guys can look up stuff. If you're online, you can look up all these things at your leisure. The Grok logo, which, according to Elon Musk, will be the first AI to live in space and answer questions from outer space. For some reason, he thinks that's important. My kids get addicted to the black hole event in Fortnite and many other things in popular culture. It's quite cool.
- 1.72** We'll be exploring these features. What makes black holes so interesting? What makes them so important to scientists and non-scientists? Here are some recent events that I've played the most microscopic role in by trying to popularize them. As I said, scientists have a moral obligation, in my opinion, to disseminate their findings to the public who pay our salaries, who pay for our telescopes.
- 1.73** In particular, the Event Horizon Telescope, led by my friend Shep Doeleman at Harvard, made this image, which looks very much like the image, the classic image from the movie Interstellar, which came out four years before the discovery of the actual image. Here's the actual image in the upper right. That's an actual image of what's called the event horizon. We'll see it's slightly overstated. It's actually imaging a shadow, and the shadow is the light shadow that surrounds the event horizon. We'll explore why it looks like that and what that reveals.
- 1.74** Now, the first image of an event horizon in a photon sphere was done of a galaxy millions of light years away, 55 million light years away in the constellation Virgo. It was called M87 for the arcane reasons I won't get into. But the one at the center of our galaxy is called Sagittarius A*. It's not a star star, it's asterisk, and that means it's a black hole. That's how astronomers designate black holes. So it was also measured. It's a selfie taken. Now, what does it mean? Are we seeing light? I thought we couldn't see

light from a black hole. So what are we actually looking at? That's going to be a major takeaway from this course.

- 1.75** Okay, so we love black holes. Physicists love them. Non-physicists love them, movie directors love them, musicians love them. They're incredibly fascinating objects. My hope is that you will not only come away with a deeper understanding, but a deeper love of these magnetizing monsters that encapsulate almost every aspect of physics, from the laws of gravity to thermodynamics to quantum information and beyond, and even the origin of the universe itself, which some suggest is predicated, is only due to the fact that we happen to live in a corner of the universe where there's a certain amount of black holes. It's a theory so bold, it's called [cosmological] natural selection. Like Darwin's theory of natural selection on the origin of species, many scientists think that you could only have life to have our existence, to have Peterson Academy, thanks to the properties of black holes.
- 1.76** It's crazy. Not biological evolution—cosmic evolution. Black holes are really the story of gravity gone amok. It's essentially without gravity, there'd really be no story for us to talk about. I call it "Gravity's End" because effectively, once a black hole forms, there's almost nothing that interesting that can ever happen to it again.
- 1.77** We'll also see that black holes only have effectively three different unique properties. We'll see they have their mass, their charge, and their spin. Those three features define every black hole that's ever existed. There's no other feature that they can have. They don't have names. They don't know they're called Gargantua or Sagittarius A*. They just have those three properties. They're like electrons or rice. Rice has its weight, its color, and maybe one or two other features about it. That's all it has. So they're commodities. Black hole, you've seen one, you've seen three, you've seen basically them all.
- 1.78** So gravity is the ultimate both creation force and destruction force for black holes. Black holes are utterly dependent on gravity, and yet they're entirely determined by their gravitational properties, their spin, their charge, and their gravity, their mass. And because of that, they're very simple on one hand, but their effects are extremely profound.
- 1.79** An electron is maybe just as simple. It has a mass, a charge, and a spin. And yet without electrons, we wouldn't have all of chemistry; we wouldn't have all of life. And so, yes, simple things can occur and appear to be simplistic, but in combination with other properties, other aspects of nature become incredibly complicated.
- 1.80** To start with black holes, we have to go back to the 1700s. The first theory of gravity was postulated by Isaac Newton. Isaac Newton had famously many different contributions to science. The most relevant to what we're talking about is this theory of universal gravitation. Universal gravitation means that the same gravitational force that affects the Moon in its orbit around the Earth also affects the falling of an apple onto

the Earth's surface. It was the next greatest accomplishment that he made after his three famous laws of motion.

- 1.81 So his laws of motion, Newton's three laws are encapsulated here. Inertia is the property by which an object in motion will stay in motion unless you act upon it with a force. His second law is that that force causes an acceleration which is dependent on the object's mass. So larger amounts of force and smaller amounts of mass cause greater acceleration. Gravity, what we call g forces in a fighter pilot or in a plane or on a roller coaster, those are accelerations. Those aren't forces; g is an acceleration, not a force.
- 1.82 Then lastly, every action has an equal and opposite reaction. When you push on something, it pushes back on you. When the balloon is released, it sprays out molecules of air which have mass. That causes an opposite reaction, a thrust. A rocket behaves the same way.
- 1.83 His universal law of gravitation has many, many similarities in the propagation of light, in the spreading out and dilution of electrical forces. It's very similar to this. But it's known as the inverse square law. It says that the force of gravity depends on the masses of two objects and is always attractive via the inverse of the square of the distance between them. If you double the distance between two massive objects, the Earth and the Sun, you reduce the force on those two objects [to] a quarter.
- 1.84 Each one exerts a force on each other. The Earth exerts a force on the Sun. The Moon exerts a force on the Earth. In fact, a huge force that plays a role in every one of our lives, at least on the East and West Coast, via the tides that occur four times a day. Those are all manifestations of equal and opposite reaction, force, inertia, and the law of universal gravitation.
- 1.85 Now, this held sway for 200 years that you could take velocities, you could take accelerations, and they would behave the same for any types of acceleration or velocity. If I was standing still and I threw a baseball at 50 miles an hour, it would go faster. If I was running at 20 miles an hour and threw it at 50 miles an hour, it would go at 70 miles an hour.
- 1.86 That held sway for literally 200 years until Einstein postulated that this cannot be true if the laws of electricity and magnetism are also true. And that is because there is a universal speed, a speed, an excess of which can never be achieved. That's the speed of light. And if the speed of light is constant, then if you're traveling at five miles an hour and you turn on a flashlight, the flashlight light travels at the same speed of light as if you're stationary. And even if you're going at the speed of light and you turn on a flashlight, it also goes at the speed of light— completely counter to any of the laws that were known before Einstein. There have been tests of this as well.
- 1.87 But Einstein didn't only think about the properties of light; he also thought about the properties of gravity and intuited that there must be a relationship between gravity and

acceleration. We take it for granted that here on Earth, we're experiencing one gravitational force, the force of 1 g.

- 1.88** I want you to close your eyes and imagine what Einstein used to do, a thought experiment, not a real experiment. He didn't actually do this. But imagine you are in space away from any object and you're in a rocket, and that rocket is blasting up from your feet towards your head with exactly the same acceleration that the Earth is producing here on its surface, 9.8 meters per second squared or 32 feet per second squared. That would create the same sensation in your backside, in your feet, in your chair. You could not distinguish between accelerating at 1 g and being in a 1 g gravitational field.
- 1.89** In fact, he visualized, if you were in an elevator and it was being pulled up at some force, and then the elevator cable broke and you start to fall freely, you wouldn't feel any force whatsoever. You couldn't tell you're moving. Therefore, you'd be traveling in a straight line. So you'd just be freely falling. And that sensation led him to the notion that gravity and acceleration are not properties of some force, but are really caused by what we call the curvature of space-time itself. And we'll explore that in great detail in the exploration of black holes.
- 1.90** The very first proof or verification that gravity is not necessarily a force like pulling on something with a rope, but is in fact the manifestation of curves in space-time came in 1919, by the man who ambushed the young Indian physicist Subrahmanyan Chandrasekhar. His name: Arthur Eddington. Arthur Eddington realized, thanks to the work of Einstein and the predictions of Schwarzschild, who had died three years earlier, that if you had a massive object close to you and behind it were millions of stars, you would see those stars move, and they'd be displaced from where they were if you took away the mass.
- 1.91** That's a very hard experiment to do, to get a giant mass in space in front of a bunch of stars. But every so often, the Sun, the Moon, and the Earth align in what is technically known as one of the best words you could ever use in Scrabble, a "syzygy." Lots of Y's, Z's, S's. A syzygy is an astronomical alignment, a conjunction of three or more astronomical objects. In this case, the Sun, the Moon, and the Earth.
- 1.92** In 1919, May 29, so it's over 107 years ago, almost, Arthur Eddington and others led an expedition to measure using not webcams and HD cameras and stuff like that. They had photographic film, which many of you in the audience don't even know existed. These are plates, glass plates that had special chemicals on them, which, when you exposed them to light, would make images on them. This is a modern image of it here, but that's nothing like it. This was taken by the Hubble Space Telescope 100 years later.
- 1.93** But what happened was the Earth on the left was looking at the Sun being eclipsed by the Moon. It happens to be that the Earth and Moon, so there's 100 and something moons in our solar system, our Moon, for some reason, is the only Moon in the entire

solar system that has the exact same apparent size as the Sun. The Moon is physically 400 times smaller than the Sun, but it's 400 times closer than the Sun, so they exactly cancel each other out, and the Moon can perfectly eclipse the Sun, making a complete shadow.

- 1.94** If you've ever seen a total solar eclipse, you know it's one of the most magical experiences, a transcendent, spiritual—it's not religious, but it's spiritual—encounter. And there's very few of them, but there's one coming up this year and there's one coming up in 2027, which I'm going to go to. I think it's visible in Western Yemen. I'm really looking forward to going to Western Yemen. No, I'm not going to go there. I don't think they'd like me there. But instead, I have seen two of these so far, and it's been an incredible experience, and I do travel to them. A lunar eclipse, I sometimes don't even leave the house. If there's a lunar eclipse, that's when the Moon gets eclipsed by the Earth as seen from the Moon. But a solar eclipse is magical.
- 1.95** And they observed that the light was displaced exactly as Schwarzschild had predicted three years earlier based on the mathematics and physics of Einstein. This is how scientists build on each other. And even though Eddington was, I'll say, possibly, allegedly a racist jerk towards Chandrasekhar, he still carried out this proof, this test, a painstaking back-breaking labor across thousands of miles.
- 1.96** He had to do one observation in Brazil, one in Africa. Imagine, 1919, long before jets and planes, and coordinated, analyzed it, and produced the final data. And it made the front page of The New York Times. And this is what caused Einstein to say, when asked, "What if it had come out otherwise? What if you had been wrong?" He said, "I would have felt sorry for the good Lord God because my equations are correct." Imagine the arrogance, but kind of deserved, right?
- 1.97** This is called "gravitational lensing," and it was the first verification, not of a black hole, but that gravity bends the trajectory of light. G

Lecture 2: Seeing the Invisible

At a glance — Seeing the invisible — detecting black holes.

- 2.1 If black holes absorb all the light that they emit, the dark star theory of John Michell and others, how is it possible we can see "evidence" for their existence? Well, the first evidence came in the 1970s. It was not from the black hole itself. It was from its companion. We'll see that black holes are never lonely. They're almost never in isolation. They always seem to come in pairs or multiples, at least the types that we can see, because we don't see the actual black hole itself. We don't even see its event horizon. We don't actually see the black hole. It is, by definition, unseeable, unknowable.
- 2.2 But everything that surrounds it in its neighborhood is subject to its behavior. Just like a bad neighbor in your own neighborhood, you can always tell when they're having a raucous party, even if you don't get the invite. In the case of black holes, when they're surrounded by either another black hole or an ordinary star or a clump of matter like a planet that had the misfortune of getting too close to it, we can see quite clearly its properties. The engines that power these events are related to the properties of the black hole in a way that's exactly predictable because they only have those three properties I spoke about last time: mass, charge, and spin.
- 2.3 Most of their energy and their dynamics comes about from their mass and their spin. We can determine their existence. We can measure their mass and their spin. We have scales, like a balance or a weight-measuring scale, and we have ways of measuring its rotation as well, only thanks in part to its companion. Again, it could be a normal star, like we'll talk about in the case of Cygnus X-1. It could be an exotic star called a white dwarf, or a neutron star, or even a pulsar, as we'll talk about, they become laboratories by which we can measure the mass, charge, and spin of the unseeable, unknowable black hole. They become our witnesses, eyewitness testimony to the properties of these mysterious objects.
- 2.4 In the '70s, this object was discovered. It had a companion. That companion was a star much more massive than our Sun, about 2.3 times more massive than the Sun. It was called a blue giant. This type of source had a companion towards it. And it was orbiting next to a black hole, as shown in this image from NASA. And it would orbit every 5.6 days. The year, if you think about the black hole as the Sun and the orbiting blue giant as a planet, its year was only five and a half days, pretty much. That's an incredible rate of speed for a massive star that's way more massive than our Sun, in orbit around a black hole.
- 2.5 Now, at first, it was thought, maybe it's an ordinary star, but the mass turned out to be much more massive, in this case, 7 to 20 times more massive than our Sun. And thanks to the work of Subrahmanyan Chandrasekhar and other scientists in the 1900s, it was

realized you could not have an ordinary star. You could not have that object that was causing the gravitational field, 7 to 20 times more gravity than our Sun, be an ordinary star or even a compact star like a neutron star. They could only be as massive as two and a half times the mass of the Sun. In this case, it left out everything.

- 2.6 Now, it doesn't prove it's a black hole, but you see an object orbiting around something else with its incredibly high rate of speed, with this activity that manifests itself by its extreme rotation. The only thing that can do that in that small enough volume, in that quick enough, short enough period, was theorized to be a black hole. Again, not proof, not definitive proof, but oftentimes we're not going to get definitive proof.
- 2.7 We're not going to go to Gargantua and stick a thermometer in it and measure its properties, its mass, charge, and spin. But we're going to measure it indirectly. And, in many cases, when you get multiple forms of evidence, that's enough scientifically to prove, at least within the boundaries of the theory, that the propensity of the evidence leads towards the existence of a black hole.
- 2.8 Now, what was causing this extreme luminosity in X-rays? X-rays are one of the most energetic forms of electromagnetic radiation. It was found to be powered by what's called an accretion disk. The accretion disk is the foam on the cappuccino. It's how you know—it's really what you're coming for, right? Let's be honest. We all get the cappuccino for the foam, not the coffee. It's how you know what's underneath exists. In our case, the event horizon is whipping around at a large fraction of the speed of light. And in so doing, it's sucking in matter from the star or the planet or the cluster of planets or the cluster of stars. They don't care.
- 2.9 The black hole doesn't care. We'll find out. Again, you can throw in Big Macs or you can throw in neutron stars into a black hole. Eventually, all that will register is mass, charge, and spin of what you put in. And theoretically, you can extract out whatever you put in. It's information. We'll find what Stephen Hawking and others battled about in the final lecture of this class called the Black Hole Information Paradox, which is quite fascinating, that theoretically, it's believed that you could reconstruct whatever fell in. But the only thing that matters to the black hole, it just takes in whatever you throw in. It's the ultimate garbage disposal. It doesn't care—mass, charge, and spin. It could be libraries, it could be Big Macs, like I said.
- 2.10 Once it gets close to the black hole before it falls beyond the event horizon, the matter itself is ordinary matter. It could be, as I said, stars or planets. As they come together traveling near the speed of light, you get incredible amounts of collisional heating, basically friction. These things are smashing together, smashing macroscopic chunks of matter near the speed of light. That causes immense heating and conversion of mass to energy in a way, mass to energy in the form of light, X-rays in this case, that is even more efficient than atomic weapons are, than an atomic bomb, even a fusion bomb, which is more efficient than a fission bomb.

- 2.11** We call these accretion discs the engines of creation because they're creating the light signature that allows us to perceive the existence of the black hole itself. Again, the accretion disk is as much a part of the black hole as the singularity, as the event horizon. You can only not get an accretion disk in a non-spinning black hole, and effectively, that's almost impossible. There's no such thing, at least in our observations, that seems to not have any spin whatsoever. It's incredibly improbable for that to happen. It's theoretically possible, but in practice, it almost never occurs. And by virtue of the spin, as we'll see later, the three most important characteristics, the three only important characteristics, we can see the engine that's powering the black hole.
- 2.12** What we're witnessing is this disk, this glowing disk, is actually the friction of the ordinary matter, the stars, the protons, neutrons, or croutons, whatever you've put in there, is going to heat up, glow, and get near the speed of light and create more energy than it would taking the equivalent amount of matter and converting it to energy using nuclear fission or fusion. It's extremely efficient, and we'll see what happens when we get to another type of black hole that is spinning with extreme rates of spin called a Kerr black hole.
- 2.13** These are the monsters that allow physics to really bend our imaginations by changing the properties not only of the matter that falls into them, but of the space and time that surrounds them, allowing time and space to get interchanged in a way that's completely foreign to our intuition. We'll talk about that when we discuss the dynamics of spinning black holes.
- 2.14** But again, there are only three things we can study, so there's only three things we can measure. There are only three different types of predictions. Even this prediction wasn't made until about 50 years after the first postulates of black hole's existence by Schwarzschild and Einstein came about. Then later, even more so was the prediction of charge. We'll see that charge is the hardest thing to measure on a black hole.
- 2.15** It leads to the main features, the only features that we'll talk about, really, because there's only these three features that matter to a black hole—mass, charge, and spin—there are only certain properties that can really be present in a black hole. The event horizon, singularity, what's called the "ergosphere." Then there are certain types of properties for things that could be present in practice. We can only visualize them, say, in the Interstellar Gargantua visualization, something called the innermost stable circular orbit, and there's something called the photon sphere. Again, we're going to talk about all of those and understand them. Because there's so few things that matter to a black hole, it actually benefits us in that we can have only a handful of ways to understand them, measure them, and that's the hard thing is actually measuring them.
- 2.16** One of the ways that we take these measurements that look like the depictions that theoreticians have predicted comes from the Event Horizon Telescope. Now, it's not a telescope you put your eye to, and you look through. It doesn't even work in the optical wavelengths of the electromagnetic spectrum. And it's not even a single telescope.

They call it a telescope, but it's a network of radio telescopes, like in the movie Contact, where Jodie Foster was sitting on her truck and listening to those giant dishes. Those dishes are actually part of the network array of telescopes.

- 2.17** It's called an interferometer, which means it takes radio waves, combines them together, and allows you to synthesize a telescope whose diameter is effectively the size of the Earth's diameter. So if you put a telescope in the north and the telescope in the south, for example, we have one at the South Pole in Antarctica, where I've been twice for a separate experiment looking for the origin of the universe, just a minor thing like that.
- 2.18** I've been there twice, the South Pole. One of the telescope network elements is there. There's some in Europe, there's some in Africa. The hard thing about it is that each one has to be pointing in the same direction, and each one gets light or radio waves from a different part of the structure of the object. It makes you have a telescope effectively as wide as the Earth's diameter. It's pretty amazing. Now, in the future, we'll have telescopes in space, and we'll have something, maybe the orbit of the Earth. We'll have a telescope, the diameter of which is the orbit of the Earth. Obviously, you couldn't build that on the Earth.
- 2.19** These data that come from the telescope are samples of voltages that are being produced by the radio waves which come from the emission from the black hole, from its event horizon, or from its accretion disk that travel through maybe billions of years, millions of light years in the case of the [first] galaxy we've measured its black hole to be, that image is called the M87 black hole, or in our galaxy center, some tens of thousands of light years away. Those waves travel for tens or millions of years, and then they get detected at slightly different times at different telescopes, depending on where they are on Earth. Scientists using mathematical algorithms can reconstruct the image, but it's a radio image of what the black hole looks like. That's what the event horizon images look like.
- 2.20** The first one to be revealed was the most massive one that we know about. It's literally a thousand times more massive than the Milky Way's black hole, Sagittarius A* that we talked about.
- 2.21** We see these features that are exactly indicative of the black hole's mass, charge, and spin. Namely, we see the so-called "photon ring." Now, the photon ring is not the accretion disk itself. We're going to see it's going to be related to the inner boundary of the shadow that is being absorbed, that is the absorbing shadow, like the black hole's event horizon, it's absorbing this light, and we're seeing a shadow of the light, not of the matter. It's a very curious type of phenomenon that we will explore. We also see brighter regions and dimmer regions, and that's because of relativistic Doppler beaming. That's due to the motion of things orbiting around this black hole. Now, what's orbiting around this black hole? It has a mass, as I said, of 6 billion times the mass of our Sun.

- 2.22** We don't find any use in measuring things in astronomy in terms of kilograms or meters or things like that. Instead—or seconds—we have to use years, millions of years, light-years, which is the distance light travels in a year. That's not even enough. We have to use thousands or millions of light-years to describe how far away these things are. Similarly, we can't measure mass in pounds or kilograms. We measure it in masses of the Sun.
- 2.23** So the first thing that they noticed when this image was first released in 2019, where the properties were exactly in accordance with the theoretical predictions of what you'd see for a given mass, a given spin, they can't measure the charge of the black hole. So they measure two of the three properties using this. Since then, they have measured properties of the charge, namely the magnetic field that this moving charge material produces. Magnetic fields are predominant throughout the universe. And in particular, when you have a large amount of current charge moving at high velocity, you get large magnetic fields.
- 2.24** These images really revealed not just qualitatively, which are the most beautiful kinds of images you can imagine for a scientist studying these types of properties, but quantitatively, the size of these features, the way that they're distributed across the black hole, the way that they will be later, as we will see, revealing the three major properties of the black hole, were completely consistent with relativity, what's called a Kerr black hole, which is a spinning black hole, which itself is a complexity associated with the Schwarzschild black hole.
- 2.25** Now, we'll see at the very end of the class that black holes, some of the largest objects, most massive objects in the known universe, have properties that are almost entirely dependent upon the smallest aspects of science, namely the properties of the subatomic particles that make up ordinary matter, namely the protons, electrons, subatomic particles, elementary particles, and photons. Those are typically described by quantum mechanics.
- 2.26** But gravity is described by general relativity. General relativity, as you have already seen, is a quite baffling theory because it involves enormous masses, many, many trillions upon trillions of individual particles going to make up, say, the Earth or the Sun. For a black hole, the properties of the subatomic particles we'll see, cannot be resolved. Their quantum mechanical properties break down at places like a singularity at the center of a black hole.
- 2.27** We'll see, for example, that when we encountered the Newtonian theory of universal gravitation, that the force of gravity depends on the inverse square of the distance between two massive objects. Well, what happens when the distance is near zero? What happens when the distance is below the size of the quantum or at the size of the quantum? Then the force would be infinite, and physics can't really treat things that are infinite. They do behave as singularities where mathematics breaks down and we can no longer do physical predictions.

- 2.28** We will also see that quantum mechanical effects occur at the event horizon itself, where some particles can come into and out of existence called particle-antiparticle pairs. Matter, antimatter can come into existence. And then one of them can go into the black hole's event horizon never to escape, and the other one can escape, and we can watch a photon from that particle.
- 2.29** But what that means is very baffling quantum mechanically and even what's called thermodynamically. We'll see that black holes can emit heat. Well, that's just weird because heat is a form of radiation, which is a form of light. How can something that's black emit heat? Turns out that's a problem Stephen Hawking solved. We're still wrestling with the implications of that because if information can fall into a black hole, maybe it gets destroyed forever. But that violates a lot of principles that are held in quantum mechanics, that information can never be destroyed. We'll see that in the very last lecture.
- 2.30** That process called Hawking radiation, depicted here from 1974, is one of the big features of the movie, The Theory of Everything. Black holes can emit heat via what's called Hawking radiation, just like a glowing coal in a cooling barbecue emits heat. Eventually, it'll come to absolute zero—it won't come to absolute zero, but it'll come to a colder temperature, the absolute temperature of the room it's in or the outdoor barbecue that it's in. Well, what temperature will black holes get to? They'll actually evaporate away, giving off radiation, and they can actually give off tremendous amounts of heat, it's found, as we'll see in the final lecture of the class.
- 2.31** These are fascinating aspects of black holes that connect the most massive objects, black holes, centers of galaxies and compact objects, with the most microscopic laws of physics known as quantum mechanics. So black holes aren't forever. They don't last forever. They're not even so black as Stephen Hawking said. They actually emit heat, they glow. But the caveat is the temperature that they're emitting at is almost absolute zero. And the rate of heat loss, the power that they radiate, will take something like 10 to the 70th years before they disappear. But there's no saying the universe won't last that long. So the question is, what happens in that deep, deep future? And should we or should we not keep paying our taxes? That, I always say, is the question.
- 2.32** And as I said, in quantum mechanics, information can't be created or destroyed, like energy can't be created or destroyed. Matter can be created or destroyed, but energy can't be. So where does the information go that falls into a black hole?
- 2.33** If I take my first book, which is called Losing the Nobel Prize, and I beam in a digital form into a black hole, does that information get destroyed? If so, that violates the conservation of information because you can't destroy information. So you can't destroy energy. You can convert it from potential to kinetic, kinetic to potential, but you can't destroy it. It goes somewhere. That's related to thermodynamics. We'll cover the properties of thermodynamics briefly, quantum mechanics, and the ultimate mystery of the information paradox.

- 2.34 Thus far has been an outline of all the interesting features of the black hole, a map of the terrain that we're going to cover in this class. Now we're going to get into some of the details of how we know what we know and how we know what we do not yet know, and how are we going to learn to fill in the gaps between what we know and what we don't know and what we want to know for many, many reasons.
- 2.35 As you can tell, we've already touched upon some really interesting aspects of philosophy, as I said, information theory and other things. But the most interesting thing is about to happen because we're going to explore the very nature of what it means to have an observer, to be traveling through space, what space and time themselves mean. This takes us back to the theory of special relativity. There are two theories of relativity. A lot of people talk about it in popular terms. What you think is relative, what I think is relative. That's not what relativity means.
- 2.36 If you remember back to Newton's first law, the law of inertia, it said that an object in motion continues in motion unless acted upon by a force. Now, it doesn't say an object at rest tends to stay at rest unless acted on by a force, even though that's almost equivalent. But the reason it doesn't say that is because there's no such thing as rest. Because I can always say rest with respect to what? Rest relative to what? Even motion, if you like, is relative to something in motion and a form of being at rest. You could say you're at rest, I'm moving around, I'm a New Yorker, so my hands move around a lot.
- 2.37 But it's relative to what? There's no absolute standard of rest, as the ancient Greeks thought. The ancient Greeks thought that the natural state of matter was at rest, at the center. The most perfect distribution of matter was around the most perfect object there was, which was the Earth. The Earth was the center of the universe. Everything tended to be at rest, and Newton obliterated all that.
- 2.38 But Einstein came along and did one better over Newton because he described, as I said earlier, what happens if you're traveling near the speed of light and you turn on a flashlight. What Einstein said is everything is relative. It doesn't mean that you can win an argument with your older brother by saying, "Well, that's just what you think." No, your relative is relative. That's not what it means. It means that motion is indeterminate. You can't say who's moving and who's stationary. You can't say that your motion is better than my motion. No, there's no such concept. There's only one thing that's the same for you, for me, for someone on Vega, for someone at the black hole near the center of our galaxy. It's the speed of light.
- 2.39 Now, what is speed? Speed is distance divided by time. It's all it is. Speed is distance divided by time. So speed involves time and space. And we'll see that in order to have speed be constant, time and space have to be non-constant. Time and space have to vary depending on certain properties relative to certain features of the observer.
- 2.40 So special relativity brings in an understanding of the dynamics of moving objects. In fact, that's what Einstein's paper was called, "On the Electrodynamics of Moving

[Bodies]." It was found that you couldn't have the laws of electromagnetism and the laws of Isaac Newton, they were not compatible with each other. Einstein said one thing: if the speed of light is constant, everything else has to change. Everything else must be relative in a very narrow range, according to physics.

- 2.41 It meant that time cannot be absolute. Remember, speed is space divided by time, space divided by change in time. If the speed of light is an absolute constant, then either space is not absolute or time is not absolute or both. And that's what is the real true insight that Einstein had. Einstein showed through a series of equations, where the math is not that much more difficult than the Pythagorean theorem.
- 2.42 Special relativity is motions, adding motions together, how time and space are manipulated by velocity. And then general relativity is the laws of special relativity near gravitating objects like planets, stars, black holes. It will be a separate course. It's very fascinating, but I have to give you the basics of each one of them now in order for us to proceed to understand the beautiful math and dynamics of black holes.
- 2.43 For light to be constant, space or time or both have to change. What it was found was that a clock in motion effectively is slowing down. You'll see that and you'll remember that from the movie Interstellar. Although in that case, we'll see gravity affects time and space as well. So motion and speed affects space and time, and then gravity affects motion, which affects space and time. That's the logic that we're going to proceed.
- 2.44 The faster you go through space, the slower your clock appears to tick to somebody else. But also to them, to you, their clocks are going to appear different to you. Just like when you're in a car, you come to a stop light and then you're looking at your phone, which you're not supposed to do, and then you see the other cars are moving, you're like, "Wow, I'm moving, right? My car, I just hit the gas pedal." No, they're moving, and it just feels like you're moving. But if you are stationary and they're moving forward, it looks like they're stationary and you're moving backward. That's why you have this sensation. So, the faster you move through space, the slower you move through time. You have to go really darn fast before it starts affecting your clock as seen by somebody else.
- 2.45 Here's a diagram. The slowest the time can tick for a stationary observer is just one second per second. But it can slow down tremendously if you're moving near the speed of light. It actually goes up in this hyperbolic asymptotically infinite time dilation.
- 2.46 If you're moving at the speed of light, no time passes for you as perceived by somebody else. Not according to you. Your heart keeps beating one time per second on average, and you get slowly gray hair. You can't stay young forever. I've tried. But you cannot do it. You keep aging at one second per second on your internal biological clock. But to somebody else, you don't age at all if you're going at the speed of light.
- 2.47 But we'll see to go at the speed of light requires an infinite amount of energy for a massive object like us. But for a massless object like light, it requires no energy. That's

what light does. That's one of the definitions of light, is that the elapsed time for light is zero. It never changes. It never ages. Photons never get older. The amazing fountain of youth.

2.48 We've measured this effect, predicted, again, over 100 years ago, 1905. Just recently—recently by cosmological standards—in 2010, so 15, 16 years ago, they took very, very precise clocks. And they had already measured very high-speed clocks. We'll see in just a second that it had already been measured. But they said, does it even work for the speed that Usain Bolt can run at? I can't really run at 20 meters per second. But they actually showed that for walking speed and the very most left part of the one meter per second, they were able to see that time slowed down for the objects that they were looking at, which were quantum mechanical arrangements of atoms and ions called ion clocks.

2.49 It slowed down by a few parts in 10 to the 15th, quadrillions of a second, so a thousandth of a trillionth of a second per second for these objects. Like walking. If you were holding one of these clocks and you're just walking along, it would slow down by trillions of a second. Now, if you're moving fast to the speed of light, which is 300,000 kilometers per second, that would cause a much, much greater time dilation such that it would be a 100% change in the speed of light or your speed of time, and you'd be basically slowed down completely.

2.50 This shows how precisely they were able to measure it. These are the error bars. An error bar tells you how much of a mistake could you possibly have made and still be consistent. You didn't make an error or blunder. It's just there's always some uncertainty. How fast is the clock moving? How accurately can you measure it? It's actually quite difficult to do these measurements, which is why it took so long to do it. But we've had measurements for over 100 years that this effect is exactly what's predicted by the theory of special relativity.

2.51 I said the speed of light is constant for all observers. If I shoot a laser beam on a train, it will travel in my reference frame, even if the train is going 100 meters per second or whatever it's doing, it will travel from my perspective on the train. Here, I'm pointing it out, it will travel at the speed of light. You on the stationary platform watching me on the train shooting the beam of light, will also measure the speed of light being the speed of light, C , 300,000 kilometers per second, pretty much. The only way that's possible for that to happen is not only does time slow down—in this case, time for light is effectively not elapsed at all. Time is zero, changes zero for light beams, particles of photons.

2.52 But what happens for the train at rest versus the train in motion, when the train is in motion, it actually is effectively squished along the direction of its motion. So time slowing down is called time dilation. Things take longer or elapse slower, time flows slower. But the lengths get smaller and shrink when they travel near the speed of light.

Again, these are tiny effects, but they're actually measurable effects. You can actually measure the effects of both time dilation and length contraction.

- 2.53** Reality, though, can only happen in one way. It can only have one true event. An event either occurs or doesn't occur. Here's a perspective that has been known about for about 50 years. There are subatomic particles. They're called elementary particles. An electron is an elementary particle. It means it's not composed of other things. A proton is not an elementary particle. It's made of quarks. Quarks are elementary particles. You can't chop them up into other things.
- 2.54** There's another particle called a muon, which is like a super heavy version of the electron. But unlike electrons, which are stable and seem to have a lifetime, at least as old as the age of the universe, muons decay really fast. They actually break apart when they form, and they live essentially for just a few microseconds, a few millionths of a second.
- 2.55** Now, sometimes muons can be produced by what are called cosmic rays— particles that come out of other stars, supernova explosions, our Sun. They travel toward the Earth. Sometimes they hit a molecule in the Earth's atmosphere. These are typically protons or electrons. That collision produces muons. Now, the muon has a clock. It's basically how long it lives on average. It's called its half-life, just like a radioactive half-life.
- 2.56** When a laboratory muon is produced, it lives for 1.56 millionths of a second, 1.56 microseconds. If we're watching it on average, not every single one will only live that long, but on average, the average lifetime of a muon before it breaks apart into other particles, it actually isn't breaking apart, it's decaying into other particles. It doesn't mean that those particles are inside of it or that it's not elementary. It's just one of the ways that it radioactively decays. It decays into other particles, and we can measure how long that takes.
- 2.57** For a muon just sitting here in the lab at rest with respect to us, it will live on average. If you have billions and billions of these muons produced, on average, they live 1.56 microseconds. When you produce them in the atmosphere from a cosmic ray hitting an air molecule, it produces a muon, and then they can live, they're traveling now at 98% of the speed of light. They can't travel exactly the speed of light. That would take infinite energy because they have some mass.
- 2.58** If you create a million cosmic ray-produced muons at the top of the atmosphere, 10 kilometers up, some of them will live until they hit the surface of the Earth, which takes more than 1.56 microseconds. Their clocks have slowed down. They have been allowed to age, on average, much, much longer. They can actually live by a factor of about five times longer, about 8 microseconds. Again, to get this extra lifetime, they had to be basically produced in a velocity of almost the speed of light. 98% of the speed of light is a tremendous velocity. Completely impractical for us to get to.

- 2.59** A jet travels at 0.1% of the speed of light, nowhere close to it. The fastest object we ever launched from Earth is a satellite called the New Horizons satellite, and it went to Pluto back when it was a planet. Sadly, it's no longer a planet. It made it there in something like six or seven years. That's not even the edge of our solar system. It took six or seven years traveling [35,000] miles per hour. That's about the fastest we've ever made it. That's a fraction of the speed of light, less than a percent the speed of light. So these muons are proof that time slows down for an object in motion when it's moving near the speed of light.
- 2.60** Now, not all of them live that long. Again, it's a very small fraction, a few percent, 5% or so, 49,000 out of a million. But that is enough to demonstrate this effect. And now it's been measured in many, many other different contexts. You can produce them in particle accelerators and accelerate them moving around at extremely high velocities. Measure this extremely precisely. They match exactly the prediction of Albert Einstein.
- 2.61** Now, if time is relative and space is relative, then now and here are also relative. Remember, I said we exist in latitude, longitude, like some latitude, some longitude, some altitude—three dimensions in space—and now in time.
- 2.62** But imagine there's another Peterson Academy, and it's moving. It's in space, and it's on the International Space Station, and there are all these brilliant students, and they're moving with respect to us at half the speed of light. When I say something right now, it's not right now for them. In fact, we would see them slowing down. They would take much, much longer to complete this class than you will do it in 8 hours. They might take 16. But the illusion of when I started speaking right now, we can't say when now is and have all observers agree on it, as this diagram shows.
- 2.63** Imagine you're in this rocket ship. The rocket ship is just cruising along. At the center of the rocket, at the center of the spaceship, you light off a firecracker. Not a smart thing to do in space, but let's say you do it, and two beams of light from the explosion hit either side of the spacecraft at exactly the same time because it's exactly in the middle.
- 2.64** But you, looking at it from Earth, see something very different. In other words, in the spaceship, if there are two people at each wall, A and B, they would say, "I see it exactly at the same time in that reference frame." People on Earth, looking at that event, would see the event first reach the left side, because it's accelerating from left to right—moving from left to right. It doesn't have to be accelerating. As long as it's moving at constant velocity, you'll see the same effect. The faster the rocket is moving with respect to the Earth, the larger the difference between the left event, A, being seen, and the right event, B, being seen.
- 2.65** For example, from our perspective, if it's moving at the speed of light, it will never reach the other side. It will never have enough time to reach the other side. It's just you would "see"—you can't see this— but the light beam would be traveling always toward the other side. But the other one would see it almost instantaneously, it would hit

event A. If it was now for both observers and stationary, it's not now for both observers that are seeing it moving. Therefore, now doesn't exist.

- 2.66** The whole notion of what's called simultaneity, simultaneous events, is a fiction, and it's an illusion of the fact that we on Earth move at very slow speed. If we were living at —very fast velocities were very common for us, we wouldn't have the same notion psychologically of what it means to be alive right now. It really would not exist. It was a life-changing, philosophically shaking, shattering the perspectives of philosophy for all time when this notion came about.
- 2.67** Now, if now is an illusion, then how can events take place? How can you say that event A caused event B? Here's just a cartoon example. You shoot someone with a laser beam, and I say the laser hit the person first and then the person died. But if the thing is moving to the left, you'd see the person die before the laser was shot at the person, right? Doesn't that confuse things? There's a concept we call protection of causality. Causality is the phenomenon where effect follows cause. So something shoots a gun, that's the cause. Effect: this person dies—sorry for the morbid sentiment, but it's just in cartoon form.
- 2.68** What is the protection that causes the cosmic events to never violate causality? It's that the speed of light is constant. As long as the speed of light is constant, you can't have things like the so-called grandfather paradox, where you go back in time, kill your grandfather before he met your grandmother. Keep it PG-13. If that happened, you could say, "Oh, well, then I wouldn't be born. But then I couldn't go back and kill my grandfather, so then there's no paradox." But then I didn't proceed to cause this causal loop to be violated.
- 2.69** So scientists introduced this concept of what's called light cones. There are two light cones, and then we exist at the apex joining point of these two cones. That point we call now or the present. The past light cone is all events within which all events could effectively affect us. We could, for example, in the past, you guys got in your car and came here. There's some event there, time, space, and position, not far away in space, not too long ago in time, to get here on time. You guys are all great students. You got here on time. You can't be that far away.
- 2.70** If you were outside the past light cone, you could never have made it here in time. If you were at the galaxy that we called M87, 55 million light years away, the only way you could have made it here today is if you left 55 million light years ago—sorry, 55 million years ago—traveled at the speed of light, and just so happened to get here exactly at this time, knew exactly where the Earth would be in rotation around the Sun. It's very improbable, but it could happen. So the past light cone expands at a rate of one light-year per year. And here, time is going down or up. The past is down, the future is up. We cannot influence things that are outside of our future light cone.

- 2.71** If you tell somebody, you've really got to take this course by Professor Brian Keating. If you told them and they're outside the future light cone, they're on M87, they're in a spacecraft orbiting around the black hole in M87, not advisable. You can't tell them unless they happen to be 55 million years in the future. You could barely tell them right then.
- 2.72** For example, we can't go back and warn the dinosaurs, "Watch out, there's an asteroid. You better build a giant space laser or have a space program." We can't warn them. They are inside of our past light cone. We see their bones and stuff like that. But we can't send a signal retroactively in the past.
- 2.73** Everything that's going to happen to you in your future is in the future light cone. Everything that could have possibly influenced you entirely, whether you knew it or not, is in your past light cone. Where we are right now is that red dot. That's that now for us. But this is just for us, stationary in this room.
- 2.74** For another classroom on the space station, they have a light cone that tilts and tips with respect to ours. There are certain things that they can influence that we can't influence, and vice versa, because they're moving with a certain velocity and they're in a different position. This is the notion. Their present is slightly different than our present. The only thing that we agree on is the angle of the cone is 45 degrees. That's the speed of light. The speed of light is how much time changes for a given amount of space changing for a light beam. Everywhere else is elsewhere, and you cannot influence it as much as you might want to.
- 2.75** You might want to go back in the past and know what happened in the Super Bowl and influence it so you can bet on some game and make a lot of money, but you can't. You can't go back and change events in the past. They are separated from you by this elsewhere feature of the light cone.
- 2.76** One of the most interesting consequences, one of the only equations we'll deal with in this class, but for those of you who like the equations, we'll deal with that a lot in our general relativity and special relativity class. I like to show you the equations because I find them beautiful. They have features that are attractive in a weird way to scientists and nerds like me, hopefully you. Some of those features are symmetry.
- 2.77** Here we have two different observers. We have a person throwing a ball on the train. We have a person catching the ball on the ground. Galileo, the great scientist who was born on February 15, [1564], and lived to the ripe old age of 80-something, 70-something. He was the first person to think of relative velocity. He did it in a very famous book called The Dialogue.
- 2.78** In that book, he was trying to motivate that the Earth orbited around the Sun. But in the 1600s, that was a very dangerous thing to do, especially in Roman Catholic Italy, which is where he lived. He postulated against the Bible that the Earth in motion could be derived from the behavior of the tides of the ocean moving as the Earth moved

around the Sun and rotated on its axis. I talked about that in my course, Astronomy 101. What happened to him, of course, is quite tragic and famous. In a way, he was imprisoned. It turns out his prison was pretty nice. I've been there three times. It's a luxury villa in Arcetri in Florence, Italy. You can see the Duomo, and it's a beautiful place to visit.

- 2.79** But he came up with the first law of relativity. It's called Galilean relativity, that velocities just add. If you're moving at 100 miles an hour and you throw the ball at 50 miles an hour, then you'll get a total speed of 150 miles an hour. But according to Einstein, that's impossible because if you're moving at $8/10$ the speed of light and you add 0.7 the speed of light, you'd get 1.5 times the speed of light, which is forbidden if the speed of light is the maximum velocity.
- 2.80** And so the equation that determines the addition of velocities is actually this equation here for V . And it's that the velocity that you see of the ball or the rocket, in the case of the spaceship, is actually the sum of the two velocities, as it would be for Galileo, just the velocity of the train plus the velocity of the ball. But you have to divide by one plus the product of those two speeds, divided by the speed of light squared.
- 2.81** Now, the speed of light—let's do it in miles per second—is 186,000 miles per second. You square that—that's an enormous number. It's certainly a lot bigger than 50 miles per hour plus 100 miles per hour. That product is incredibly small. So for ordinary velocities near human speeds, VP and VS are much, much less than the speed of light. And when you square something that's much less than something, it becomes even much, much, much, much, much more less. So for ordinary velocities, the velocity addition reduces from Einstein to Galileo, VP plus VS .
- 2.82** But near the speed of light, you get strange behaviors. To get something moving, if you were to add 0.8 and 0.7 , you get 1.5 in the numerator. That's true. But in the denominator, you get 0.8 times 0.7 — 0.56 —divided by c^2 , and so it'd be $0.56 c^2$ or something. I'll call it 0.5 . You square that, you add it to that, and you get one point, and in the numerator, you got 1.5 . In the denominator, you'd get 1.56 or 1.6 or something like that. In the denominator, you would get something that's closer to one-half the speed of light, or maybe a little bit more 0.7 , 0.8 the speed of light. So that's less than the speed of light; therefore, it checks out. As long as you don't violate velocity addition, greater than the speed of light, you're okay.
- 2.83** The last consequence is that to make something massive travel near the speed of light, you need infinite energy. There is a very, very well-known factor to physicists called the Lorentz factor. It's related to what we called before the Lorentz-Fitzgerald contraction, and it has to do with how much momentum and energy you need to supply to something to accelerate it to a given velocity. As you see here, it's asymptotically— asymptotically means it approaches infinity. To get close to the speed of light for any mass, even a speck of dust, let alone something much heavier than that, you would

need an infinite amount of energy because you're effectively dividing by something that will go to zero. This is a consequence.

- 2.84** All six of these consequences came from Einstein's famous work on the theory of special relativity. We haven't gotten to general relativity yet, which governs massive objects like black holes, but we will. Rest assured, even though there's no such thing as absolute rest. We call stuff rest. We call it rest mass energy. The most famous equation in all of science is $E = mc^2$. It came about in Einstein's famous 1905 series of four papers. Each one is incredible and should have won the Nobel Prize, but again, didn't because he was a German Jew. The basically Aryan scientists of the day didn't allow him to win it until 1922. It says, we won't derive it, but it says that if you take an equivalent amount of mass, you can convert it to energy by a simple product of the speed of light.
- 2.85** Now, there's a funny cartoon where it shows Einstein at the chalkboard, and he writes down $E = ma^2$, crosses that out. Then he goes, $E = mb^2$, crosses that out. Then c^2 — c is just the speed of light. C stands in Latin for celeritas. That's where it comes from. The speed of light is a constant. Most famous constant there is. And so this equation is going to play a role in this class. Not that we're going to derive it, but that we had this association with how much energy could you theoretically get from a given amount of mass of any form.
- 2.86** It doesn't say mass in the form of uranium. It doesn't say helium. It just says mass of any kind. Again, you could throw in a basketball, you could throw in a Big Mac, you put this in there. That's how much rest mass energy theoretically could be converted to energy. It's impossible for any mechanism to convert all the mass to energy as far as we know for macroscopic masses.
- 2.87** Now, we do see this effect in many different instances in physical conditions, like particle accelerators and the most massive explosions there are in the universe, which are supernova and these cosmic ray production and X-ray accretion disk that we talked about.
- 2.88** Here's an example. To create a tiny amount of matter, the lightest fundamental particle is the electron. It has an antiparticle called the positron or e^+ here. It's the positive electron. To create a positron and an electron requires crashing together two photons with enough energy twice their rest mass. So twice the rest mass is called 1.022 million electron volts. That's the mass of it. And you need to create this. Now, there's almost no source of energy that can produce this. In fact, it's never been observed in this form.
- 2.89** The opposite process where a positron, an electron with a positive charge, hits an electron with a negative charge, produces two gamma rays, that has been observed many, many times, and we'll talk about that later in the class as well. It's a mind-bending concept that the laws of nature depend on things that are not absolute, that are relative. Now, again, not meaning everyone's entitled to their opinion. That's not

true. You're not entitled to your opinion on the mass of an electron. You're not entitled to your opinion on how many different properties a black hole can be manifesting. But you are entitled to your measurement in your own reference frame.

- 2.90** Then it's up to us to compare the predictions of Einsteinian relativity to the observations and then deploy them into extreme circumstances. Things moving near the speed of light, things with large masses, things with high rates of spin, things with enormous charges. That's what astronomers use. We can't build enough of them here on Earth, so we use the cosmos as our most perfect laboratory. And so far, we see the implications of Einstein, of Schwarzschild, of Chandrasekhar, of Andrea Ghez. We see them manifest in every observation to date.
- 2.91** Does that mean they'll be right for all time? Absolutely not. A good scientist hopes that he or she will be disproven because otherwise we know the answer. If Newton, as brilliant as he was—Einstein thought Newton was the most brilliant person who ever lived. It's funny to think Einstein was so arrogant, but he had the imposter syndrome when it came to Isaac Newton.
- 2.92** But Isaac Newton had the imposter syndrome. He didn't think he was worthy. His hero was Jesus Christ. He said, "I died completely unworthy of my hero, Jesus Christ." In fact, he tried to be like Jesus. The only way he could be was to die a virgin, and he did. Some say that it was because of his personality. That's another thing. We'll talk about that later in a different Peterson Academy class. But everybody has idols, and they have people they feel like they're imposter of.
- 2.93** That means that no one's the final absolute, right? What if we stopped at Galileo and said, "Oh, Galileo, he's the final word," then we wouldn't have Newton. What if we stopped at Newton? We wouldn't have Einstein. What if we stopped at Einstein, we wouldn't have somebody out there, one of you maybe, in the audience today.
- 2.94** Relativity is not to say anything is possible, everyone's relative, you can do whatever you want. No, it says very specific, quantifiable, measurable, predictable, and possibly falsifiable, but not yet, predictions about the physical world. So it emerges a new universe from these simple postulates. Remember, all it says is that one number is constant. From that, we get six different consequences, some of which will then be applicable to the theory and application of black holes. Oh, let's take questions. Yeah, please.
- 2.95** I wanted to ask you, based on what we understand here of time, two questions. One is, if theoretically we could go faster than the speed of light, would time be going backwards? Another question is, do we, based on what we know of time as of today, know exactly how long a moment of time is?
- 2.96** I'll answer the second question first. A moment is just what humans call some difference in time between events. Let me first say that time is not really well-defined.

Time has multiple definitions. There are as many definitions of time as basically there are branches of science.

- 2.97** For example, as I said, biological time, how you age. There's psychological time: God, this lecture is taking a really long amount of time. But it's the same hour for everybody in the room. But for some people, it might feel like it's long. That's called psychological time.
- 2.98** Then there's a concept called the arrow of time. The arrow of time attempts to classify time as the byproduct of another physical phenomenon called entropy. Entropy is a measure of how things change, but change with respect to what? Usually change with respect to time. And so it becomes circular, the definition of time. The best definition that I've ever heard came from Nobel Prize winner named Frank Wilczek on my podcast. And he said, "Time is what a clock measures." Again, it's circular. It doesn't really answer the question of what time is. A moment is a difference in time between two events. The notion of absolute now, again, exists relative to a person.
- 2.99** You are experiencing your now. Your now is slightly different than her now because you're moving slightly different because you're on a different surface of the Earth, you have a different velocity. Technically, your times—even the person right next to you. There's actually a difference in the flow of time between your left ear and your right ear, but it's imperceptible. Time, for it to be useful, has to be connected to something observable, measurable, and quantifiable.
- 2.100** Now, we can measure tiny increments of time. But what scientists have done is say that we can't measure tiny changes in speed. We say speed is a constant; the speed of light is a constant. We can't measure it. If you try to measure the speed of light, you're actually measuring how the rulers that you're using to measure the length and the clocks that you're using to measure the time are. But you can create a ratio between time and space that cancels out the subjectivity of it.
- 2.101** Some people try to connect time to entropy because the laws of nature aren't asymmetric. In order for time to flow in one direction, it must be that there's some asymmetry between the past and the present, or otherwise, you wouldn't notice anything different. We know that for our bodies, our kids grow up and our beards grow, whatever. But the laws of physics are time-reversible. They're time-symmetric. If you look at the motion of the Earth around the Sun, you can't tell where you are. If you looked at it upside down, it would look different.
- 2.102** Or here's another example. You ever see a grandfather clock, a pendulum? If you don't see the clock face, you just see the pendulum going back and forth, you can't tell if it's going left or right, or if you started it off right to left. It's completely arbitrary. The laws of physics don't care about the direction of time. But the macroscopic manifestation of trillions and trillions and trillions of particles in our body seem to, and our brain seems to, and other people seem to.

- 2.103** To answer your second question, there's no definition of a moment other than the elapsed time differential as measured by a single observer in a single reference frame. But that's a completely idealized concept which has no absolute definition. So that's question number two.
- 2.104** Question number one, what happens if things move faster than the speed of light? So faster than the speed of light travel has been postulated for a long time. And the consensus is that it is theoretically possible, but for such particles that move faster, massive particles, actual particles, not fictitious things or not massless particles like photons, which can't move faster than the speed of light, by definition, that those things always move faster than the speed of light.
- 2.105** They can never slow down to below the speed of light or slow down so much; they're only traveling at the speed of light. Sounds weird. They're called tachyons, and they have lots of different problems. One of them is that they can affect causality in certain particle collisions. You could have a nuclear fusion event occur before the particles were produced. There are strange aspects of that. So most classes of physics treat it as a purely theoretical phenomenon.
- 2.106** We'll see in the very last lecture, there are ways to travel in time that actually—or travel faster than the speed of light using what are called wormholes that traverse distances that are truly astronomical in a finite amount of time, which means that you're traveling faster than the speed of light compared to some observer "at rest." But they're great for movie concepts like Contact and Interstellar and 2001: A Space Odyssey, my favorite movie. But they don't seem to be theoretically well motivated nor physically possible.
- 2.107** This has got me pondering on the imagination of Western civilization and how Newton and Galileo have shaped our culture, as science is a part of culture, like you said. I, as an observer here in this lecture, I'm finding it even difficult to transition my mind from the old way of conceiving the now and time to what we have been discovering. I'd love to know what challenges you see the Western mind or the Western imagination facing as our culture shapes around these new theories.
- 2.108** When we think about science, we think about classification. There's things that are like "-onomies," like astronomy, and then "-ologies," like geology, and they have to do with certain types of classification. But typically what you'll see in the tree of knowledge depicted as, you'll see the trunk of the tree is physics. It's like the observational physical properties of matter, of energy, of space, of time. That'll be the trunk of the tree.
- 2.109** Then the branches will be chemistry. In a sense, chemistry is like applied physics, the physics of the atom, but applied to purely molecular and chemical considerations. It doesn't mean it's easy or simple. Biology is applied chemistry in a certain sense. It's much more complicated. Then you look at human sociology and physiology. Those are farther downstream from biology. Then you get to culture, and then you get to art,

then you get to civil—It's so hard. It's much more simple just to study the trunk. The trunk is just one thing, it sticks out of the ground. The ground is math. The ground is math and philosophy and logic. That's what everything has to be rooted in.

- 2.110 Things happen in civilizations that aren't purely rational, and that's a good thing. As I said, it's not rational to look at a piece of art and say, "Well, this is going to stimulate these photoreceptors, which is going to cause this emotional reaction," because it's very difficult to predict the downstream consequences of physical phenomena.
- 2.111 I would say that besides the misapplication of the notion of relativity in the popular parlance that everything's relative: "Well, that's what you think, relative, okay, big brother," the misapplication of it, I would say that the combination of that, as I said, with art in the early 20th century, Picasso, the Cubist movement, things like that, poetry and lack of structure, and the decoupling from Renaissance art, which is more realistic to Impressionism, those things may or may not have happened without the developments in science. But it's so complicated. It's so above the pay grade of a scientist to think, what would be the downstream consequences of understanding the black hole information paradox, which we will encounter at the very end of the course?
- 2.112 Because that really, again, is challenging our notions, again, of we've already challenged our notions of time and space. We now have to think about information. We have to also think about things like the ultimate extent of our existence of time itself coming to an end. So now, gravity comes to an end in the black hole. Gravity is this combination of space, time, matter, energy, curvature, all these different properties that we'll explore next.
- 2.113 But the concept of how that then propagates downstream into culture is so fascinating. But it's also so hard to predict because it's dependent on things themselves, which are highly nonlinear, namely our human minds. But I definitely think that there's no limit to what influence you could have. Seeing the influence on music, on art, tells me the more that we learn about these mysterious things, the more that they fascinate the public.
- 2.114 It's ironic that we have this notion that it's boring or it's tedious or I don't know math. Einstein used to say, "Don't tell me about your problems with math. I have my own problems with math." But in that sense, I do think it is relative, relative to our imagination. As you said, this curiosity that we have is what impels us to make new discoveries in physical sciences. But it's untold what will happen beyond that.

Lecture 3: When Gravity Bends Time

At a glance — When gravity bends time.

- 3.1 We've already come quite a long way. We've encountered new definitions that required a revitalization of how we understand time and space. Now we're going to extend the bizarre mental machinations into what matter, gravity, energy, and space-time itself actually is.
- 3.2 To go back to the first quantitative explorations of gravity, we have to go back to Newton, who, in order to understand the laws of gravity, had to first invent a little thing called calculus. It was found to be impossible to understand gravity and even space, motion, and velocity without an understanding of how tiny changes in both position and time can change how we understand space and motion.
- 3.3 So Newton, as we've already mentioned, came up with the first law of universal gravitation, meaning that the same laws that apply on Earth apply in the heavens above. That was quantified by this law called Newton's law of universal gravitation, which is that the product of the two masses of an object, divided by their distance squared, times a small constant, gives us the force between those two objects.
- 3.4 That is the law of universal gravitation, which has some features in common with other laws of nature, including the law of electric charge and electric attraction, which is called Coulomb's law, which is also an inverse square law, and even the propagation of the intensity of light, which also obeys an inverse square law, in that if you move twice as far from an illuminated object, the intensity of the light goes down by a factor of four.
- 3.5 That's a consequence of the dilution of force fields and lines of force over the area of the sphere that bounds the system that we're talking about. Mars, for example, is about twice as far from the Sun as the Earth, and it also has a weaker gravitational force from the Sun by a factor of four. The application of calculus to the laws of gravity and the observational facts that were known empirically— falling masses, moving apples, moving planets and moons—led to the adoption of this law.
- 3.6 Now, there was no understanding of where the law came from, but the proof of the laws of reality explained many previously understood and measured phenomena like the planetary orbits and periods of planets, for example, that Kepler had come up with almost 200 years or 100 years before the laws that Isaac Newton had proposed.
- 3.7 So where this comes in is that gravity, if it's a force, must produce an acceleration on all masses, via Newton's other second law. His second law is $F=ma$. That means that to double the acceleration on a chunk of matter, a baseball, a beach ball, a black hole, you need to double the amount of force that you apply to it.

- 3.8** So the question was whether or not the same resistance to motion that is called inertial mass in the equation on the left, lower left, is that the same as the gravitational mass of an object? It's actually not known, or why it may be, that the force of resistance to changing in motion, changing acceleration, changing velocity, rather, why that should be the same as the gravitational mass that we associate with this term on the right, $m(g)$.
- 3.9** The puzzle is that they all behave as if they're the same. There's no intrinsic reason for that. But at all levels of precision to parts per million, even better, trillion, we measure the equivalence principle of these masses to be the same, which also means that an object freely falling in a gravitational field will fall at the exact same rate, independent of what it's made of and even independent of its mass. Therefore, a feather and a bowling ball will fall at the same rate in vacuum, as Galileo showed in the early 1600s.
- 3.10** Galileo wasn't the last word on it. Actually, Newton didn't really think too much of the proposition beyond what he had derived in the law of universal gravitation. He explained the mathematics of it. But Einstein it took to really relate the relationship between gravitational force and acceleration to ordinary accelerations, say mechanical accelerations. We've discussed this before.
- 3.11** If you have a clock in Newton's universe and you have a mass that you drop, it will fall at a rate of 9.8 meters per second squared at the surface of the Earth. Clocks will tick at one second per second. However, if you go into space far away from the Earth, star, planets, anything else, black holes, you will feel the same 1g of acceleration. If a rocket is accelerating, you'll feel the same force. The object will drop at the same speed per second, and clocks will tick at the exact same rate.
- 3.12** Einstein called this the happiest thought of his life. Physicists are unusual in what makes them happy, but this was actually verified many, many times throughout. And there's a wonderful, yesterday, it turns out when we were filming this, it was Galileo's birthday, his 462nd birthday. And I thought back to, there was an Apollo 15 Moon landing, when the astronauts landed on the Moon. They did this experiment in the perfect vacuum on the Moon's surface. The Moon has gravity, but no [atmosphere]. And they dropped a feather and a hammer, and they hit the ground at the same time, just as Galileo predicted it. It's pretty cool to watch that.
- 3.13** So Newton's law, the cosmos became predictable. It became calculable. They called it a clockwork universe. Spoiler alert, that only held for a couple hundred years until the invention of quantum mechanics, relativity, special relativity, and general relativity. But it was useful for predicting the exact motions of planets and projectiles, which are very important if you're a medieval kingdom trying to wage war with cannons and archers and stuff like that. The tools and technology of astronomy became used in weapons of war quite frequently. But that's not the fault of astronomers; that's just what typically happens. But it did predict and explain, rather, mathematically, why Kepler's laws held and why various objects had the orbital dynamics that they had.

- 3.14** Now, what happened next was quite fascinating—using only Newton's laws of gravity, the discovery of a new planet. So planets are very rare things. At the time of the discovery, basically from the ancients, from the ancient Greeks, up until the 1700s, as we'll see, we only knew about planets from Earth, Mercury, Venus, Jupiter, Mars, and Saturn. So we only knew those planets. And so, even though it was possible to detect other planets, they weren't discovered for a long, long time after that. For various reasons, it's hard to detect the planets when they get farther out, the planet Uranus, for example.
- 3.15** But actually, the orbit of a new planet was detected via its gravitational force on an existing planet called Uranus, which was known about. And that planet was discovered via the gravitational pull. The implication was that there was an unseen chunk of matter which was pulling on this planet, causing its orbit to speed up or slow down from what it would be in the absence of that planet.
- 3.16** So, it was a theoretical prediction based—again, it was circumstantial evidence. It was eyewitness evidence. It wasn't actual physical evidence, going to the planet Neptune. "Oh, we detected it." It was seeing there's some anomaly in the planet Uranus' orbit. "Oh, we can explain that, if there's another planet with this mass at this distance," and astronomers found it basically the very next day. And so, that was used to correct our understanding of the entire solar system.
- 3.17** I said planets are rare things. There's only a few in our solar system. The word planet itself in Greek means wanderer, like plankton in the ocean, the same root; plankton are wanderers, planes, fly around us, fly around in. Those are also the same root. So there's very few things that wander. What do they wander against? They look like stars, but they're moving with respect to the stars that are really, truly in our galaxy far, far away. So, there are very few of those things, so discovering a whole new one was quite astounding.
- 3.18** And it actually led to an interesting event, which I'll talk about in the discovery of Einstein's proof of general relativity and the effect of matter on space-time. So, up until this time in the 1840s, so 150 years after the law of universal gravitation, this astronomer, in particular, his name is Urbain Le Verrier, predicted, based purely on gravity laws, the math of gravity, thanks to Newton, the existence of a new planet, a spectacular, spectacular accomplishment.
- 3.19** Now, before—yeah, so I want to talk about a concept that comes directly from Newton's law. It's called escape velocity. And you could think of it in the following way. Imagine you have a cannon that's shown here sitting on top of Greenland. Maybe it's defending Greenland, who knows? But it's on top of the Earth, say, and it's shooting out cannonballs. And if it shoots it at different velocities, different things will happen to the cannonball.

- 3.20** If it shoots it with a very low velocity, the ball will land a couple thousand miles away, 100 miles away, maybe. But it will land, and these are all shot horizontally. Nothing changes in the angle of the cannon. It looks like it's pointing up, but it actually could be just pointing parallel to the ground. If you shoot it with some velocity, it'll eventually hit the ground. Shoot it with a slightly higher velocity, it'll hit the ground at a slightly greater distance.
- 3.21** There is a velocity that if you shoot it at that speed, it will enter into an orbit around the Earth at exactly the height of the cannon that it was fired from. That orbital velocity, when it escapes the gravitational pull of the Earth, no longer impacts the Earth, is called the [orbital] velocity. It will formally enter into an orbit around the Earth, and the velocity by which you have to fire a cannonball is independent of the cannonball itself. All it depends on is the Earth's gravity.
- 3.22** In the case of the Earth's gravity, you would have to fire a cannonball at about 25,000 miles per hour. It's quite astounding velocity. If you're on the Sun, you'd need to do it something like 60 times faster than that. So the Sun has a stronger gravitational field, much more mass, much bigger escape velocity. Now, what we talked about last time, or earlier in the course, is that John Michell thought, well, what if you had so much gravity that the escape velocity got higher and higher until it reached the speed of light?
- 3.23** Now, he didn't know at the time that the speed of light was the maximum speed. They actually didn't know what the speed of the light was within a huge uncertainty. They knew it was extremely fast. They knew they couldn't measure it using any of the tools of the time. They knew it was extremely fast. They thought, well, if it's less than infinity, we could measure it someday. Obviously, it's less than infinity. We measure it today at about [300,000] kilometers per second.
- 3.24** So if you had a star who was so massive that its escape velocity was [300,000] kilometers per second, then you would have light shot out of a light cannon, a laser beam, that would just orbit around that dark star, and so it would never escape. We'd never be able to see it. He had visualized the light going up and coming back down. But in reality, we believe it would enter into a circular orbit.
- 3.25** And we'll see later, we have a name for that concept now, and it's visible and it's imprinted upon black holes, as we'll encounter black holes. So this is sort of related to black holes, a dark star, but it's not exactly what we think of as a black hole today with a singularity, an event horizon, an accretion disk, and what's going to be called an ergosphere in just a bit. So it's quite fascinating.
- 3.26** So, for 200 years, you could make predictions, you could discover new planets, you could project new concepts that wouldn't be discovered, like black holes, dark stars, for a long time. But we needed to have a better understanding of the physics of gravity. Just as we saw that time is relative, that the concept of now is relative, that lengths of

things and space is relative, but speed of light is not relative—the speed of light is absolute—gravity also affects the concepts of space, of time, of moments, of now, as well.

- 3.27** The first inklings of this were done almost 100 years ago now in an experiment which showed that a photon, a particle of light with pure energy associated with it, no mass, shot with some frequency, wavelength, if you like, when it was shot from a lower-level emitter, say, a laser at the floor, and then it was detected by a detector that could measure its frequency at a high altitude above it, again, on Earth, that it would actually have a redshift.
- 3.28** Energy would shift and get lowered by the reduction in energy that it experiences climbing away from the Earth's gravity. It still travels at the speed of light, but its frequency or wavelength would shift. So, red light travels at the same speed as blue light, as far as we can tell. Physicists do try to measure discrepancies of that kind, but the frequency doesn't affect its speed. So at the top, you could actually envision, if you like, this thought experiment.
- 3.29** If you had an emitter of light that could then convert into matter, you could take, for example, a photon, collide it with another photon, produce a particle and an antiparticle, a positron and an electron. They have mass now, so you convert a pure energy to pure mass with no light associated with it. Then you could let that mass fall freely down to the ground, and then it would be converted back into light, and that would emit energy, and you'd get a perpetual motion machine.
- 3.30** Now, scientists soon found out that that was impossible because it does lose energy as it climbs. It doesn't lose speed, but light increases its wavelength, decreases its frequency, becomes redder and redder. That's called redshift. This particular phenomenon is called gravitational redshift. You start with high frequency, short wavelength near the Earth's surface. You go up, climb out of the gravity well, the potential pulling on everything from the Earth. It actually lowers its frequency, increases its wavelength, and therefore its energy goes down. So you can no longer create, recover, perfect amounts of energy, even if you could convert it purely from energy into matter.
- 3.31** So, because frequency—you could imagine a clock, which is an atomic clock or a light clock, that you count the number of cycles per second in a wave train of light. You shoot a frequency of light, and you count the number of times a given peak or trough of a wave passes a given location. That's how certain types of clocks can be predicted to work. Then you'd actually find that time runs slower near massive objects.
- 3.32** You'd count longer and longer periods between a peak of a wave and the peak of the next wave. Therefore, the clock would be running slower, just like we saw with time dilation when things move faster. So, strong gravitational fields are equivalent to high velocities, if you like, except that in the velocity case, velocity is relative. So if I see you

and your clock, and you're moving near the speed, I see you, I'm moving near the speed of light according to you, but you're moving near the speed of light according to me, that's relative. But gravity is not relative. Acceleration is not relative. Acceleration is absolute, but velocity is relative to the observers.

- 3.33** So, time dilates. It passes at a slower rate near massive objects. That has big implications and is used frequently in Hollywood devices, as we already saw with *Interstellar*, when Matthew McConaughey—again, spoiler is hard to say 12 years after the movie came out—but nevertheless, and the same thing happened in *Contact*, where there's this 17-minute gap where you can't explain where Jodie Foster was, and the cameras on Earth were running, but there's 17 minutes' difference, and that's, she experienced a couple of days on this planet near the star of Vega.
- 3.34** So, it's not only useful in movies, it's actually responsible, at least for my— I'm not that great at direction. So I need the GPS in my car, basically wherever I go; and I've muted the voice whenever I'm driving with my wife, because I don't like to have two women yelling at me at the same time. But the GPS has to be corrected for the effects of time dilation. So light travels one foot per nanosecond. That equates to 186,000 miles per second. If you convert that to feet in seconds, it's one foot per nanosecond.
- 3.35** So these satellites in orbit around the Earth in its gravitational field, if they don't correct for the gravitational field of a satellite that happens to be closer or farther from the Earth—you need multiple satellites to triangulate your position on Earth—you would actually not be able to locate where you're going to within tens of meters, and that would get larger and larger the longer distance you have to go. So eventually, you'd miss your destination completely.
- 3.36** So this is a relativistic effect that's used in your phone, in your car, wherever you are, by pilots flying around. And you also have to correct it for a velocity as well, even though it's a tiny correction. If you're guiding a missile to go somewhere, a microsecond is 1,000 nanoseconds, which means you'd be off by 1,000 feet, if you didn't correct a microsecond, of the gravitational time dilation. These are all taken into account in the clocks that go on top of satellites.
- 3.37** You see the different effects here, 7 microseconds due to the speed of the satellite—it's really small— and 45 microseconds due to the effect of the clocks running faster on the surface, in the satellites. The net effect is 38 microseconds, or 38,000 nanoseconds, or 38,000 feet, which is 7, 8 miles. So this is a huge effect if you didn't take account for it.
- 3.38** I told you about this genius guy named Le Verrier, who predicted the existence of a planet called Neptune, based on the gravitating influence of the planet, on the planet, Uranus. So he was pretty satisfied with himself, and that was a pretty awesome thing, to predict the existence of a new planet. Remember, a scientist should try to predict as much new stuff as possible that could be wrong. If he said there's definitely a planet there, if there's not, Newton's theory is wrong, and they don't find the planet there,

that's a huge discovery. And it would be a death blow, perhaps, for Newton's theory. Of course, it didn't turn out that way.

- 3.39** But then he said something—he kind of got full of himself. I can't blame him. He said, "Let me apply that successful technique to other planets in our solar system." At the time, it was known the innermost planet, Mercury, has a bizarre orbit. It orbits in an ellipse as predicted by Kepler's laws. Kepler's laws were just empirical laws, explained exactly, mathematically, by Newton's law, why the planets move in ellipses. But the ellipse that they orbit, that planet Mercury alone orbits in, does a weird thing.
- 3.40** Every year, the ellipse, it still looks like an ellipse, but it rotates a little bit, such that the closest position called the perihelion, when it gets closest to the [Sun]—it's shown here as the opposite, the aphelion, but ignore that—the point of the ellipse that's farthest away or closest to the Sun rotates ever so slightly. In fact, it's so small, it's less than the width of a human hair held out at a distance of a mile. That's how small it changes. But astronomers are really smart people, and they were able to detect this thing back hundreds of years ago; it wasn't, like, discovered recently.
- 3.41** There was a problem. This weird behavior doesn't exist for any of the other planets but Mercury. So what do you think Le Verrier did? How do you think he tried to explain it? Curious if any of you could guess. How did he explain the bizarre orbit of a planet?
- 3.42** There's another planet.
- 3.43** He said there's another planet, exactly. He called this planet Vulcan, because it had to be close to the Sun, and it must have been a hellish inferno. And what was it doing? It was pulling on the orbit of Mercury, just so to make this thing happen. Perfect. Why not? It works well. They say, if all you have is a hammer, you treat everything like it's a nail. And it turned out that he said, "You can't see it because it's so close to the Sun." It never is visible because the Sun is too bright, so we'll never see it, but he made the prediction.
- 3.44** Fortunately for him, the second time was not the charm. He was right the first time, postulating the existence of unseen matter, like a black hole, if you will, a dark matter, whatever you want to call it. But he was wrong this time. It actually has nothing to do with the behavior or the influence of the gravitational force of another planet. It's a great idea, but it had nothing to do with that. It has to do with the fact that gravity, the force of gravity, if you will, is affected by the gravitating mass of objects warping the fabric of space and time.
- 3.45** In general relativity, the curvature that planets will orbit in, is influenced by the gravity of the sources nearby to them. We won't prove that here. We're going to talk about that conjecture, how it came about, in our course on special relativity and general relativity. But I'll summarize some of the results in just a bit and give you a little bit of the flavor for the beauty of the mathematics. That's really what I want you to appreciate in this course, the beautiful and terrifying mathematics of black holes.

- 3.46** So if mass influences matter to have strange orbital dynamics, then the more massive the object you see, the more influence it will have on the gravity surrounding it. And the more influence on the gravity surrounding it, the stranger the effects on the matter that orbits in the gravitational influence of that large mass.
- 3.47** There are objects in our universe, in our galaxy, called pulsars, which are highly magnetized stars called neutron stars, which themselves are the endpoint of a stellar evolution process that can either lead to a black hole, lead to a red giant, lead to a white dwarf. We'll talk about that towards the end of the course. How does a black hole form? We can talk about the properties, how we observe them, but how do they actually form physically in the universe? That'll come soon.
- 3.48** What two scientists discovered in 1974 was that there was a pair of stars, very odd stars, and they were orbiting around each other. And instead of changing by the width of a human hair one mile away, they were changing 4 degrees, which is eight times the diameter of the Moon as seen from the Earth, so that's half a degree. Eight times that is 4 degrees. They were changing 4 degrees, not thousandths of a degree, so massive was their gravitational force field.
- 3.49** And this was the first evidence that there was this really rippling effect of gravitational energy and the warping of space-time that we now call gravitational waves. There was a prediction that Einstein, again, never thought would be detectable. He predicted it, but he never thought anyone would observe it. He was a little bit, again, sometimes he was very arrogant, sometimes he was way too humble. In this case, he was way too humble. He predicted this effect called gravitational radiation, that when objects orbit around each other, they give off energy. We'll see later how that's used to prove, again, not prove, but give evidence for black holes' existence.
- 3.50** But what was causing these things together was this warping of space-time. Energy was getting radiated away, just like light from a warm object gets radiated away. And the predictions exactly match what Einstein predicted 100 years before it was detected. And soon we'll talk about the modern-day experiments that can measure things a fraction of the width of a proton, used to detect the collision of black holes together, billions of light years away. So it's one of the most stunning achievements of humankind.
- 3.51** So here are the different pillars of general relativity. I talked earlier about special relativity, advertisement for the class. I mean, you get all these courses for the same price, right? You get special relativity, you get general relativity. I'll throw in cosmology, astronomy, maybe quantum mechanics. So, there's all these courses you can take here. It's phenomenal.
- 3.52** We talked about the consequences of special relativity, time dilation, length contraction, energy, and momentum transfer. We talked about the simultaneity, light

cones, all those features. Those are the pillars of special relativity, concerned with a motion at constant velocity, but high velocity compared to the speed of light.

- 3.53** Now, these are the pillars, the consequences of general relativity, the effect of time on time, on space, of matter, and of gravity. What are they? Light bends due to curvature in the universe. Curvature is caused by gravity. Gravity is caused by masses. Time slows down just like it does for moving objects. Time slows down due to high gravitational fields. We don't have the same concept of length contraction from gravity.
- 3.54** We have what we'll see later, tidal forces, which will result in things like spaghettification, in just a bit. But we'll talk about the different types of warping of space-time have these different consequences. When you have dynamical systems like binary stars or pulsars or black holes even, they can radiate energy in space-time ripples called gravitational waves. These were only detected 10 years ago.
- 3.55** On all these things, I have topics on my podcast, on my YouTube channel, Dr. Brian Keating, interviews with the people that won the Nobel Prize for all of these consequences, all these measurements, in general relativity. I have conversations with 22 Nobel Prize winners, some of whom have built the most phenomenally precise atomic clocks ever made. Some of them have tested quantum mechanics in severe situations. Others have measured these black holes colliding together near the speed of light in galaxies a billion light years away. It's an incredible resource and honor that I get to talk to these people, and we share them with you guys as well.
- 3.56** The summary of what the consequences of general relativity are can be summarized in one pithy sentence by the man who coined the term "black holes." His name was John Archibald Wheeler. He was a brilliant, brilliant scientist at Princeton. He said, "Space-time tells matter how to move, and matter tells space-time how to curve." So you can see, it's sort of a nonlinear [circular] claim. You don't have curvature without matter. Once you have curvature and matter, matter will move differently than if that matter wasn't there.
- 3.57** These are called highly nonlinear systems. When you change a tiny, tiny bit of something, you get exponential or geometric growth in some kind of outcome. We'll see later how that comes into play for the behavior of, say, the gravitational radiation from a black hole that crashes into another black hole. If you increase the mass a tiny bit, you get an enormous output change. That's the pithy statement. That's what we talk about, that gravity affects space-time, and matter causes gravity, effectively.
- 3.58** So the statement is, "Space-time tells matter how to move; matter also, it tells light how to move." Light gets bent due to the curvature of space-time itself. Okay, so that's the statement. That's one of the most famous phrases in all of science. He had a lot to say about science, as we'll see later on—again, the person who coined the term black hole. "Space-time tells matter how to move; matter tells space-time how to curve."

- 3.59** The distribution of how matter, and because of Einstein's $E=mc^2$, you could say pure energy has a material, almost material curvature-like effect, that will be manifest in the observational signatures that we'll see from black holes. The more extreme the matter, the more extreme the curvature, the more extreme the effect on other matter that orbits around it. I like to show you these things because they're beautiful, not because I expect you to be able to appreciate them, although if you take the course on special and general relativity, you hopefully will.
- 3.60** And that is called a tensorial equation. Really, what this equation says, the left side describes geometry, the world of the Greeks, the perfect curves and spheres and Platonic solids, the beautiful world of geometry. The right-hand side, this strange T-looking object with two Greek letters, describes the distribution of matter in the universe. This is actually, each one of these objects, these are called tensors. A tensor looks like a matrix, which is like a table of numbers, like a giant Excel spreadsheet.
- 3.61** But imagine you had an Excel spreadsheet, it's four rows and four columns. Each column or row in each element, there's an equation. That equation is this nasty equation called a differential equation, which could have dozens of terms in it and parameters involving mass and gravity and curvature and change in velocity, all those things. So, imagine this 4-by-4 Excel spreadsheet, and each entry in there is a giant enormous equation, and they all have to link together, and they're all coupled together.
- 3.62** If you change column A1,1, you get, or whatever it's called, it's AA, I think Excel calls it, it actually has an implication for column DD or whatever it's called. So, it's a very nonlinear system, but it's a very beautiful system. It has symmetries built into it. And because of this, it's really saying what Wheeler said. The left side is geometry, curvature. The right side is matter, energy. It's called the stress tensor, not because it stresses in the south, but because you can have actually—you can push on a type of matter like this, purely in the vertical direction. And some matter, it will cause a shear force to move in the perpendicular direction. There's all sorts of weird, bizarre, exotic, and quite beautiful matter in the universe.
- 3.63** And then there are constants—Newton's gravitational constant from 250 years ago, and then the speed of light, which is a constant, as we know. And then there's the number pi. Why should pi occur in here? You can actually imagine something really interesting. If your friend goes out, and he measures or she measures the curvature of a gravitational, of a space, space-time, in this room, say, and then you tell the person, that person tells you what the matter is in the universe, how much mass we weigh, how much energy is coming from the photons in this room, and someone else told you what the speed of light is, because it's a constant, you could use that to measure the constant pi.
- 3.64** In other words, things that are purely physical, you can't hand somebody pi. First of all, it's an irrational number, has infinite number of digits. It's a theoretical concept. It's just

a ratio of a circle's circumference to its diameter. Why should that care about the matter and energy and the curvature in this room? It's wild. That'd be a very hard way to measure pi, but you could theoretically do it. There's thousands of ways to do it, and I just think it's beautiful.

- 3.65** Turns out, some of these things are symmetrical, some elements like D_{11} and A_{44} , they have to be the same. If they're not, you're in this weird space-time. So it's called a symmetric tensor. And then the diagonal elements, A_{11} , B_{11} , C_{11} , and D_{11} , they all have certain properties. They have to link to—so there's actually only 10 out of 16. You think they're 4 by 4, 16. It's actually only 10 things you have to solve. Again, they're nasty and difficult. They're nonlinear and they're coupled, and it's the hardest equations to solve. But with the challenge comes the glory.
- 3.66** And this was then used to predict why Mercury had that weird spirograph pattern of its elliptical precession. It was explained by Einstein. Interestingly, he knew about this effect for 100 years. They knew about it for 100 years. So he didn't predict it. He didn't say, "Ah, my theory predicts it." He explained it, which is fine. That's a totally appropriate scientific thing to do. Just because somebody else discovered it, now you explained it. That's very powerful.
- 3.67** Also, there was an explanation by Le Verrier—there's another planet. That was proven wrong. This was accepted to be correct. So now we understand why Newton is wrong and why you don't have to have another planet inside of Mercury's orbit, if you have space-time itself bending. And again, to be explored more in the course of general relativity. But this effect is minuscule. It's so tiny. This advance of the perihelion, it was a triumph of astronomy. It made people start to think the Einstein guy might be pretty smart. This is in 1915. Again, he didn't win his Nobel Prize for another seven years. Again, there were reasons for that that were not scientific.
- 3.68** But in addition to looking at, say, the orbit of Mercury around the Sun, why look there? Let's look at two enormous massive stars, many times the mass of the Sun, and compare their orbit and what they do to planets or to other objects around them. In fact, the very first planet found outside of our solar system was discovered orbiting around a pulsar. It's crazy. And that's because pulsars are the most precise astronomical clocks. They emit very precise beats of radio energy, typically at one-millisecond periods. So every millisecond, thousand times a second, they emit a pulse of energy.
- 3.69** And it's because these are these highly charged stars—these are not black holes—highly charged, rapidly rotating, that produces enormous magnetic fields, which produces enormous bursts of light. And every so often, if you think about it like a spinning top with lasers shooting out of it, it just so happens to be, given the large number of pulsars in our galaxy, that some of them will point towards us. Every so often there's a pulsar that happens to have lined up with our direction in space. It doesn't have to be that way.

- 3.70** And when they go around, it's like a lighthouse flashing. So we see these lights flashing 1,000 times per second in the radio, and they're used to measure processes that are occurring due to the extreme gravitational fields of objects in motion, emitting these bursts of light. Now, if you have two of them, now you have two clocks, and you can start to compare them to each other, and what the predictions are of general relativity versus if the mass didn't affect the curvature of space and time.
- 3.71** And that process, these binary pulsars, also not only tested the effects of general relativity on the orbit of the pulsar, but also how much energy it loses by shooting out this much energy every second, how that occurs, and then in the form of gravity. We'll see later, black holes do this in a much, much more astonishing fashion. So, the predictions of general relativity have been borne out.
- 3.72** Here, you can see, for example, there are these bars. We call, scientists call these error bars. That's how much the deviation could be in the measurement, due to the precession of these pulsars as they orbit around each other. And there's a maximum error here, and you can't even see, say, for example, in this period, these measurements here, the error bar is so small, it's almost the thickness of the curve that matches the theory so perfectly. There's almost no measurements in science, let alone astronomy, that match prediction to observation so accurately.
- 3.73** And so, as I said, they measured this deviation, which was 4 degrees, 35,000 times larger than in our solar system, because the stars are so much more massive, and their orbital velocity— Mercury's orbit is, like, 88 days. These pulsars have, like, a one- or two-day orbit around each other, so you can get many, many cycles per year. And they measure this to less than a hundredth of a percent, not 1%, not 1.3%, a hundredth of a percent, [~ 1 part per 10,000], deviation between—that's the maximum variation you could have between theory and observation. It's incredible that these are measurable.
- 3.74** And then later, we're going to encounter the measurements of a much, even more precise experiment called the Laser Interferometer Gravitational-Wave Observatory, or LIGO, is the name of it. It measures how two different black holes collided together and were behaving as normal black holes until they collided together. They had mass, charge, and spin. When they collided together, one of them had a mass that was 35 times greater than the mass of our Sun. One of them had a mass that was 30 times larger than the mass of our Sun. When they combined together, they produced something with a mass of about 60 times the mass of our Sun.
- 3.75** So something happened to five solar masses of energy. Now, you multiply that by the speed of light squared, you get an enormous amount of energy greater than all the light put out by all the stars in our galaxy. And that was measured on Earth from a galaxy about a billion light years away. We don't actually know where it is, but we know that it happened, and I'll explain why later. This was one of the most phenomenal and slam-dunk cases for the existence, not only of black holes, but of black holes in pairs,

colliding together near the speed of light, and a theory matched the experiment to even better than what we just saw.

- 3.76** Okay, so we've talked a lot about the properties of space and time. That's because black holes distort space and time so severely that we can measure it and learn about the properties of the black holes before they came together and after they came together, for example. So, general relativity remains the most precise and accurately tested. Accuracy is how close it is to reality. Precise is how well you know a given number, how many decimal places you know, a given fact about nature, too. And so, we know it, we've measured it in many different circumstances.
- 3.77** Physicists break this down into three different field regimes—weak, strong, and medium, if you like, or highly dynamic. So strong field could be a single mass with a lot of dynamical motion moving together, or it could be an enormous mass affecting, or multiple masses coming together. Dynamic. And we've tested it in all these regimes. All seem to agree astonishingly well with our understanding of general relativity, which again is predicated on special relativity as well.
- 3.78** The one thing that's missing is how you reconcile these at very, very small distances. We've been talking about things the size of stars. Stars are much larger than the size of the Earth. Sun is about a [1.4 million] kilometers in diameter, 700,000 kilometers in [radius], compared to 30, 40,000 kilometers. We've been talking about massive and huge objects. What happens when you get down to the size of an electron? Can we think about gravity at those scales?
- 3.79** It may or may not be possible, but we do seem to believe that the effects of quantum mechanics do manifest themselves in one circumstance above all. In gravity, it's at the event horizon of a black hole. That is where we are going to turn to next. Let us go deeper, armed with our understanding of how mass distorts space and time. Each point on a grid, as shown here, shows an event. You shouldn't think of it as a two-dimensional surface, but our minds are very advanced, but we are incapable of envisioning things, even in three dimensions.
- 3.80** If you think about it, you think that we do live in three spatial dimensions, but we can't see into objects' third dimension, can we? We're not X-ray machines. We see surfaces of things. We have retinas on the back of our eyes. Those are two-dimensional, phenomenally engineered, designed, or produced by Mother Nature, whatever you want to say. They're two-dimensional surfaces. We have two two-dimensional surfaces, so we synthesize a three-dimensional perspective, but we don't see into the third dimension, let alone the fourth dimension.
- 3.81** So you should think of this as a very highly abstract and, really, just a cartoon, but it's as good as we can do. No human being can visualize third dimension or fourth dimension. People say they can, but I think it's nonsense. But we can make projections of four dimensions onto two dimensions, and then our brains can conceptualize what this

must be like. And then we can actually generalize up to infinite numbers of dimensions, if we like.

3.82 So, what we think of as an ordinary star, like the Sun, it bends the relationship between points in space and time, meaning that it can affect how they behave, as we saw in the eclipse of 1919. The Sun distorted the trajectory of light, curved the path of light, called gravitational lensing. It bent the stars from where they should have been in the absence of the Sun—in fact, where they were when the Sun is not there. And you could see those stars because the Moon eclipsed the Sun in 1919. And you can see it nowadays. We measure these phenomena every time there's a total solar eclipse.

3.83 But that's the weak field. The Sun, even though it's massive, is actually considered quite paltry compared to some of the objects we're going to discover and encounter in this class. The Sun bends, warps space-time, such that the trajectories of objects orbiting around them changed in different ways. The positions change in the eclipse photographs. In Mercury, it comes to a different point in a different time. Time shifts, so it's time at which it comes back to the same point, is distorted purely gravitationally. If the Sun wasn't there, it wouldn't orbit at all, but it would not experience a gravitational time shift.

3.84 So we've seen that every intersection is not just a place in space, it's a point in time, too. Gravity is the curvature of space and time. It's weird to think about the curvature of time, but we talked about time dilating, stretching; that's a form of curvature, if you like, of time. What we want to talk about is, if you had a huge star, a massive star, such that the warping was extreme, what would happen to the path of light and matter around such an object?

3.85 The first person to do that, as we talked about before, was Karl Schwarzschild. Again, his name is an aptonym, which means "black shield." And he is credited with predicting the existence of the event horizon, which wasn't really observed, still technically has not been observed, but we have so much circumstantial evidence for it, it's as good as we can hope to do, at least with our technology now.

3.86 And that's that, if you have a large enough mass warping space and time, effectively, the walls, the steepness of the curvature imply that, to get out of there, you would need something traveling faster than the speed of light. So around the Sun, there's a redshift that makes the Sun attract and slow down time and increase the wavelength and decrease the frequency of light. But you could have an object that makes the star completely invisible, as the concept of John Michell.

3.87 What Schwarzschild did is, took those 16 equations that I described before, and he solved the four on the diagonal, more or less, of those, the $A_{1,1}$, $B_{1,1}$, $C_{1,1}$, and $D_{1,1}$, and came up with what's known as the metric or the properties of geodesics, called the Schwarzschild solution. And that's what we're going to turn to now. It's the simplest

black hole, simple but not easy. It only has mass. It has no charge, it has no spin. That will come about in the next lecture.

- 3.88** The question that he asked was, what would happen to a poor, unfortunate traveler, not in Hollywood, but in reality, coming close to a very, very massive object, an object that would later be called a black hole by John Archibald Wheeler, but not for 60 years. And we consider the perspective of two individuals, the person, the astronaut, the traveler, and the observer, us, far away from the black hole, watching this person come near it, eventually traverse it and go into it. And then what do we see? What do they feel? What do they experience? That's what we're going to encounter, our first journey to the black hole.
- 3.89** So we have to define some things because time and space are both relative. Let's define the coordinates and the properties of the observer, of us, watching this person on his or her perilous journey. Then they have a clock, and they have rulers, and they have compasses, and they can measure the properties near them. We'll talk about them experiencing things and us experiencing things. What we measure, we're going to call coordinate time. What they experience, they're going to call proper time.
- 3.90** We can measure their heartbeat, they can measure their heartbeat, and then we're going to compare the two. We already know what's going to happen near the gravitational maximum potential of the black hole or any massive object. We know, physically, their heartbeats are going to slow down, but how do we observe their heartbeats slowing down? What do we actually measure? And is it consistent with what Einstein and Schwarzschild predict?
- 3.91** We stipulate that they're going to experience something, and we're going to experience something different. We measure their time dilation using equations. Again, we approach them and we talk about them in our other course. But from a great distance away, far, far away from the black hole, 10 times the units of what we call the event horizon, that's very far away, the change in time experienced by them as we measure it compared to our clock, in other words, how fast their heart is beating, let's say it's one second, one beat per second, 10 times away, farther away than the event horizon, is nothing. They don't have any change relative to us. It's one second; their heart beats every second.
- 3.92** Then they get closer. They get half the distance, travel half the distance that they were at the start. They're at five units. It's changed a little bit, maybe 25%. Now, when they get very close, two, three, and eventually, when they get close to this maximum here, it's going to go towards infinity. In other words, it's going to be at about two times the event horizon. For every second we experience, they're going to be receiving five seconds. So it's more than two and a half times the time dilation.
- 3.93** The equation that governs the rate of change of their clock relative to our clock is given here. Δt , it's like one second between the events, by our watch, by the traveler's watch,

is Δt here. We're going to see that divided by this quantity here. The more massive this quantity is, as this quantity gets closer and closer to 1, in units that we'll talk about in just a second, this goes to 1 minus 1, and then the square root of 1 minus 1 is the square root of zero, which is dividing by zero. So Δt , as we see it, goes to infinity.

- 3.94** As they approach this special position for a given mass of the black hole, where this radius is going to govern that distance, we see them basically freeze. We'd see their heart stop. If we were looking at their electrocardiogram, it would be stopped. At one point, it would just stop. It would be very slow. First, it would start off just normally, then it would get a little slower, a little slower, a little slower, and then it would stop. Other weird things would happen, too. The light from the electrocardiograph screen would get redder and redder and redder as they carried it with them, going into the black hole. But that's what we would see.
- 3.95** Now, what would they see? What would they feel, when they get to this point here, when they get to, eventually, to the Schwarzschild radius? According to us, their hearts would be taking an infinite amount of time to beat. What do they feel? Would they be dead? Would they be frozen? How do they experience it? So this property, this boundary, is a boundary in time as we observe it. It's not a boundary in space. The singularity will be a boundary in space. But the boundary in time is separated from the boundary in space. It's part of the weirdness of black holes.
- 3.96** Where we normally think of, say, a singularity is where both time and space might come to an end. But that's not what happens. Time, as we perceive it, comes to an end for us far away from the black hole. For them, nothing happens, but we can no longer see them. They can no longer see us, and we can no longer see them. And their fate is sealed. There's no way for them to avoid the singularity once they cross through the event horizon; that's why we call it an event horizon, sort of the last event that we can see, and it's the last thing they ever control their fate by which.
- 3.97** Here, the event horizon, we're watching it; we're the observers. The redshift turns completely red. We can't see them. Actually, it goes to—it's not even red anymore. It's infinite wavelength, zero frequency. It's not even electromagnetism at that point. It's just DC. Time for them appears to freeze. We see them doing something, they're waving, and they're just freezing. This is the point from which they can never return to us. But nothing happens to them. Physically, nothing happens to them. Mentally, biologically, they keep on living. They don't see us doing anything. Also to them, they see us as frozen, infinitely redshifted as well.
- 3.98** The formula is given at the bottom. That's called the Schwarzschild radius. That is the distance of the event horizon from the singularity. We'll get to the singularity in a little bit, but for now, you can think of it as the center of the black hole. Remember, this is just a plain vanilla black hole. Zero charge, zero spin, and very large mass. It has to have a mass, as we'll see. It has to have a mass at least two or so times the mass of the Sun.

But in practice, it typically comes from a star that's about eight times the mass of the Sun or much bigger.

- 3.99** Now, what physical distance does that correspond to? Well, if you could have a star like the mass of the Sun turned into a black hole, first of all, anything could technically be a black hole. You could be a black hole if you were compressed beyond the size of a quark, because it's just a gravitational, it's just an amount of gravity. But in stars, stars won't form this due to the stellar processes that we'll describe in lecture six. They can't actually form those processes. They can't actually form the black hole themselves.
- 3.100** But theoretically, everything has a Schwarzschild radius because it's just given by this equation here, and you solve for R . If you tell me mass, it could be a kilogram, it could be 100 kilograms, it could be a trillion, trillion, trillion kilograms, it doesn't matter. But for a solar mass black hole, if such a thing existed, which they don't form from stars, it would be three kilometers.
- 3.101** So imagine the mass of the Sun, which is 700,000 kilometers in [radius]. If it were squished down to three kilometers, and somehow kept its mass—I don't know how you'd do that—it would become a black hole, and it would have its event horizon. Actually, it'd be slightly smaller than this. It would have an event horizon three kilometers from its center—not 700,000 kilometers, three kilometers. Incredibly compact. This tells you about the density of these objects.
- 3.102** Now, how does the astronaut, what does she experience, or Matthew McConaughey, what does he experience as he goes through the event horizon? Actually, nothing happens to them for a long time. Here, it's shown on the vertical axis is the radius and distance from the event horizon. Here, it's shown here. When you solve for it, it's 2 in these units. And let's say they start off at some time, let's call it 5 in units of this clock that's measured in terms of the gravitational constant and the mass and the speed of light.
- 3.103** So you're starting off far on the right, 5 sec 5, whatever you want to call it. And eventually you're going to hit at the center, say at this point, we're starting on the far right, we're eventually going to hit, in 20 units of time, whatever that is, 20 black hole clicks or whatever, you're eventually going to hit the singularity at radius of zero here. It's going to take from 15 seconds before that, 5 seconds after that, you reach a radius of the singularities, r of 0. You cross the event horizon in these units at 2. That's 2 here. That happens about 17 seconds after you were 10 units of distance away.
- 3.104** So you start really far away. Your heart's beating one time per second. And then by the time you get to this beat, it takes an infinite amount of time, as we see it. But as they see it, nothing else changes. Their heart keeps beating and beating as they go through the event horizon. So here, between this point where the circle is on this dashed line, there's still some time. There's two units of time here, before it gets to the singularity.

In that time, their time is just completely going on as normal. They don't rip apart, nothing happens to them, they don't die.

- 3.105** But the point is, as soon as they go through the event horizon, they can't tell, but they also can't escape. And no light from them can escape. No signal traveling at the maximum possible speed, which is C , can ever reach an outside observer. So, how is that possible? Again, we had the same conundrum when we talked about special relativity.
- 3.106** How can an observer see one thing, and then an observer who also thinks that they're stationary, sees something completely different? They see a gunshot killing somebody before the gun goes off, for example, or theoretically, the gun can reach a certain destination. Because we see one event in one frame is simultaneous, the hitting on both sides of the train car. But moving relative to them, they don't see it simultaneously. They see one event preceding the other event.
- 3.107** Scientists talk about these different types of singularities. There are singularities in this problem. The singularity, with a capital S , when they hit $r=0$, the center of the black hole, that is a true singularity. There is no avoiding the physical laws that break down there. But at the event horizon is a singularity in time, but it's not a true singularity for space and time. That's called a coordinate singularity.
- 3.108** You're familiar with that if you've ever seen the Mercator projection of the Earth. The Mercator projection of the Earth, for example, Greenland looks enormous. It looks bigger than most of the United States, bigger than Australia, bigger than Europe, but it's actually much, much smaller than that. That's because the Mercator projection takes the right side of the globe, and projects a line through each point on the globe, and then projects that to a flat plane. It's the only way to map, it's one of the ways to map out a spherical boundary onto a two-dimensional surface.
- 3.109** It'd be pretty annoying to have to carry around a globe everywhere you went, but that's more accurate. But we can approximate that locally as flat. You know that the Earth looks flat. We navigate by latitude and longitude. We rarely use the curvature of the Earth here. So you don't have to do it as long as you're not going too far. If you're going from here to the North Pole, you do have to take into account that curvature of the Earth, and there the Mercator projection is wrong. The Mercator projection takes the whole North Pole and stretches it across the entire top of the map.
- 3.110** So it would look like you could actually—and it is sort of true in a sense—if you started from any point here, say we're in North America, if you go along one of these lines of longitude directly north, you will hit the North Pole. No matter what line you go to, you'll hit the North Pole. But the North Pole here looks like it's also stretched out along the entire diameter of the Earth, just like the equator. Here, the North Pole and the equator look identical. They're just horizontal lines of latitude, and that's obviously not physically real.

- 3.111** So that's called a coordinate singularity; it's not a real singularity, but in some sense, it helps to visualize things, as does this analogy. Again, we can't see four dimensions, we can't see three dimensions. We can see two dimensions and make analogies of what a two-dimensional creature might see in three dimensions. Here's an analogy that physicists sometimes use. It's called a sonic or hydrodynamic black hole.
- 3.112** You can think that, far away from a black hole, imagine you see this fish, and you're this fish's friend over here on the left, and there's another fish in the middle, and then there's a fish far away in the right. This fish over here, if it's traveling in this direction, it's heading towards the event horizon, but theoretically, the flow of water is not faster than the speed of sound. Here we're assuming that, once you get past this boundary here, the waterfall is faster than the fish could ever swim. It's faster than its sonic speed or its light speed, equivalent for it.
- 3.113** All the fish are moving at the same speed. This fish here could send a sound wave like, "Help me, I'm heading towards the event horizon." But once it fell past the event horizon, nothing really bad would happen to it. Until here, it would reach the singularity, maybe it reaches the bottom of the waterfall. That's like the analogy of a singularity for a fish.
- 3.114** So, space-time itself, if you like, is flowing in a direction near a black hole. And once it gets past this point here, this event horizon, rather, the fish has no chance to escape, but nothing has happened bad to it yet. Eventually it will, and it can't avert its fate. There's no way to get around it. So space is effectively flowing in it. And this brings up the last slide for today's lecture, this lecture, has to do with those light cones that we talked about before.
- 3.115** Far away from the black hole, nothing really bad is happening. Your past and your future are basically orthogonal to each other. But once you get inside the black hole, there's a little bit of your space. When you try to move in space, you're actually moving in time and vice versa. When you're trying to move in space, you're actually accelerating your trip towards the singularity, and you cannot avoid it. So, that is the traveler's fate.
- 3.116** The last thing I'll say, rather, is, something bad eventually does happen due to the gravitational distortions of space-time, and that is called spaghettification, that the gravity is stronger near your feet than it is near your head. As we said before, something high up experiences less gravitational force than something close to the surface of the Earth. But on a black hole, when you're amplifying that by trillions of times the human's mass, the gravitational forces go not as the inverse square, but as the inverse cube.
- 3.117** So as r goes to 0, close to the singularity, everything rips apart. Your atoms cannot be maintained by chemical forces alone. And your survival will depend on how fast you're

moving, and there's no way to avoid spaghettification. But luckily, none of us are in danger of getting close to a black hole. Thank you.

Lecture 4: Anatomy of a Black Hole

At a glance — Anatomy of a black hole.

- 4.1 We left off in the last lecture with a map of a black hole, which is remarkably boring and quite unlike what we'll see in some popular feature films. We effectively characterized the black hole as having two properties: a singularity and an event horizon. The properties of each are determined by just three properties of the black hole: the mass, spin, and its charge. Again, the simplest black hole, which is by no means easy, is purely massive. No spin, no charge. We'll get to both of those later on. In fact, we'll encounter spin in just a few minutes. But the structure of the singularity is more or less irrelevant at the event horizon itself.
- 4.2 Here are some depictions of how they measured it using this worldwide, literal world-diameter telescope called the Event Horizon Telescope. I have an interview on my podcast channel with the leader, Shep Doeleman. This is right after the discovery was announced and we were just so enthralled. This is the second-ever image of an event horizon from the Event Horizon Telescope. This one's in the Milky Way galaxy. The previous one is in a galaxy 55 million light-years away called M87.
- 4.3 This one is in our own home galaxy, a few tens of thousands of light-years away, called Sagittarius A*—similar types of images, the features of which we're going to get into and describe and discuss. But this is far more accurate than the obviously much more dramatic true Hollywood ending, the explosion at the event horizon. I hope now you know nothing of the sort would happen as one traverses the black hole's event horizon. Again, not that you want to do it, not that there's a lot of travel agents lining up to take your money to go there.
- 4.4 It is a boundary in time. There is a boundary in space-time, which is the singularity to which we will soon turn, and we briefly hinted at that last time. But a black hole's event horizon is nothing but the division between points once from which a traveler, an astronaut, unfortunate as he or she is, could no longer send information to us, far, far away, comfy and cozy, watching Peterson Academy. That's all the event horizon delineates. Of course, the future for each one of the observers, us and the black hole traverser, is very different.
- 4.5 Once we get through the event horizon, there will eventually be a point of not only no return—we cross that point of no return at the event horizon—but there will be a point of termination of that individual's fate, as you'll see. But the main thing that occurs is that past and future are no longer so cleanly connected along what we and the outside observer's frame see as these perpendicular directions between space and time, where we can move freely in space and time, one not affecting the other one. Outside the event horizon, do what you want. Inside the event horizon, all paths take you to the same destination: the singularity at the heart of the black hole.

- 4.6 It will become obvious to us that we are not actually observing the event horizon, but we will be observing the closest thing possible to it. To do a more detailed analysis of that, we have to encounter and turn on the spin parameter of a black hole. We'll get to that in just a second. But I do want to point out that there will be dramatic consequences of going through the event horizon eventually, but they involve getting closer and closer to the singularity, where space and time come to an end, not just time from the perspective of an outside observer, but space and time from the perspective of the traveler inside the spacecraft.
- 4.7 It has to do with what's called tidal forces. Tidal forces involve not just gravity, but the gradient, the change in gravity perpendicular to the radius of the object. Here you see a black hole. You see its singularity, perhaps at the center there. And you see these lines of force that affect these two different spacecraft.
- 4.8 Because of the inverse-square law, the force on an object's outermost region is less by the inverse of the square of the distance between them. But actually, the horizontal force has also come into play, tangential to the radial force, and they go as the inverse cube, such that if you're twice as close to the black hole singularity, once you get beyond a certain point, you'll have eight times the gravitational tidal force, not four times the gravitational force that you feel.
- 4.9 So gravitational tidal force is a squishing and a stretching. The gravitational force is a pulling. The pulling force goes as the inverse squared. The stretching and spaghettification process, that goes as the inverse cube. So it's much stronger the closer you are to zero: $r = 0$. None of these really come into play for most black holes until you're much, much closer to the singularity than the event horizon, certainly not outside.
- 4.10 Here's a graph of that inverse-cube law, and it shows the tidal force dependence. So it turns out that the smaller, the less mass of the black hole, the farther away from its singularity, you will get ripped apart. And truly, the force depends on the inverse square of the mass, not the distance. So for an object like the black hole at the center of our galaxy, which is about [four] million times the mass of the Sun, you'll actually be able to travel much, much farther before you get ripped apart than a low-mass object, say 10,000 times the mass of the Sun. And certainly for something the mass of our Sun, it's not even on this chart. It's way off the side. You don't even ever experience a tidal spaghettification force.
- 4.11 You enter large distances from the singularity in units of the event horizon. So on the far right, you travel twice the event horizon distance. You travel along this line, the yellow line, for a 10,000-solar-mass black hole. You start feeling something unpleasant, maybe between—or you start noticing something: your feet start to feel a little heavier than your head.

- 4.12 That starts to occur at about 60% or 70% or 1.7 times the distance of the event horizon. And then when you get to 1.5 times the distance, you get ripped apart. So you actually get ripped apart both farther away and over a very, very tiny distance, tiny change in mass. For a very large solar mass black hole, many, many millions of solar masses, you can actually make it far within the event horizon before you would get ripped apart by the black hole's spaghettification, tidal forces. It's contrary to what you'd think.
- 4.13 In the movie, they show this giant black hole and you're going to get ripped apart. Gargantua, you wouldn't get ripped apart. And that's actually taken into account in the film Interstellar. You actually would not get killed in Gargantua. It's much more, millions of solar masses. Nor would you in the black hole M87. And if you get up to billions of solar masses, you might have to be essentially at the singularity before you get ripped apart. So good news, everybody. Good news. You want to go to a massive black hole.
- 4.14 Now, what are you actually seeing in these event horizon images that I showed you from the Event Horizon Telescope? You're not seeing the event horizon, but you are seeing something called the "photon sphere." Remember the notion that John Michell had in the 1700s, that if you fired a cannon with enough velocity, and that velocity was the speed of light, the light would go into orbit around the black hole. That is effectively not a circular orbit. It's a spherical orbit because those orbits the light could travel in any one of the entire 4π steradians of angle around the black hole. That is called the photon sphere.
- 4.15 It's a critical distance. It's 50% larger, farther out in radius than the event horizon. Light gets captured at that point into a circular orbit. Anything slower than light, of course, is going to get captured much, much farther out. You see the event horizon is here, and light cannot escape from the event horizon to larger distances, certainly at the event horizon. But even for the photon sphere, you enter into a circular orbit. In other words, if you launched a photon ray, a beam of light here, and it was going outward, and it's a little bit beyond the event horizon, it will eventually reach an outside observer, just depending on the angle.
- 4.16 But no angle is possible within the event horizon. So if you're in the event horizon, you could shoot it perpendicular, it's not getting out. But if you're at the photon sphere boundary at the distance of the radius of the photon sphere, if you launched it at a parallel angle, it will just go in orbit around the black hole, even light, let alone matter. That's what we are effectively seeing. We're seeing the shadow of the photon sphere. Actually, we're not seeing the photon sphere. It's the matter that's heating up to create light that's just barely at the boundary of the photon sphere.
- 4.17 Remember, the accretion disk is where all this matter is colliding together, heating up to superheated temperatures, and that's illuminating infrared radiation, heat, light, gas, gamma rays, et cetera. There's a shadow that's larger than the event horizon, and that's what those images are showing. They're not showing the event horizon. You

can't see the event horizon, but you can see the shadow effectively where no light could escape if it's in a tangent potential orbit.

- 4.18** If you visualize it—what we see on the right, my friends at the Event Horizon Telescope made this image—these are radio waves that can traverse the region outside the light shadow—this isn't even the light shadow. This is actually the innermost orbit of which the accretion disk can occur here. Now, why does it look like this? On the left, why does the simulation look like this? They don't actually look very similar. I mean, they're both colored the same, but you could make it rainbow-colored. The colors are irrelevant. They're just to guide the eye.
- 4.19** Why does it look like this on the right, where you've got this strange-looking hat on top of the black hole? We have the photon sphere [outside] of the event horizon. So you see the event horizon in white, this thin white line. And then between the event horizon and this first bright patch over here and over here, you see the photon sphere, the outermost radius of the photon sphere. And then you see this bright stuff on top, and then you see this disc in front of it. What are we actually seeing here?
- 4.20** Well, in reality, the black hole doesn't have this hat on top of it or below. The black hole is actually—we can model it as a spherical spinning object. In this case, it is spinning. It has an accretion disk like Saturn's rings. Imagine Saturn's rings are chunks of ice going around Saturn in orbit around it. The accretion disk is orbiting around the black hole. But Saturn doesn't have strong enough gravity to bend light, but the black hole does. So this part on the top is actually the back of Saturn's rings, if you will, getting bent gravitationally, by gravitational lensing over the top of the black hole's photon sphere.
- 4.21** So you're actually seeing through the black hole. Basically, you're seeing something at the top here that you couldn't see if it was a normal non-black hole object. If it didn't distort gravity severely, you couldn't see this. This is like the backside of Saturn's rings. So if this is like Saturn's rings in the front, you see those, and the back is getting bent. There's photons that are coming up at an angle such that they'll get pulled in by the gravity and then bent over and they'll eventually make it to our telescope. It's pretty amazing. That's what they're imaging.
- 4.22** And you see this here, it's upside down, but that's the exact same thing. You're seeing the back half of the accretion disk pointing towards us on Earth from a galaxy, 55 million light-years away. It's astonishing. And we've seen this now in two different black holes. This one has billions of times the mass of the Sun. Our galaxy has millions of times the mass of the black hole— it's [four] million times the mass of the black hole.
- 4.23** So again, the features of the black hole—go to the doctor—one of the vital signs of a black hole, the photon sphere, the event horizon, the singularity, those are all manifestations, just like your height, your weight, your cholesterol, those are all manifestations of your body. You go to your doctor. What are the properties that really

make you up? It's the number of atoms you have in your body, what chemicals are in your body.

- 4.24 Here for a black hole, there are only three [properties]: mass, charge, and spin. It's very simple, but they have very profound and, crucially, observationally significant, detectable, provable, but more importantly, falsifiable. It could be wrong. If it looked completely different, it looked like some perfect sphere, that would not agree with general relativity.
- 4.25 We'd have to revise our predictions about black holes, perhaps. This is predicted and then measured. The first simulations, like the one on the left, were made about 50 years ago with very painstaking, almost hand calculators back in the 1970s. But the original picture looks almost like this, except it's black and white, and it's on what's called a dot-matrix printer, a very old-fashioned printer. It looks more or less—you could recognize it. It's amazing that the predictions from 50 years ago and then updated with modern supercomputers and Hollywood special effects look identical, more or less, in their most important features.
- 4.26 We have these two different perspectives, falling in and watching from far away. Time stops from our perspective. Nothing happens from the observer's perspective. Time goes on, their hearts beat until they reach close to the coordinate, close to the event horizon, close to the singularity itself, in which case they do become gravitationally ripped apart. At that point, then the physics, the coordinate, and the time, and space break down.
- 4.27 Okay, I said that black holes have these three properties. You're probably sick of hearing about it. But in physics, it's so rare. First of all, black holes are rare. We're obsessed with them today. There are very few of them that we've ever measured.
- 4.28 Secondly, there are very few properties, but that's very good for us because it allows us to make predictions and test our theories of physics, which we will then see can be used to test our theories and predictions about thermodynamics, gas physics, stellar physics.
- 4.29 How do these stars turn into black holes? We haven't crossed that divide yet, but we will. That's what makes it so fascinating. We can test everything from the theory of stars, which the Sun will never be a black hole. But we can learn about things in terms of the properties of the Sun compared to other much more massive stars and much less massive but more dense stars, things like white dwarfs and things like that.
- 4.30 The Schwarzschild solution is great. It allows us to make predictions. Schwarzschild took the newfangled theory of Einstein, 1915, turned it into a prediction, 1916, which wasn't verified for 100 years. So it's pretty fascinating that he was able to do that. But it was the most simplified realization of the consequences of Einstein's theory.

- 4.31 There's an old joke about a physicist who's called in by a farmer to help the farmer have the cows produce more milk. And the physicist starts by saying, "Assume a spherical cow." Physicists love to make approximations. This one turns out to be really good. It can explain a lot of principles, but unfortunately, these don't exist in nature because there's no such thing as a not perfectly stationary black hole. If they did, we would have a very tough time detecting them. They would not make the accretion disk that leads to the luminous feature that the black hole ignites and fuses and makes energy out of. That turns out to be observationally apparent to us, so we can actually test and see if these black holes have photon spheres and other properties.
- 4.32 So black holes: simple properties for a non-spinning, uncharged black hole, perfectly spherically symmetric, perfectly uniform in all directions. It has this property that it has the event horizon structure of space-time around it. Remember, that's not part of the black hole. People in the movies, it's to depict the black hole as the event horizon is part of the black hole. No, the event horizon is a consequence of the black hole's existence. Just like our circular or elliptical orbit around the Sun is a consequence of the [Sun's] gravitational force as perceived by this—made by the Sun. It's not a property of the Sun, our orbit. It's a manifestation of the fact that we orbit around a massive star.
- 4.33 We have a radius, which has units of length, which is given by Newton's constant. [We have] the mass of the object. It could be any object, could be you, your phone, it could be a city, it could be an entire planet, or it could be a supermassive star. And then divide it by the speed of light. So there are two constants, the number two, and just the mass of the star. There's almost nothing else that goes into it. Very, very simple, but very profound.
- 4.34 Outside the horizon, space-time is curved, but it's not changing. Inside, all paths that you take direct you to a single location in space, the singularity. There's no way to move. The faster you move, the quicker you get to the singularity. In fact, your moving through time is really moving through space to eventually get pulled into the black hole. Not like a vacuum cleaner, but as an inevitable, inexorable effect of the gravity distorting space-time around the black hole.
- 4.35 In 1963, so 50 years essentially after Schwarzschild, a New Zealand physicist named Roy Kerr predicted what would happen if you put the black hole in motion. Still not charged, still massive, but moving and rotationally about a symmetric axis. So you assumed it was spinning. So it's called angular momentum, just like a spinning ball. Later on, we discovered that there's only one type of solution. It's called the unique solution. And this started the real elevation of black holes into a serious way to test the properties of space and time, physics itself. It was a revolution for theoretical physicists. Again, decades before it could ever be measured, and even the existence of black holes was known to exist. This is right around the time that Wheeler even came up with the term "black hole." Again, this is over 60 years ago.

- 4.36 As I said before, there's something called the no-hair theorem, but it's just a fancy way of saying we can't really tell what made the black hole. Here it's potatoes, it could be stars, it could be whatever you like. It only has these three properties. I've told you that numerous times before. And the potatoes or the stars, as they fall into a pre-existing black hole, can amplify its characteristics. Certainly, it amplifies its mass. But once it's a black hole, it never stops being a black hole.
- 4.37 There's no way to put antimatter into the black hole and have it come out. That's not a possibility. Antimatter has positive curvature. Ordinary matter has positive curvature. That wouldn't do it. So if you throw something into it, it can also have those three properties. It could be charged, spinning, and massive.
- 4.38 So you throw in more mass, the Schwarzschild radius gets bigger, the singularity, the tidal forces, get lower. Remember, the more massive, the lower the tidal forces, the longer spaghettification takes or longer it's delayed closer into the singularity. You could throw in things with charge. But most things, it's almost impossible to have stuff in space moving without spin, without rotation. Because to do that, you'd have to imagine orchestrating every single potato to come exactly matching its mass and momentum along an exact radial path as it enters the black hole.
- 4.39 Now, you couldn't do that with potatoes. You couldn't even do that with baseball. I'm just taking every single baseball, have a hundred baseball players throw to first base or to home plate with exactly the same speed, with exactly the same baseballs, exactly the same. It's impossible to do that. So there'll always be some net imbalance, some asymmetry, and that net asymmetry will be accretive and additive, and it'll keep adding to the spin mass and charge of the black hole.
- 4.40 No matter what you put in there, it'll leave a remnant on the black hole, but that's all that will leave a remnant on the black hole. If you threw in magnets, just pure magnets, you would never know that you threw in pure magnets. If you threw in pure charges, you would know it would change the charge of the black hole. If you threw in things that were not spinning, you'd be able to detect it. But if you threw in things that have other properties, you threw in something that's red, a red potato versus a white potato, you could never tell a difference. Only if it's spinning, massive in charge.
- 4.41 Now, when spin occurs in ordinary space, if I spin something really fast here, it doesn't affect the properties of things around it. In fact, the Sun is spinning. The Sun spins about once per month. A day on the Sun is how fast it rotates about its axis. It takes about a month. Galileo observed that with a very primitive telescope in 1611. He took the telescope, pointed at the Sun, and projected the light onto a wall or into a piece of cardboard, and he saw there were sunspots.
- 4.42 He was so meticulous, he mapped the pattern of the sunspots, and it took about a month for them to reappear. He was amazing. He discovered the moons of Jupiter. He discovered the rings of Saturn. He did a lot of great stuff. He also made a lot of

mistakes, one of which was to assume that the tides on the Earth were caused by the Earth going around the Sun because he wanted that to be true. It is true, but he wanted to prove it, so he was right for the wrong reason.

4.43 The tides on Earth have nothing to do with the Earth revolving around the Sun. They have to do with the Moon pulling tidally—inverse cube law—on the Earth's oceans, which squishes it around the equator or around the direction close to it, and then squashes it in the direction perpendicular to that. That's why we have tides four times a day. At least in California, where I live, you can see four times a day, not twice a day. So he was right. The Earth does go revolving around the Sun, but not because of the reason that he claimed it was due to.

4.44 Ordinarily, you can't detect rotation. The Sun's rotation, if it stopped today, we wouldn't notice it. But very, very close to the Sun—or like Mercury—Mercury is affected slightly by the rotation of the Sun. It happens to be the Sun is rotating very slowly, spinning very slowly. So the metaphor is that from supermassive objects, the connection between space and time gets dragged along with the rotating object exactly as a bowling ball spinning in a huge vat of honey would pull the honey around it with it. So you could measure distances inside the honey.

4.45 Let's say you're a bee trapped in there and you're measuring distance, you're a little tiny bee ruler and your little tiny bee compass, and you're measuring position and angle. You could measure things. It would get distorted, but you would also be distorted, so you wouldn't be able to measure the difference. You're getting swept along with it. But far away, we'd say, "Hey, your bee, your ruler got squished." It's made of honey or whatever, so it's getting squished, and you better take into account if you're trying to get a certain length of—I don't know what bees would want— but you do a certain length of honeycomb for your hive, you've got to take into account the spinning effect.

4.46 We call that frame dragging. So space-time is fluid; it behaves like those fish we saw before. It's flowing along at a certain rate. It's also spinning. The spin has a very important consequence. It effectively channels, funnels, distorts, compresses the matter and other things that are falling into the black hole. So you get this bulging effect, just like you can imagine a spinning ball of Jell-O in the International Space Station. It would start to bulge out, squash out a little bit, and you'd have a difference between where the event horizon is and the pattern of the matter as it flows around in what's called the ergosphere.

4.47 So there's a thing called the ergosphere, which depends on the rate of spin. So in the ergosphere, you cannot stand still with respect to an outside observer. You're being pulled along with the matter and the rotating of the black hole, and space-time itself is rotating, like the honey being pulled along as you rotate. Space-time is moving faster than light, very close to the black hole—space-time itself, if the spin is large enough.

- 4.48 So, there's a difference between the ergosphere and the event horizon, which is that stuff can escape the ergosphere. Because of that, we can see the "event horizon," which is really the photon sphere shadow against the accretion disk. The accretion disk is being channeled by the ergosphere, by the matter and energy. It's actually compressing the matter perpendicular and expanding it circumferentially. So lucky for you, if you're there, you can get out. Again, it's not the place you really want to be in.
- 4.49 So now we've added a third landmark on our map of black holes. We've covered a lot in just four lectures. The map from the singularity to the event horizon. Now a new phenomenon called the ergosphere. We also encountered the light sphere, photon sphere, and soon we'll encounter the innermost stable circular orbit, or ISCO. We'll talk about what that means in a second. It's a new property or a new characteristic on the terrain of black holes.
- 4.50 Inside the ergosphere, which is larger than the event horizon, you can get in, you can get out. You still have extreme distortion of space and time. You have this tipping of the future light cone, which basically means that your space-time dimension is mostly moving you in time towards the singularity, but not quite exactly towards the singularity as you would reach inevitably if you went through the event horizon.
- 4.51 You cannot change the spinning. Here's the black hole again. This is the ergosphere. The event horizon is way inside here. This is the accretion disk, and the accretion disk will be dominated and structured by the properties of the rotation of the ergosphere.
- 4.52 There's only a finite amount of rotation that you can have for a black hole. It's interesting. You can't just arbitrarily bring it up to infinite amounts of spin. There's something called the spin parameter, which goes from zero, which is a Schwarzschild non-rotating black hole with no spin, to maximally spinning with a of one [$a = 1$]. And the spinning is inversely proportional to the event horizon.
- 4.53 So what ends up happening when you have this behavior for a spinning black hole, we'll have a smaller event horizon than the same mass black hole that's not spinning. So that's good news, right? Because you can get closer to the black hole singularity without getting trapped forever. So again, explored in the movie Interstellar, they actually could go into the ergosphere near the accretion disk, close to the accretion disk, which could be partly outside of the event horizon. And you can get closer to the singularity.
- 4.54 If you get closer to the singularity, what happens? Time will slow down even further. So that's how they use this device to get closer to the center of the black hole, to the singularity, slow down more, but still stay outside of the event horizon so they can escape, come back to Earth, which is what happens. Not all of it's science fact. There are properties of space and time that are accurate, but actually traveling to it, coming back from it, those aren't accurate, as much as I like the physicist that made the movie.

- 4.55 So for a given mass, a spinning black hole has an event horizon that's smaller than a non-spinning one. So a spin changes the dynamics of the event horizon. It's pretty interesting. And that in turn changes the influence on the matter that falls into it, which affects how easy or hard it is for us to observe it and theoretically visit it, perhaps, someday. So we've added a new parameter, spin. The spin can only take on a certain maximum value. The higher it is, the closer you can get to the event horizon, or the closer you can get to the singularity because the event horizon shrinks.
- 4.56 You can have a massive particle like a spacecraft, and it can go around a black hole, too. And that black hole could either be spinning or not spinning. But a black hole that's spinning, you can get closer to the event horizon, or get closer to the singularity because the event horizon is shrunk down to a much smaller value.
- 4.57 So the orbit that a particle can keep going around without entering the event horizon, after which it cannot escape, it won't get ripped apart immediately, but it can never escape. We know that for sure. That's called the innermost stable circular orbit. Stable means you can keep going forever. If it's light, it goes in a perfect circle exactly at the boundary. If it's ordinary matter, it could go circular as well.
- 4.58 For example, a black hole with exactly the same mass, one solar mass here, M . A Schwarzschild black hole, the radius will be six times larger than a black hole that's spinning at the maximum amount of spin, which is called Kerr maximum with $a = 1$. Because matter—not light only, but matter—can go close to it, it can convert more of its mass to energy because it rotates faster the closer you get to the singularity.
- 4.59 You have more collisions, more heating, more thermal energy can be produced, more light can be produced. So counterintuitively, it's easier to see a spinning smaller black hole. Because its event horizon is smaller, it means the velocity of matter that's going to heat up and convert into energy is higher. So it gets a higher conversion factor. It can convert up to 40% of the matter into pure energy.
- 4.60 Nuclear fusion can only convert [0.7%] of the matter into energy. Again, [0.7%] of mc^2 is a huge, huge number. That's why we have such terror about atomic weapons and such hope for atomic energy in the form of fusion. Fusion is [4-7] times more efficient than fission. This is saying a black hole, a Kerr black hole, spinning at maximum rate, you can extract 40% of the energy of mc^2 type form into energy. So it's quite amazing. And that makes it easier to see.
- 4.61 So counterintuitively, you can see smaller black holes from much farther distances because they're much brighter. We don't see them by their physical size. Actually, we saw the black hole in M87, 55 million light-years away, years before we saw the black hole in our own galaxy, tens of thousands of light-years away, because it was more massive and it was spinning much faster.
- 4.62 It had a much more compact disc and emission properties than our galaxy's black hole. Here's what I said a second ago. A non-spinning black hole can get down to close to

nuclear fusion, but actually a little bit higher than nuclear fusion, about 6% mass energy conversion, seven times larger for a maximal spinning black hole. So seven times more efficient in converting matter into energy.

- 4.63 And that's the reason that was discovered for the power behind these mysterious objects called quasars. Quasars were discovered in the '50s and '60s. They were super bright and they were super far away, and that made no sense to astronomers until two of my colleagues at UC San Diego, Geoffrey and Margaret Burbidge, a husband and wife team, predicted why this would occur, and they showed that these black holes provide the energy to power the sources of these galaxies that are billions of light-years away. Not millions, billions. That explained why we could see them across the vast distances of the cosmos. It's an incredible, incredible discovery.
- 4.64 Okay, again, let's summarize our trip through black hole zoology. We're going to continue in just a moment with how these black holes can even form in the first place.
- 4.65 Schwarzschild, our entree into black holes, predicted from the depths of trenches in World War I by a young scientist who wasn't really a scientist. He was a patriotic German Jewish citizen who volunteered to fight in the Great War and died not long after, tragically. He got his message to Einstein, who didn't suffer fools lightly. He hated when people would, say, make predictions about his theory, especially when it came to cosmology, my field and a related field. But he immediately saw that Schwarzschild was right. It was incredible.
- 4.66 And then it took 50 more years until Roy Kerr at New Zealand made a prediction on what would happen if we started to set these things in motion: that you'd actually make them easier to see, more bright, more luminous, and perhaps, in fact, the only type of black hole that actually exists. But that's scientific reasoning. You start from something "simple." None of this is simple. You start from something more symmetric, more idealized, more simple, more elementary. Then you build upon that, add new features, and ask what the consequences are. Then it turns out that you probably could never see a Schwarzschild black hole. It would never be visible, but it became one of the most powerful analogs and models in physical reality for us to understand relativity.
- 4.67 We can see these objects, including quasars, that are powered by black hole accretion disks themselves, the byproduct of the Kerr black hole spin. We can see them across cosmic distances, which allow us to use them as lighthouses to map out the structure of the cosmos, to understand how did the cosmos perhaps begin and how will it evolve, including its elements that make it up—among them, the most mysterious forms of matter known to exist, but unknown in composition, called dark matter.
- 4.68 As we have time towards the end, we'll talk about propositions that suggest that black holes could make up most of the matter in the universe. We'll explore how black holes could have formed at the very earliest moments of the universe, what are called

primordial black holes—black holes the age of the universe, by far not the most common black holes that there are in the universe, including the black holes we've seen so far in our galaxy and in the M87 galaxy. Those are formed by "ordinary stars," and that's what we want to understand next.

- 4.69 How does an ordinary star, living a peaceful life like the neighbor you never suspected won the lottery, just living a quiet life, right? All of a sudden they're billionaires or millionaires. How ordinary stars become black holes. That's a fantastic story that links together astrophysics with general relativity, and that's what we're going to turn to now.
- 4.70 Black holes are the most outrageous objects in the known universe. Perhaps less outrageous than wormholes, but we know that black holes exist. We don't know that wormholes exist. Wormholes are better fodder for movies and TV. We'll get to them towards the end of the course.
- 4.71 How does an ordinary, quiet, workaday star turn into a black hole? Where do they come from? We've been postulating, assuming that they exist, stipulating that their existence is trivial. But how do they come to be?
- 4.72 I've come to learn that astrophysics in general is a battle between matter and everything else in the universe, between gravity and the other forces of nature. There are four forces of nature. We know and are familiar with gravity. We know and are familiar with electromagnetism and light. There are two other forces far more powerful than gravity.
- 4.73 One is called the weak nuclear force, and one is called the strong nuclear force. They're responsible for nuclear fission, nuclear fusion, radioactive decay, far more powerful. In physics, however, the strength of a force is inversely proportional to how far it extends out into space. Black holes, gravity, they extend across the entire universe. The nuclear force barely traverses a billionth of a meter in diameter, but it's far more strong on those scales.
- 4.74 Gravity battles for attention, but it actually not only determines the fate of individual stars, but of the universe as a whole, as I describe in both Astronomy 101 and my Introduction to Cosmology, both of which already are available on Peterson Academy.
- 4.75 Physicists discovered, thanks to the work of the pioneering astrophysicist, Subrahmanyan Chandrasekhar, who we met in lecture one, that there are delineating landmarks or statistics, physical properties of stars that determine their fate. And the real deciding point for what determines the future of a star is its mass. Stars are determined almost entirely by their mass as well as their spin, but not so much their spin as their mass. We talked about the spin of the Sun. It's almost irrelevant to almost every astrophysical process.

- 4.76 When the eclipse of 1919 happened, the Sun's rotation came into it— not at all. So it's completely irrelevant for most forces and even in astrophysics itself. However, for things like why all the planets in our solar system orbit in the same plane, more or less with the same direction, that is caused by the Sun's spin in the pre-existing solar nebula. We'll talk about that as well.
- 4.77 So the Sun exists as actually a somewhat below-average star, and what astronomers call the main sequence of stars. So we talk about stars. They really have two observable properties that we can use to test their theories of how stars are born. That's their temperature, the color that they emit at, their black body temperature, and their luminosity, how much energy they put out.
- 4.78 You could see two stars that have the same color, say, as the Sun, but they might have very different masses. But most stars, if they're in the so-called main sequence, they start their life at the top with very high luminosity and very bright blue colors. Over time, they slowly cool down. They slowly cool down from tens of thousands of Kelvin to a few thousand Kelvin, eventually way below that and become what are called brown dwarfs and red dwarfs and so forth.
- 4.79 But our Sun is in the middle of its life. It's gone through these hot early phases. It's on its way to eventually becoming a red giant. And that process is the time scale in which the Sun is converting light elements to heavy elements using nuclear fusion, converting protons and protons together and say, hydrogen to make helium, then from helium to make carbon, nitrogen, neon, oxygen, etc. They continue on their lifespan for that period of time. In some cases, they can live for billions and billions of years.
- 4.80 Its mass, the star's mass and its age, determine where it is on this line, which is not a sharp line, but it's a band of stars called the main sequence. There are stars that aren't on the main sequence. They're called dwarfs and giants. There's white dwarfs down here. This is the type of star that Subrahmanyan Chandrasekhar studied, which have low masses, and they are quite hot. They have low luminosities, and they're quite hot. They all seem to have a [maximum] mass of about 40% more than the Sun. Then there are supergiants, some of which can be red, but very luminous and very massive.
- 4.81 But the preponderance, say more than 50% of stars lie on this line here, and then tens of percent are supergiants. Many, many thousands of stars are dwarfs, and the most common star in our galaxy, at least, is actually a red dwarf or a brown dwarf. These are very low, lower mass stars, lower temperature stars.
- 4.82 So we say that the density is destiny, and here the mass is related to its destiny: the larger the luminosity, the faster the star will die. So it's like stars in Hollywood. If they go super, super bright, their careers are usually pretty short. So the stars, the more massive they are, the shorter they live for, because gravitational energy is pulling in matter and causing it to fuse faster and faster, and there's more and more of it. That creates a hotter and hotter temperature, which makes it glow bluer and brighter in

luminosity. So what astronomers do is they normalize the mass of a star to the mass of the Sun, because that's our most well-studied star. And the relationship is the lifetime of a star that's the mass of the Sun is about 10 billion years.

4.83 If you increase the mass of the Sun, you increase it by a factor of roughly 100, say, it will live roughly 10 to 100 times shorter in duration. Or you increase it 10 times, it'll live about 100 to 250 times less. So its lifetime is shorter the more massive it is. Typically, it's converting mass to energy, which is heat and luminosity. The frequency of that luminosity determines the color, and the amount of matter determines how much it burns for, and that determines inversely how long it lives for.

4.84 Now, observationally, it was found that there is a dividing point between stars that will live a relatively benign life and end up cooling and slowing down for all time, and then stars that will eventually collapse in a violent death called the supernova. And then the supernova can terminate in one of a few different options, one of which is a black hole. So you see here the different bifurcations, the two roads that are diverging on the main sequence. If they're more massive than eight solar masses, they have potential, but not a guaranteed outcome of becoming a black hole. If they're less massive than eight solar masses, they're guaranteed to not become a black hole.

4.85 So that's comfort for us around the Sun. The Sun will not become a black hole. Some really bad things will happen, but again, keep paying your taxes and take Astronomy 101 at Peterson Academy to learn more.

4.86 So a low mass star like our Sun, our Sun has one solar mass, one is less than eight, it will start its life on the main sequence, burning hydrogen to helium, etc. Burning, releasing heat, releasing light, eventually swelling up, and gravity will start to lose to the photon heat that's produced in this battle between gravity and heat and thermal expansion. Gravity will eventually cause the fusion inside of it to terminate. And at that point, what can happen is the star will have this uneventful, semi-uneventful burst, burp, where it just basically its outer layer gas basically expands out into the surrounding parts of the interstellar medium, and you get what's called a nebula, a planetary nebula.

4.87 And after that, there is a core left over, which in some cases is pure carbon. Another way to say pure carbon is a diamond. Basically, you have an analog of a diamond core glowing at thousands of degrees Kelvin, much brighter than the Sun is now—much hotter than the Sun is now.

4.88 As we saw in this diagram, the white dwarfs have temperatures of 30,000 degrees. So it's a chunk of diamond about the size of Phoenix, Arizona. And it's glowing and it has the mass 40% of what the Sun had. That's the white dwarf endpoint. Our Sun can't do that because it's less than 1.4 solar masses. It's less than eight also, but it won't end in that way. It'll just make this production a planetary nebula.

4.89 So a higher mass star, say two solar masses to begin with, burns its life. It turns into a red giant, belches out this gaseous cloud called a nebula. That core is left behind 1.4

times the solar mass, and that's a white dwarf. Now, it turns out that the volume of the star, as I said, is quite small. This diameter is tens of kilometers or less. It's actually essentially a pure, if you like, a gigantic atom supported by the pressure of electrons within it. And there's no way you can do it if this star has a mass greater than or less than 1.4 times the mass of our Sun. That's called the Chandrasekhar limit, after a friend Chandrasekhar.

- 4.90** The most famous one happens to be around the brightest star I've seen from Earth, which is called Sirius. Sirius is actually a double star, and there are two stars in it. One is a main sequence star that's super hot, super bright, super blue. That's why it's so bright. You can see it most winter nights if it's clear. And there's something you can't see, which is a tiny white dwarf, which is also super bright, super hot, super compact, but you need a very powerful telescope to see it.
- 4.91** Here is where it is depicted here. And here you see the size of it for a slightly bigger version of it. When down to 1.2 times the mass of the Sun, these stars cannot form a white dwarf, but this one can if it has 40% the mass of the Sun, 40% more than the mass of the Sun.
- 4.92** Okay, that was path one for stars that are less than eight times the mass of the Sun. What about stars that are more than eight times the mass of the Sun? Well, they don't stop at the simple white dwarf endpoint as we just depicted.
- 4.93** Here's a cutaway cross-section of a massive star. Towards the end of its life, it's already fused out hydrogen. It starts by fusing out hydrogen. Then when it runs out of hydrogen, it makes helium. All the helium is gone. It makes carbon, neon, oxygen. So the things that it made in the beginning get pushed out to the outside like an onion. At the center is the newest thing to be made.
- 4.94** When stars try to make the element iron, something strange happens. When iron is fused from lighter, say, silicon, if you take two silicon nuclei and you add them together in a fusion process, you get iron. But all these other processes, when nuclear fusion takes place, it always gives off more heat, more energy than it took to build them together. It's a net exothermic reaction, produces more heat than was used to fuse it together. But iron, the amount of energy required to produce iron to fuse it is equal to the amount that it produces. So it's the point of no return. After that, there's no more possibility for iron to produce or anything heavier than iron, like molybdenum or gold, to produce more heat and keep the star burning. That's a very dramatic process.
- 4.95** When that happens, the star collapses. It has too much gravity to be supported by the heat that's being produced. There's still heat being produced, but it's much smaller compared to the amount of energy that gravity is pushing on it to produce.
- 4.96** What happens is you first start making iron. As soon as that happens, you get a core collapse one second afterwards. It starts burning for millions of years, maybe a billion years. All of a sudden, one second after it started to make iron, it collapsed. And it

collapses with the mass of eight times the mass of the Sun. So it's an enormous amount of energy. There's a shockwave that occurs that causes it to blow out and go backwards. That shockwave releases the iron into the galaxy that the star is living in. And that's where we get what are called metallic meteorites.

- 4.97** And for those of you here in the audience and online, if you're in the United States, I send pieces of these meteorites out to you when you go to this site, briankeating.com/pa, Peterson Academy. I will send you, again, sorry for my international viewers, the US Postal Service doesn't like these things being shipped. They're slightly radioactive, much less, though, than a banana, so don't worry about it.
- 4.98** So it releases as much energy in one second that an entire galaxy is producing of ordinary stars in a century. The most luminous objects in the entire universe after the Big Bang. And you can see them across the entire universe. You can see these objects from billions of light-years away. We use them—as I talked about in our class in cosmology—to measure the size and expansion rate of our entire universe. And that's the endpoint of these types of stars—not all of them, but many of them that are greater than eight solar masses.
- 4.99** It ejects the material that's heavier than iron, including the iron that eventually made it into your blood. Because the Earth has a tremendous amount of iron from the nebula that the Earth condensed out of—as I talk about in both Astronomy 101 and Cosmology. So find more information there. So it's interesting, the death of the star produces life on Earth. Without it, we wouldn't exist. It's fascinating. All those things are the result of giant massive stars that are a little bit too heavy, like some of us, I suppose, speaking for myself.
- 4.100** So there is two different things that can happen after the iron core has been rebounded and compressed. One thing that can happen is it can produce a neutron star. This is a giant atom, if you like, with 10 to the 50th atoms in its nucleus. It's basically one enormous neutron sitting in space. It's actually many, many neutrons, but it's like the nucleus of an atom. It's pure nuclear density. One tiny sugar cube of it weighs more than the Earth. That's the density of it. It's an incredibly bizarre phenomenon.
- 4.101** Not only that, for stars that are, again, less than 20 solar masses, but greater than eight solar masses, they can create pulsars. They can have net charges on them because the electrons get blown away. They might have net negative charge on them or net positive charge on them. And as they rotate at these enormous speeds because the implosion wasn't perfectly symmetrical, they start rotating at fractions of the speed of light. That creates a moving charge in motion, creates a magnetic field, the Faraday effect, magnetic induction. That means they can produce pulsars. The pulsars will be repeatable atomic clocks that will last forever and be some of the most precise clocks that we used earlier in the class to talk about the slowing of gravity near a massive gravitational force field.

- 4.102** Then we come to why we're talking about this in Peterson Academy Black Holes course. If the star was greater than 20 solar masses, it becomes a black hole. It's too powerful for even neutron pressure to stop. It collapses into a place of pure curvature. There's nothing left. There's no matter, there's no neutrons. All the energy has been converted into curvature of space, a perfect hole in space.
- 4.103** It's completely unlike any other form of matter there is. I mean, all these things are. The giant diamond of a white dwarf, the giant neutron of a neutron star, the incredibly charged magnetic field of a pulsar, and now a black hole, a place of no matter but infinite curvature. It's incredible. So that, we call a stellar-mass black hole.
- 4.104** And it is observationally shown that we don't see black holes, and all the black holes we've seen, we've seen hundreds of them, as we'll talk about next time. We don't see black holes less than 20 solar masses. So the theory that went into this, which is a combination of gravitational physics, general relativity, special relativity, mechanics, and astrophysics of stars on the main sequence—it's a tremendous amount of scientific evidence—leads to a prediction that could have been proven wrong if we observed a four-solar-mass black hole, but we don't.
- 4.105** It's an incredibly powerful confirmation. You can never prove something in science the way you can in mathematics, but you can get as close to having evidence for it as possible and be open to the fact that if evidence came in to the contrary, you'd have to revise your scientific hypothesis. That's what a good scientist does.
- 4.106** We see these objects here. We see solar-mass objects down here. We see white dwarfs in this region over here. We see neutron stars. You never see them greater than about 2.3 solar masses in this band over here. This is all consistent. You could see them as much as four solar masses, but you never see them. Then you never see black holes, in this case over here, spanning in this region over here for stars that came to be black holes in the original objects.
- 4.107** The gap—this is called the mass gap—is the proof that the initial mass determines the fate of the object. It's a powerful confirmation of gravity, of nuclear physics. Remember, we had to have a nuclear explosion to lead to this. It's incredibly messy, not really well understood until just the past few decades when the confluence of theoretical physics, general relativity, astrophysics, computer simulations, all converge with observations like the Event Horizon Telescope and the stellar-mass black hole observations that we're going to discuss next time for black holes in our galaxy. We're going to talk about that next. I thank you, and we'll take some questions then.
- 4.108** Hi. You talked a lot about energy, and I wanted to know a little bit about the types of energy that there can be and that there are gravitational, electromagnetic. What kinds of energies are we talking about here just generally?
- 4.109** Energy is, of course, one of the core concepts of physics. Energy is one of the most important principles in physics because energy cannot be created or destroyed. People

think that matter can't be created or destroyed, but no, you can. You could take two photons which have no matter, no mass, collide them together, and they will produce matter in the form of electrons and positrons, which have honest to goodness mass. You can weigh them on a scale if you have enough of them. So matter is not conserved, but energy is conserved. There are very few things that are conserved.

- 4.110** When we talk about energy, it's the interconversion between the various forms of energy. Generally speaking, there's two different types of energy. You mentioned there's electrical energy and so forth, there's gravitational energy. Those also come into two different forms as well: kinetic energy and potential energy.
- 4.111** These were first really discovered and cataloged around the time of the invention of thermodynamics, when it was shown that mechanical energy, kinetic energy, and heat were related to each other. It used to be that people thought heat was a property of the object. An object had a certain amount of what was called "caloric," which is where we get the name calories from, and that you stored it, and that was something intrinsic to it, or something that was flammable or wouldn't burn. Those were intrinsic properties of the object, before we knew about things like oxygen and combustion and things like that.
- 4.112** Generally speaking, there's only two different forms of energy: kinetic and potential. Kinetic is the energy associated with dynamical motion, and potential is the ability, I think of it as the ability to be converted into kinetic energy. So gravitationally, if you have a mass like this water, you could take this water, which has some density, some mass, raise it in a gravitational field, that takes energy. Now, I've stored potential energy in the system. If I drop it—I'm not going to drop it—if you drop it, it will generate kinetic energy.
- 4.113** As its height goes down, its potential energy goes to zero until it hits the floor, but its velocity goes up to a maximum when it hits the floor. Same with a pendulum. You take a pendulum, you pull it all the way out to one side. Maximum amplitude has maximum stored potential energy, zero kinetic energy. Release it, it has maximum kinetic energy at the bottom where it has minimal potential energy.
- 4.114** What was found is you could raise a weight like this up in a gravitational field, and then you could let it fall. As it was falling, it would turn a turbine, a windmill, a water wheel in this case. Then you could measure the temperature of the water, and the temperature would go up. So the temperature of what we call heat is associated with the thermal agitation, which was converted from potential energy to kinetic energy.
- 4.115** And there's electrical energy that behaves the same way in a battery. You can store potential energy. And then a current flowing when you have dynamical energy, can have kinetic energy. You can then convert the kinetic energy and you can store it into potential energy.

- 4.116** That's what a generator or electric motor does. Electric motor where you supply energy from a battery, which is pure potential energy. It's not doing anything. It's just sitting there. You let some trickle out. It goes into a motor that turns a wheel that does kinetic energy. A generator does the opposite. You pull the wheel down. Instead of turning a water wheel to heat up a cup of water over here, you have a generator which stores electrical charge, which you then use into a battery. So they're interconvertible, but not perfectly. There's always some loss.
- 4.117** And what I was saying about gravitational energy is a black hole will convert the gravitational potential energy of chunks of matter. It could be planets, stars. We'll see enormous clouds of gas, the size of our solar system, getting gobbled up and converted into light. We'll see that in the work of Andrea Ghez in the next lecture.
- 4.118** Thank you very much for this lecture. I have a question continuing on the Wheeler statement of how space-time tells matter how to move, and then matter tells space-time how to curve. Then right after that, you threw in, matter tells light how to move. Later, you talked about the effects of gravity as well. I'm curious to know, for clarification's sake, in these conversational Wheeler terms, how does that continue to light?
- 4.119** How does light get manifest? Yes. When we talked about the pattern of light in all these different situations here, what we're talking about is that mass is affecting the separation between these points in space-time. But effectively, it's similar to when you take a flight on a plane. If you go from where we are here to Europe or something, from America to Europe, you don't take a straight line to get there. You actually go up north towards the North Pole, then you come back down. That doesn't look like the shortest point, even on a flat map, but it is on a spherical map, on a globe. That's called a geodesic.
- 4.120** A geodesic is the path of shortest distance or shortest time. Remember, light travels only at one speed, never slower, never faster in vacuum. And so light, the analog of that is that if you're traveling along this object here to get from point A to point B, the straightest path is not going underneath through the bottom of this, wherever this well is. It could be a black hole. It could go on forever. So that's not the shortest path. To get around it would be the shortest path. That's a very curved path.
- 4.121** So light takes the path of shortest time. That's called the principle of least time or least action. And the reason for that is it attempts to minimize it. Actually, in some concepts, it actually knows about every possible path it could take, but knows to take the path of the shortest possible elapsed time. And that's how things behave.
- 4.122** That's why in, say, media like this, the shortest path, if I put in a pencil in here, the shortest path for some of these rays would not be—it would look like the pencil is broken, right? You ever put a pencil in water? It would look—because the shortest path

through the water is different than the shortest path through air because light slows down in the water.

- 4.123** Similar to light slows down, if you like, gravitationally because it takes more time to go the same distance in space. So it's as if space acquires this lens-like behavior, that's why we call it gravitational lensing. There's no lens, giant lens in space, but the path at which light comes from does get bent exactly like water or a refractive medium, like a lens.
- 4.124** When we are looking at these stars explode or these black holes, we're looking at it from many millions of light-years away. So what is happening to that star, that black hole in the now and of its location in space?
- 4.125** Yeah. Here's a supernova image of a supernova after it exploded. In the beginning, it was just a normal star, and it was just going about its day, except it had 20 solar masses, maybe it was more, and it would collapse. After it exploded, we actually don't know. If we look at any star, for example, if we look at Sirius here. Sirius is about [8.6] light-years away. We don't know if it's there. It could have blown up. It could have actually been a supernova. In this case, it could be a type 2 supernova. We don't know. We won't know for [8] years if it happened right now.
- 4.126** Horrifyingly, if the Sun disappeared right now, we wouldn't know for 8 minutes because light travels a finite distance in a finite time. It's the fastest speed known to exist, but it's still finite. We can't tell the influence of something being there or not being there any faster than the speed of light. It has to occur within that light cone. Things that can travel at the speed of light can be as far out on the light cone as possible, but not beyond it. So it could be there, it could not be there.
- 4.127** After these events occur, the black hole, the neutron star, the white dwarf, that all was set into motion when the star was born by its mass and how much it created from wherever it existed from. I didn't tell you where the star originally that became a supernova came from. That's part of astronomy. That's astrophysics.
- 4.128** But the connection is quite startling that it does play a role in the physics of gravity in black holes, because if it formed with more or less mass, we would know about that and be able to refute or prove ourselves wrong about how stars form and evolve in that main sequence that I started off this particular lecture with.
- 4.129** So again, how these stars form is a topic for another class here, but where their endpoints end up is exactly relevant, especially when they become black holes, neutron stars or white dwarfs, because their gravity is so intense. They're so compact and they have such high masses that they affect space and time around them. By virtue of their initial conditions, the state that they were born with, the mass, spin, and charge that they had, we can reveal the properties that they had before they died.

4.130 And then the question will be, what if we take something whose properties we know very well, like a book, and beam it into the black hole and take our genome and digitize it and put it into a black hole, does it get erased forever? That's the actual topic I will conclude the course with at the very end. Yeah, that's it.

Lecture 5: Observing Black Holes

At a glance — Observing black holes.

- 5.1 We started way back in lecture one with an exploration of the very first black hole that was assumed but really not proven to exist, back in the 1970s, discovered, known as Cygnus X-1, whose influence was felt on a neighboring ordinary star. That turned out to set in motion a revolution that had started hundreds of years earlier with the predictions of John Michell, later Karl Schwarzschild.
- 5.2 But we humans are greedy. We like to see, touch, hold things. Of course, black holes are a little dangerous to hold, so we can't get quite that close to hold and cuddle up with a nice black hole. But what we can do is get more and more different avenues of evidence for their existence. The explosion, the literal explosion, of evidence mounting from all different types of tools and technology that we're going to cover today, from literal sight to literal sound waves, have basically confirmed beyond any shadow, photon shadow of a doubt, that black holes are real.
- 5.3 Today, we're going to get up close and personal with the very heart of the Milky Way Galaxy, the closest true supermassive black hole to the Earth, and the second one whose baby picture was taken, if you like, with a telescope called the Event Horizon Telescope that we talked about yesterday. But after that, it had an image taken of its gravitational dynamics, courtesy of optical light. The Event Horizon Telescope uses a massive array, a network array of telescopes spanning the globe, synthesizing a telescope with the diameter of planet Earth.
- 5.4 In contrast, we don't have to go to such great lengths, literally, when you go into the optical regime of the electromagnetic spectrum, because the wavelengths are so much shorter. And a telescope can see something better, the shorter the wavelengths that the light is being produced and the bigger the telescope. So, for the millimeter waves and radio waves produced that image the Event Horizon Telescope, those were needed to pierce the dust surrounding the black hole in the center of M87 and in our own galaxy.
- 5.5 However, we don't need to pierce the dust, at least not as well as we do with the Event Horizon Telescope, if we're actually imaging not the black hole's event horizon, but the dynamics of other objects that have the unfortunate fate of getting a little too close to the black hole.
- 5.6 Now, they actually survive. We're going to be looking at the survivors of cosmic collisions today that have been imaged for decades, such that we have not a still image, as we have from the Event Horizon Telescope of the black hole, Sagittarius A*, in our galaxy, but a movie, an honest-to-goodness movie, of objects orbiting around the black hole. Now, it's not as exciting as Gargantua from Interstellar or Event Horizon or The

Black Hole movie, or any of these other things. But you'll see that it had much more impact on science and the history of science as well, as we'll explore today.

- 5.7 So this is called The Case of the Missing Mass, and it's really a detective story looking for what's unseen. There's a famous Sherlock Holmes story, where the criminal is caught by Sherlock Holmes because of the dog that didn't bark. And so, how could a dog that didn't bark reveal the presence of the criminal and the identity of the criminal? All because the dog must have known the criminal. Our culprit today is gravity in the black hole. We're going to see the stars that seem to pirouette when nothing else seems to be affecting their motion.
- 5.8 Our cosmic backyard explored in, if I do say so myself, some beautiful depth in Astronomy 101 right here at Peterson Academy— please do click on that for your next viewing enjoyment—our cosmic neighborhood is a galaxy, an assemblage of perhaps 100 to 200 to 300 billion stars, arranged in a rotating spiral shape known as the Milky Way Galaxy, galaxy being Greek for "milkiness," Milky Way. So our galaxy is kind of redundantly named the Milky Way, Milky Way, if you speak Greek, at least.
- 5.9 It has a radius, at least out to where we are in the Milky Way's disk, in the plane of the Milky Way, of about, we'll call it 25,000 light years. That means it takes light 25,000 years to go from the center of the Milky Way out to the edge where we are, near the edge. We're not quite at the edge. Now, the closest major galaxy to us that people have as their screensavers is the Andromeda Galaxy. It's about 10 times that distance. So galaxies are about, on average, 10 times smaller than separation between galaxies.
- 5.10 They have this pinwheel shape, some of them. The type that we live in is a spiral galaxy. Astronomers have a whole taxonomical nomenclature for the different types and structures that appear in different types of galaxies. That doesn't concern us right now. But suffice it to say that we live on the edge of a spiral galaxy, not too close to the center, and the galaxy is rotating, and we are being carried around the galaxy at a pretty feeble pace. No Usain Bolt references here. It actually takes billions of years for us to make one, or hundreds of millions of years, to make one circuit around the Milky Way center.
- 5.11 Now, many people think we're orbiting around the center of the Milky Way, just like the Earth orbits around the center of the solar system, because that's where all the mass is. The Sun is located there. It's not exactly true. We're not rotating around the black hole. The black hole is paltry compared to the galaxy. It's almost a thousand times less massive than the galaxy itself. Most of the mass in the galaxy is not in the form of stars; it's in the form of dark matter, as you'll understand when you take cosmology with me at Peterson Academy.
- 5.12 So we're able to see a lot of different types of objects in our galaxy and in other galaxies. In particular, we're able to take a view towards the center of our galaxy, and it's actually quite polluted and smoggy. Being from San Diego, I say it reminds me an

awful lot of Los Angeles on a smoggy day. But infrared radiation, which is just slightly longer than the wavelength of red that our eyes can see, can pierce through cosmic dust. It did so in the case of the Event Horizon Telescope, allowing us to also image a static image of the event horizon.

- 5.13** Now we're going to see a region of the black hole surrounding the black hole at the center of the galaxy, far, far larger than the event horizon. Or even, as you know, we didn't really see the event horizon. We see the photon sphere shadow cast upon it. But that also has a lot of information, the glowing disk around the accretion disk. That's all only possible and visible in not the infrared, but in the radio, what's called millimeter wavelength light, about a thousand times longer wavelength than what the human eye can see.
- 5.14** To pierce the dust, it's like a fog lamp. Fog lamps on your car pierce particles of water vapor. Well, they need to be longer wavelength in order to traverse that, so that's why they're yellowish, orangish color of fog lamps typically. And so, too, if you go to the redder end of the electromagnetic spectrum, the invisible light, you can pierce the dust that would otherwise obscure our view.
- 5.15** And that's what scientists like Andrea Ghez and Reinhard Genzel did, starting decades ago, using massive telescopes in Chile in one case, and then in Hawaii in another case, that will be showing the output, the movies, the cosmic baby pictures and home movies of our galaxy's ginormous black hole. Again, it's a paltry black hole compared to the one in M87, but the virtue is, it's much closer to us. And so, we can hope to resolve and image the stars that lie near the black hole and ask what's happening to them.
- 5.16** And then, as good scientists, we always want to compare to what our predictions would suggest that we would see if general relativity is true, if the outrageous predictions of Schwarzschild, of Kerr, and of Einstein hold up, or if they meet their untimely fate, as many theories do, often in science. The job of a scientist is not to prove anything. I don't get paid by a theorist to prove Einstein right; I'd like nothing more than to show he was wrong and, in fact, transcend what he was able to do.
- 5.17** It's just darn hard because he was so smart, and the people that came after him have done so much work to confirm these findings. But as scientists, we want to look to refute things, not to prove things. So far, we haven't been able to refute general relativity, gravity, but we haven't reached the end of gravity. In our last lecture, we'll talk about the boundaries of things that nobody knows. We don't even know if we don't know if we don't know. So it's a quite spectacular situation to be in.
- 5.18** So, this light gets dimmed tremendously in the visible by dust. It's just like smog or soot. It absorbs the light by a factor of a trillion. And we don't want the precious light that we do receive, not that much, to be absorbed by dust particles on the way out of their creation, or on their way out of the black hole that's going to produce the images that we hope to use to test the theory of relativity. So we use infrared light.

- 5.19** When you do that, you get a much different image. It looks very different. This is a close-up image of the very center of the galaxy. It's in the constellation Sagittarius. Stars are typically not named after the constellation that they're in. Bright stars have their own names—Vega, Sirius. We've talked about those, Betelgeuse, famous Betelgeuse star. But other objects have very, very arcane, abstruse names that only their progenitors really understand.
- 5.20** This is an image of the center of the galaxy, false-color image, colorized, if you will, to show what the galaxy would look like if you could see radio waves. This is, again, from large baseline interferometers, like the one Jodie Foster used, in the very first lecture I showed you, from the movie Contact. That synthesizes a telescope not quite the size of the Earth, but the size of a good fraction of a county in America.
- 5.21** There are different attributes. There are different astrophysical phenomena that are observable here, including supernova remnants, SNRs, these threads, these arcs, they're actually reflections of light, echoes. And this kind of gives us a finder scope view — before we zoom in, spend a lot of time, decades, in fact, looking. You want to use your finder scope to identify, am I in the right place? Because it's happened more often than not, at least to me as an astronomer, oh, I did this great observation, but I forgot to take off the lens cap. I mean, if you've ever been an amateur photographer, you know the pain of this.
- 5.22** So, when you zoom in from the picture that we just showed, we zoom in, expand that on the left, you see, again, now you see in another image of this radio source, they were surprised to find there was a very powerful and variable, meaning a change with time. Most things in life, in astronomy, rather, are constant. The Sun puts out more or less a constant amount of energy. It changes a tiny bit if there are sunspots or whatnot. The Moon, the planets, they all basically look the same, day after day, night after night, literally.
- 5.23** But something different was happening at this object, and it was changing at a rate that is sort of indicative that there were some processes going on that looked different than they would look if they were near us. Remember, one of the foundations of special relativity was a feature called time dilation, that moving things, their clocks internally speed, slow down compared to us outside at a distance from them. Similarly, when things move at high speeds, they contract, they compress.
- 5.24** Were these features, these threads, these arcs, these bubbles, are they related to special relativistic effects? Are there effects of gravity that also slows down time and also reddens the light, making it more and more red, more and more easy to see through the dust? Could we verify those two predictions of general relativity before we dedicate, literally, tens, hundreds of millions of dollars in highly precious taxpayer money and, even more so, tens of years of individual students' time in their lives to get their PhDs?

- 5.25** My average time for a student in my laboratory to get a PhD, I've graduated about 20 PhDs over 22 years, I think, as a professor, and it's about 6.5 years. It's a lot of time you spend with these people, much longer than an undergraduate education, far longer than Peterson Academy, until we start having our own graduate program here, which I hope to be a part of, if that ever does materialize.
- 5.26** What is this thing that's burping out material, that's belching out radio waves, that's making these bubbles and threads? It's quite beautiful. We still don't have enough resolution to see what's going on. We still don't have the time domain sampling of this object to make a movie. That changed with the work Reinhard Genzel and his competitor, Andrea Ghez. We'll talk about both of them in a bit.
- 5.27** Now, if you zoom in the optical and look at the stars nearby, you see the Sagittarius A*. Again, A*, it doesn't mean it's a star. It doesn't mean it's a star. It's just a designation for an object that we think is the black hole. Cygnus X-1, actually, if it were discovered today, it would probably be called Cygnus A*. It was called X-1 because it was the first X-ray source that was observed with this particular satellite in this region in Cygnus, Cygnus being a constellation.
- 5.28** Here it's in the constellation Sagittarius, which you can see on a summer night from North America, and it's quite spectacular to look at it. The constellation is very rich. You can see these clouds of dust with your naked eye. You can see the clouds of stars that make up the Milky Way there.
- 5.29** What happened was, colleagues, including Andrea and Reinhard, they started to track several of these objects that were at distances not too much bigger than our solar system, several times the distance, say, out to, say, the Oort Cloud, which is kind of the boundary of what we call our stellar envelope, within which we would consider it to be our solar system, influenced gravitationally by the Sun, beyond which the heliopause, you start to get influences from other stars, in particular, the closest stars to the Earth outside of the Sun, like Proxima Centauri, they are four light years away.
- 5.30** Our Oort Cloud is less than, say, a tenth of a light year, maybe a twelfth, or maybe a fifth of a light year distant from the Sun. The farthest object ever sent by human beings was the Voyager 1 satellite. It's one light day from the Earth. When we want to communicate with it and get some information from it, it takes a day for us to send it a message. Can you send me your current heading and velocity? Then it will send back a day later, so it's a two-day round trip. That was launched before I was born, in the 1970s, and it has only gotten one quarter of 1/365th of the distance to the nearest star, so it's incredibly far. These things are really far apart.
- 5.31** But near the galactic center, they're very close, as you'll see. So, Sagittarius A* and all these different objects were identified as possible candidates whose velocity could be measured. And once you know the velocity of something, you can measure the mass of what it's orbiting around. And we did that exactly with these objects here, and they

have different names now. I'm sorry to make it a little more confusing. They start to name the different objects based on their different— they start to name the different objects based on their proximity and their position around Sagittarius A*.

- 5.32** When this was first started to be measured back 31 years ago, in 1995, they started to measure the positions of these objects—stars, some of them are stars, some of them are globs of gas. They were able to trace their patterns as they were orbiting around this unseen object at the center. You cannot see the black hole. This technology, invisible light, it is truly a dark star. So tracking the individual stars over decades traced out not ellipses; they trace out broken ellipses that are moving and not coming back to their same location precisely, just as what happened with Mercury, but on a far, far stronger scale. It's much more advanced.
- 5.33** Now, there is one particular, one of these 20 different objects or so. They're called S0. They're different objects. I think it might stand for Sagittarius Object or something like that. This one, S0-2, has a very, very close approach to the center. So its closest approach, which is called this perihelion, it comes as close as possible and has been measured as long as possible. So we've mapped out an entire orbit of this object, and I think we know a little bit about its astrophysical properties—its gas, its content of dust, and other particles, what luminosity it has.
- 5.34** And so, the orbital path of this object, if broken into an ellipse, measured over time, time starts at the top, 1992, so even 34 or so years ago. It's been orbiting, and each one of these, say, is a year, close to a year. Now, it got very close to the center of the black hole in 2013. It didn't go into it. Sorry, it got really far away. This is what's called the aphelion. Then it was predicted to come back again in 2002, so starting this way, coming back in 2002 and make its closest approach.
- 5.35** It was the one that got the closest around it every 16 years. That's actually, let's call it two and a half graduate students' careers. That means you could start a graduate student, and then she could learn everything about the technique. Just before she graduates, she teaches everything to a new graduate student, and then he learns about it, and then he barely graduates, and he teaches. So it's actually, if it was like a thousand years, even I'm not that cruel to keep a graduate student in college for that long.
- 5.36** So it got to a velocity of about 1% the speed of light, owing to the gravitational pull of the putative black hole. And what we're able to measure a distance away from it is not only how fast it seems to be moving like this, around, just projected onto the sky, but we get some motion towards us and away from us that allows us to extremely accurately measure the Doppler shift, which is a topic I cover in cosmology, quite in-depth.
- 5.37** But it's essentially the phenomena that you've all witnessed when you hear an ambulance at some distance away from you. And as it's coming towards you, the

frequency of the siren gets higher and higher in frequency, and then it gets lower and lower in frequency. That happens in sound. It also happens with light. So if you have a blue laser and you run away from me with a blue laser, it'll start to get redder and redder. If you run towards me, it'll get bluer and bluer. Blue shifts when it comes towards you.

- 5.38** And that's what's seen here. The colors indicate the blue shift or red shift. And then when it's exactly moving tangentially across your plane of vision, either to the right or to the left, it has no radial velocity; it only has tangential velocity at that point. So it's easy for astronomers to make these measurements, and it's been done since the year 2000, actually since the year 1992. So they were able to capture this as it got to its closest approach.
- 5.39** This makes it a perfect laboratory for testing the extreme gravity predictions of Einstein's theory of general relativity. So I want to show some videos made by both Reinhard and the team at UCLA. As I said, Andrea Ghez was only the fourth woman to ever win a Nobel Prize, despite what her high school teacher thought. I wonder what that high school teacher is doing nowadays, but Andrea has won the Nobel Prize.
- 5.40** The star is where we think the black hole is. These are all the different blobs of gas and dust that have been measured, some of which are stellar, and some of which might just be clouds of gas and dust, the really faint ones. They are orbiting, it looks like they're orbiting in a plane, but of course, they're orbiting in three dimensions. Some of them are coming out of the screen, into the screen. And so, astronomers have to be very accurate when they model the trajectories around it.
- 5.41** But every single one provides a data point. So every single object provides exactly what I showed you for S2, S0-2, namely an orbit of radial velocity and a tangential velocity, and the speed relative to the speed of light that can exactly be inferred from its distance, the mass of the black hole. So each one of these estimates the mass of the black hole, and they all seem to agree. It's a powerful confirmation in science. You don't look to any one person. You look to a team of people or a team of ideas.
- 5.42** This exact experiment was replicated across the planet in Chile by a German-led team by Reinhard Genzel, who found exactly the same results. He also won the Nobel Prize. So it's a spectacular, spectacular accomplishment. So here's a video of it from the National Science Foundation, supported research. Sometimes these objects come close and they seem to hit each other; that sometimes, that's an optical illusion.
- 5.43** But you can see, if we just look at one of these, let's see if we can see S0-2 on here. Here's S0-1. Here's S0-2. It's the biggest, brightest one here. Watch what happens to it over time. It looks like it hits the black hole, but of course, it didn't actually hit the black hole. This is an exaggeration. This says, on the left was the scale in tenths of arc seconds. An arc second is $1/3,600$ th of a degree. So a tenth of an arc second is $1/36,000$ th of a degree.

- 5.44 So imagine taking a degree. For reference, the Moon is half a degree wide. It's only as wide as your fingernail held at arm's length. That's half of one degree. And now divide that by a factor of 1,000, or 3,600. That would be half of an arc second. Divide that again by five, and you get one-tenth of an arc second. It's almost amazing that they were able to do this for every single star shown there.
- 5.45 Now, I also interviewed—this is my podcast—with Reinhard Genzel, who won the Nobel Prize along with Andrea. And he's an amazing human being, great scientist, worked on many different things, was almost an Olympic javelin thrower. And he just has an incredible ability to explain things. And he and I talked about this on my podcast many years ago, maybe four years ago now, right after he won the Nobel Prize. And he has the same types of images. This is from his team.
- 5.46 These show different artifacts, things they have to contend with. To correct the seeing, what they had to do is make a telescope that had, effectively, a rubbery mirror that could counteract the distortions of the Earth's atmosphere. Both Andrea's telescopes, which are owned by the University of California and Caltech on the Hawaiian mountaintop called Mauna Kea, both her telescope and Reinhard's telescopes are on Earth. So they have to look through this dirty window called the Earth's atmosphere, which has a lot of turbulence and other properties, dust and particles in it, too.
- 5.47 To correct that, a discovery was made 50 years ago that you could compensate for the Earth's twinkling of the stars, what we see as twinkling of stars, by having a mirror that was compensated exactly for how much vibration the Earth's atmosphere has, canceling it out, like noise-canceling headphones. They developed and used, in these cases, they used light-canceling telescopes that vibrate to cancel out the distortions of the Earth's atmosphere, called adaptive optics. It's an incredible technology.
- 5.48 So on the upper left and right, you show what the pattern would look like. If you didn't correct for these effects, the lower right shows the clean, crisp, more circular-looking blobs and stars that they got afterwards. And then on the lower left, they show the animation that their team made in Germany, from his team in Germany. So it's a spectacular accomplishment that they were able to make.
- 5.49 And what I love about it, and we'll see in just a second, that, when we examine the evidence and the data that came from the Event Horizon Telescope, it will also be shown to agree with the properties of these optical and infrared radiation, so visible light, near-visible light, compared to radio wavelengths, and they all agree. That's extremely powerful in science.
- 5.50 It's like you're studying dinosaurs, and you happen to have a time machine, and the time machine tells you, yes, exactly on this day it died because there was this huge asteroid that hit the Earth's surface, and that's what caused it. And then you look in the Earth's crust and you see certain compositions that can only come from asteroids. And

then you have geologists and you have dinosaur evidence, and you have plant evidence.

- 5.51** The more evidence you have that agrees, the better. It's not true that one discrepant fact can completely kill something because it could have its own problems. But as I explained in my astronomy course, in the case of the dinosaurs and the meteor impact that seems to have killed them, there's overwhelming evidence that that occurred, and that happened 65 million years ago on Earth. We're never going to have eyewitness evidence, but this is as close as you can possibly get.
- 5.52** These data are incredibly precise. They agree with the model that predicts that there is a mass at the center of our galaxy, of four million times the mass of the Sun, essentially concentrated in a point, point being much smaller than, say, the radius of the Sun. You can't see the actual black hole in these types of images, but you see them orbiting and coming within a distance that's comparable to, say, the size of our solar system. Again, that's much bigger than the event horizon, but we'll see how these things can be lined up with one another.
- 5.53** The ultimate confirmation will come when we can detect things in multiple wavelengths with multiple telescopes, most importantly, different people, because people have their own biases for what they want to detect, what they try to detect. If you ignore the black hole that we seem to have there, the enclosed mass that we would see is far, far less. It's completely inconsistent with the data points here.
- 5.54** The data points, if you exclude, if you say, only what we see is visible mass, in other words, there is no black hole, the data points are up here, they're 100 times too high. In other words, we see the mass infer the mass; it's 100 times higher than there would be if you only counted the amount of visible light that you can see. So if your model is, "I don't believe there's a black hole," you're 100 times wrong. It's incredible. That almost never happens in science.
- 5.55** So there's only an explanation that we think is possible for all these different objects. Remember, if it was just one, that would be interesting; it might be enough to suggest there's a four-million solar mass black hole. You could actually do that. But there's 24 of them. And then there's two different teams that measured 24 of them with completely different technology. So it's very hard to explain that. It doesn't stop people from trying to explain it and write me emails that all these people are wrong, and I should work with them, and they'll give me the Nobel Prize if I just give them the math. I get that a lot.
- 5.56** Here's the scale of that object, S0-2 or S2. It's about, say, maybe twice the diameter of our solar system. It's highly elliptical. Its precession of its ellipse does change. Here's our solar system, and this is just to show you what we're talking about here. But the mass of the black hole, where the black hole is, is far, far smaller than the region that's shown in our solar system. There's about a four-million solar mass chunk of mass or

amount of mass equivalent; and it's within, it's much, much smaller than the radius of Pluto, but it is definitely, all we can say based on the orbit of these objects, these stars, is that we don't know any other object that it could be, besides a black hole.

- 5.57** So this is looking at, again, you're seeing, imagine if you could only see bees. For some reason, you could only see yellow and black things, and you see they're swarming around this object, and they seem to go into it, and then they come out with honey on their legs or whatever, and they go in with pollen on their feet. But for some reason, you can't see the hive. You could infer its existence from the behavior of the things that are surrounding it. That's what's happening here.
- 5.58** So, with all this understood, now that there's sufficient evidence that there is a black hole, it became a laboratory to test extreme gravity. You can't do it on Earth, to create a four-million solar mass object on the Earth, and then let's play around and see what happens to light as it goes near this object; does it get sucked in or not? So, in 2018, it made another close approach to about four times the distance to Pluto from the Sun, and it got to a speed of about 3% of the speed of light, two and a half times the speed of [its earlier pass].
- 5.59** At this time, now you could test, due to the proximity to the black hole, the strength of gravity predicted by the orbit, so that told us the mass of the black hole, and the velocity— how much time dilation should this object experience from its relativistic motion, both special relativity, moving close to the speed, moving a fraction of the speed of light, far greater than we could ever travel or have traveled, and the extreme gravity.
- 5.60** The prediction was, it would be a test of the gravitational dynamics, and it did. It held up. It proved that, very close to this pericenter, which occurred here sometime in late May, early June of 2018, the predictions for how the orbit would behave and precess over time, those exactly matched the predictions of Einstein's general theory of relativity.
- 5.61** Now, it's extremely luminous, but it's not that active. It's not really ripping apart things like you see in the movies. Again, the more massive the black hole, the less extreme the spaghettification is, or the longer it takes before you get ripped apart, or matter gets ripped apart. That's true here, too. These objects are hundreds of times the Schwarzschild radius away from the center of the black hole. They're in no danger of being spaghettified. But still, we thought, well, maybe we'll see some fireworks as these things come together.
- 5.62** It actually has a luminosity far less than the very first black hole ever discovered, Cygnus X-1. We discovered Cygnus X-1 because of its intense luminosity in the X-ray region. Here, this thing is putting out less X-rays than a typical star might put out, let alone a black hole. So, that's kind of a question. This thing doesn't seem to have a very powerful ergosphere. It's not accelerating, compressing, and then illuminating the

galaxy with all these X-rays. We don't know why. It's not really anyone's fault. It could just be an accident of how it came to be. We're learning more about it.

5.63 But we can also see things about a black hole at the center of our galaxy, not using light that we see from the stars today, but light that has had enough time and enough distance to travel millions or thousands of years in this case, away from the black hole, reflect off something else like a giant cosmic mirror, and then arrive at our telescopes on Earth. And these are called light echoes. As I say here, it's called archaeoastronomy. You can look back in the past and make a movie not just of how the stars have behaved over 30 years, but as it behaved maybe hundreds or thousands of years ago.

5.64 So, astronomers used the reflection of, say, the black hole if it flared up and did emit a lot of X-rays, as we believe it did in the past. The way we know that is because we see the shadow, we see the reflection of light, X-ray light, off of these molecular clouds that act as tiny mirrors, and then bounce light to our telescopes here on Earth. And from that, very talented astronomers have been able to show that today's luminosity is perhaps 10,000 times less luminous than it was just 100 years ago.

5.65 So, if we had these telescopes watching Sagittarius A* 100 years ago, before the invention of most of modern astronomical telescopes and spectrometers and things like that, it was a million, hundreds of thousands to millions of times brighter, we would have seen a very different sky. This does then start to set limits on how fast black holes can evolve. It actually shows they can evolve really fast. They can change their eating habits. They can change some of their local gravitational structure.

5.66 Perhaps the ergosphere can change on time scales of just a human lifetime. That's very fast compared to the distances involved in what we're talking about or the formation period, at which it takes a star to turn into a black hole, as we talked about in yesterday's, and as we talked about in lecture four. So, our galaxy flared up. We don't know why, but it starts to tell us interesting things about astrophysics.

5.67 So, when we look at these objects and we zoom in towards the center of our Milky Way, we start to see some features that were indicative of the classic black hole accretion disk phenomenon, namely that there seems to be a perpendicular jet of material. Actually, it's more like two bubbles back to back. That has been observed with a different type of satellite called the Fermi Gamma-ray satellite that observes these huge structures that are about the diameter of the Milky Way, each one of them, and there's two of them; and one is in, sort of pointing in the north hemisphere of the galaxy and one's pointing in the south hemisphere.

5.68 We don't exactly know why they occurred, but it's part of the mysteries that hopefully new students will be able to explore and understand. Why does the galaxy have this structure that's very different from a classic accretion disk phenomenon, which itself comes from an ergosphere, which itself comes from a spinning black hole, which can be anywhere from maximal spin to minimum spin, have a maximum, in which case the

maximum spin has the minimum ergosphere diameter, much brighter and more intense beaming in jets. Ours doesn't seem to have that. We don't know why, but it's fascinating to think about.

- 5.69** So the next thing that really came up and is fascinating to me is how astronomers went from these images that we just showed you, these home movies, and then, essentially immediately afterwards, we retreated to taking a still picture. Now, normally, you'd think a picture is worth a thousand words. The movie must be worth a million words. So this movie must be worth a lot more. In fact, it is, for understanding the strong gravitational limit close to the black hole, but not quite near its event horizon.
- 5.70** But just a few years after this series of images were completed, this home movie rather, was completed in the late 2010s, came the Event Horizon Telescope that we talked about in a previous lecture, which was designed to image the effects of—here's a picture of the Milky Way Galaxy, as you see with your human eye, say. It looks milky. This is in the constellation Sagittarius.
- 5.71** These dark bands are not regions where the stars are absent; they're regions where there are tons of stars, billions of stars, tens of billions of stars, but they're obscured by the dust that I talked about before. Dust comes about because of the cosmic explosions that produce supernova that produce the meteorites, which I did bring for all of you in the audience, predating the Earth. The meteorites that you'll get are 4.5 billion years old. They're older than the Earth is. Earth is about 4.3 billion years.
- 5.72** That's because they formed out of the same stellar remnants, the ashes of a star that used to live in this stellar neighborhood. So she's talking about this, this discovery, which came a few years after the discovery of the giant black hole in the giant galaxy known as M87. This is a much less massive, about a thousand times less, 500 times less massive black hole in our galaxy, but it's literally a million times closer or let's say, a thousand times closer than this object is.
- 5.73** If you do the math, it turns out that they're about the same visibility, something that's 55 times brighter, bigger, brighter, a thousand times bigger and brighter, but that's 55 times farther away, $1/r^2$. You start to get a factor that makes them comparable. They discovered that they could actually image the event horizon. Again, it's kind of a misnomer. It's not exactly the event horizon. The event horizon is somewhere in here, about the size of the arrow here. The event horizon is much smaller than this photon sphere is, at least by a factor of two.
- 5.74** Then the [photon sphere] is at the boundary of the ergosphere, which, for the Milky Way, is actually quite unimpressive. As I said, it's not really spinning that fast. It's not dragging space-time like a ball rotating in honey, as we discussed in lecture four, but this is a still image. Now, imagine you take this image and then you superimpose the data from the work of Andrea Ghez and Reinhard Genzel's teams. You see the image on there would be tens of hundreds of times bigger than this image here.

- 5.75** This is smaller than, say, the diameter of the Sun or about the event horizon of the Milky Way's black hole, Sagittarius A*. It's much smaller here, so you really wouldn't see it on the same scale, but you can start to make predictions. Is this consistent, the structure of these lobes? And can we start making a movie of the black hole's event horizon? People are working on that, too, changing its structure. Then also looking at it, because it's so much closer than the other one, we can now start to look for the magnetic field of the black hole's event horizon.
- 5.76** That's important because magnetic fields are caused by charges in motion. In most magnetics called electromagnets, they are caused by a charge. Moving creates a magnetic field. And so, if you can image this over time, you have a change in motion and you have charge, you may be able to see the black hole's event horizon in motion, and also what its magnetism would be, and that would be part of, associated with, its charge.
- 5.77** Remember, there's only the three properties of black holes. We measured its mass from the stars that are orbiting it. We've measured its spin from the ergosphere and the properties of objects that are orbiting around it and the structure of the event horizon. Then the charge could be measured by the dynamics of the event horizon, as seen in a time-lapse movie of this. That's what people are working on. It's a fascinating thing to think about.
- 5.78** It dovetails nicely into our next subject, which is the study of not seeing black holes, but listening to black holes. We built an enormous cosmic ear. Scientists have been working since the time of the late 1960s when black holes first came about, to think about, would it be possible to image or hear or see the effects on space-time of black holes? It turns out that a single black hole sitting by itself produces no gravitational waves.
- 5.79** To get gravitational waves, just like to get electromagnetic waves, electromagnetic waves are caused by charges in motion; either charges oscillating back and forth in a wire can create electromagnetic wave, a charge oscillating in an antenna can produce or detect radio waves. That's how antennas work. And so, it was found that you needed charges in motion to make electromagnetic waves, and gravitational waves, according to Einstein, are just like Maxwell's electromagnetic waves. They travel at the speed of light. They are caused by masses in motion.
- 5.80** It was theorized that if you had a sensitive telescope that could detect tiny vibrations, namely sound waves, if you will—sound waves not in audible terms, but in terms of the vibration of space and time itself—you could detect masses in motion. The easiest masses in motion to detect are the most massive moving at the highest speeds with the greatest accelerations. That's where gravitational wave astronomy came into play.
- 5.81** Einstein predicted it as a consequence of his theory of general relativity. Again, more about that in a general relativity course at some time. They propagate, they travel like radiation, like light. They travel at the speed of light, c , the same exact speed. That was

an astounding finding. Einstein thought, like many of the things he predicted, no one would ever detect these things. Lo and behold, exactly 99 years later, they were detected, and we'll talk about that.

- 5.82** What creates these waves? They can't be created by a single isolated mass. The mass has to be in motion, but since motion is relative, it can't just be a single mass. Typically, to detect these things, we need binaries or even triplets of masses. Those masses could be a neutron star colliding with another neutron star. It could be two pulsars orbiting around each other, as we saw in earlier lectures. It could be two black holes. It could be a black hole and a neutron star. There are several different permutations.
- 5.83** More or less all of them have been seen using this new type of telescope called a gravitational wave observatory. The first prediction that was confirmed was that, when these objects rotate around each other, when two compact, massive objects orbit around each other, according to relativity, they must emit waves of gravity, very weak and feeble, even if the masses are huge, until they get extremely close to each other, move at extremely high velocities, and eventually coalesce and collide, going from two independent black holes into one black hole, which then afterwards emits no gravitational waves.
- 5.84** What we'll see is that, for perhaps millions of years, they orbit around each other, two black holes, in this case, two pulsars or a pulsar and another star, a neutron star. As they were orbiting around each other, they have not yet collided. They've just been slowly dancing around each other over time. Eventually, they will collide because if you radiate energy away, there's no way to prevent you from gravitationally collapsing and condensing into one another.
- 5.85** These were detected by looking at the orbit, the Doppler shift, of these objects over time, and noting how much shift of the orbit changes over time, and then overlaying Einstein's prediction of what two masses in accelerating motion, orbiting around each other, what would they lose energy at, and how would their orbits change exactly over time. You see the data from 1975, 51 years ago, all the way up; now it's gone way past this mark over here.
- 5.86** They made multiple measurements of these pulsars, and the period of the pulsars, using radio telescopes, and they were able to use this as a very, very precise clock. Not an atomic clock, not a digital clock, but a pulsar clock. And these pulses are very accurate ways of testing out how close or far are these objects behaving, compared to the predictions from relativity.
- 5.87** At this point here, in 1993, they won the Nobel Prize because this had been over 20 years' worth of data. And the data were collected when one of these gentlemen, Russell Hulse, was a graduate student working for his advisor, Joe Taylor. So he was a young person. He did the work when he was in his 20s and then won the Nobel Prize

when he was in his early '50s. And often that takes a lot of time before you can actually detect these things.

- 5.88** So this object, this system, it's a binary system, was the first unequivocal evidence that gravitational radiation, a key prediction of Einstein's theory, was correct. There's no way to avoid it. It's the simplest possible explanation. Now, can you detect these waves of gravity directly? We're looking at these objects indirectly and inferring the existence of gravitational radiation from the loss of energy as these things are orbiting around each other.
- 5.89** Can you directly detect it in your laboratory like a telescope, not watching them behave, but actually seeing the waves of gravity themselves, detecting them? The answer is yes. But the challenge took decades and hundreds of millions of dollars to accomplish. The way it was done was to ask the question, what is a wave of gravity? If you were standing on a scale in this room, and a wave of gravity came into this room, what would happen is that space would squish and rarefy. It would compress and retract, compress and [retract] like this.
- 5.90** It would do so at some frequency, depending on the frequency of the gravitational wave. The gravitational wave would propagate through the room such that, if you were on a scale over there, you would see it a few nanoseconds later than I would. Your weight would change. First, it would go down. Oh, that's great. As the space-time squeezes and pushes me up, I separate from the scale ever so slightly, my weight goes down. Then you, since you're 10 feet away—light or gravity waves travel at the speed of light; they travel one foot every nanosecond; you're 10 feet away from me—10 nanoseconds later, your weight would go down. But then by that time, my weight would have gone up. It's not a great diet plan, unfortunately.
- 5.91** What these waves do, if you had a ring of masses in this room and a wave of gravity comes through it, first it gets squished like this and squashed out in that direction. That's the compression that would be pushing down towards either the center of the Earth, the whole Earth would be getting squished as well. But these are imperceptible, even from supermassive black holes, imperceptible. We'll talk about the discoveries that were made.
- 5.92** Then at a different point in phase, 90 degrees of phase later, you get the opposite behavior. I would get lighter, space-time would get squished in this direction and compressed like this in this direction. That's the way that these waves of gravity work. Now, the amount of deviation is one part in a thousand, million, million, million, or 10 to the 21st. So if you weighed a kilogram, if you have a mass of a kilogram in the Earth's gravity and you weighted a pound or 2.2 pounds, you would change your weight.
- 5.93** Initially, you'd get lighter by one part in 10 to the 21st kilograms, and then you'd get slightly heavy as the wave passed through at one part in 10 to the 21st kilograms as well. But if you have something that has a mass of 10 to the 21 kilograms, well, now it's

going to change by one kilogram. But that's not exactly something easy to really represent. The Earth does have masses comparable to that, thousands of billions of kilograms. But that's not how things work.

- 5.94** Instead of measuring the mass using a scale, instead, we said, well, that in addition to time dilating, we also have space compressing, remember, from relativity. If space compresses by a tiny amount, then if you had something that is extremely long, let's say you had a ruler that was 10^{21} meters long, and then it was—which is enormous, that's much, much larger than our solar system—then that object would shrink by one meter as the wave went through, and then it would stretch by one meter. The ruler would stretch and shrink by one meter as well.
- 5.95** So we can't get quite 10^{21} meters, but we can get 4,000 meters, and we build two telescopes that are each 4,000 meters long, 4 kilometers long. And then you compare the two of them on different parts of the Earth's surface. And this is called the Laser Interferometer Gravitational-Wave Observatory or LIGO. It was built over about 40 years. It was different places where it was being built. They built two of these objects. These are these L-shaped instruments called interferometers. They don't look like telescopes, but they are.
- 5.96** When a gravitational wave comes in, say a gravitational wave comes in from above in this picture, it would initially stretch the distance between the laser and the mirror on the right, and it would squish the distance between the mirror on the left and the photodetector. That causes what's known as a phase shift. As a gravitational wave comes in, over 4 kilometers, it shrinks it by 4 kilometers, divided by 10^{21} power. It's fractions of the distance of a proton's width, but we have billions and billions of protons, and it happens many, many times.
- 5.97** The technology of lasers and detectors are so sensitive that they can measure these minute shifts over these enormous distances extremely accurately. The technology is, it required advances in laser power. You have a huge power laser, like a megawatt power laser, just enormous power lasers. The laser is going back and forth in a tube between two mirrors that are otherwise reflecting back and forth, just doing nothing, until a gravitational wave comes in and starts to modulate the distance between them. That causes interference of the light, and that causes detection in this detector, which is not enough to claim you've detected a gravitational wave.
- 5.98** Halfway across America, there's an exact copy of this exact same interferometer in Louisiana; the other one's in Washington state. We know the distance very accurately between Washington and New Orleans. If you don't see exactly the time delay between the speed of light traveling from Washington state to Louisiana state, if you don't see that exact behavior, you throw out the signal.
- 5.99** It could be somebody driving a truck that's causing a vibration that's causing the mirrors to go like this, mimicking the signal. But that truck is very hard to do at the

same time in Louisiana and Washington. So it's a powerful guard against the most important source of error, which is me or you, the experimentalist, seeing what you want to see.

- 5.100** In fact, one of the main leaders of LIGO, Barry Barish, and Kip Thorne was one of the members of this team who also won the Nobel Prize for this detection, they told me that one of the main sources of noise in Louisiana were hunters that would like to shoot the tunnel, these tubes, which are held under extreme vacuum, by the way. It's the longest amount of vacuum on Earth. It's the largest chamber of vacuum, if you will, on the Earth, two 4-kilometer-long tubes completely sucked out of all the air molecules, so the laser doesn't waste energy heating up air inside of there.
- 5.101** And the hunters would come by and say, "Billy Bob, let's shoot that tube over there." I'm sorry if you're from Louisiana. That's my Louisiana imitation. And it would create a huge signal. And they were like, that's okay. It doesn't matter unless we see another hunter exactly knows the exact time of flight of particles, of waves of gravity, between Louisiana. So, it never happened that they saw two events that were accidental, if you will.
- 5.102** And what they see is not something that you see. They actually hear the signal. The interference can be changed into a waveform, a sound waveform. And what happens is, these two black holes came together about 1,400 million light years away, so 1.4 billion light years away. We know only very vaguely where these two black holes were. We don't know the galaxy they were in. We can't really accurately point or know where the telescope is pointing because, effectively, it's just governed by where the Earth is and what the time of arrival is, in these two different detectors separated by a couple thousand kilometers.
- 5.103** But both of them saw this exact same waveform, and I'm going to play it for you in a second. So it's a strain. They measure the amount of strain, which is the amount of gravitational force that you'd feel. Again, it's below microscopic. Again, it's a thousandth of a millionth of a millionth of a millionth of a meter per meter of distance. What happens is, these two objects orbiting for maybe a billion years, these two black holes, one was 36 times the mass of our Sun, one was 29 times the mass of our Sun. Put those together, it should have been that it was 65 solar masses.
- 5.104** And what they're able to measure is how long it took for these two different things to come together. They make this signal called a chirp, which I'll play for you what it sounds like. A chirp is just like a bird. It increases in amplitude and increasing in frequency. As these things get closer and closer, they get faster and faster. Their orbits get smaller and smaller. Then they collide. They release a tremendous amount of energy when they merge, and then they coalesced into a single black hole.
- 5.105** You take two black holes, put them together, you get a bigger black hole, but you don't get exactly the mass that they were; the sum of the parts doesn't equal the final total.

A lot of that energy, well, it's not a lot. It was three solar masses' worth of energy went into shaking up space and time, and that emission of energy went out in all directions. Some of it happened to point towards us, got diluted by the inverse square law, where the inverse you're squaring is 1.2 billion light years.

- 5.106** So it's a fantastically small amount of energy, of reverberation, and yet they were able to detect it. They were able to do a lot of processing to confirm that this was right. They kept it hidden. Every now and then they'd inject in a fake signal into both detectors that only one person on Earth would know. He or she would stay up and, at some point at night, push a button on their computer, and it would simulate on both detectors a gravitational wave.
- 5.107** Sometimes they'd do it, a merger of black holes, and then they ask, "Well, what was it?" And everyone would get excited because nobody knew it was fake. And in fact, the first one ever detected, they just assumed it was a drill, because they had only gotten drills up until that point. And they just assumed it was nothing. They almost had turned it off for the night, basically. And then this was the detection that eventually won the Nobel Prize for three of the people, it turns out, became friends of mine eventually.
- 5.108** And so, they had to do a lot of things to ensure that they did not trick themselves into believing what they wanted to believe. And this is the characteristic signal. We call this listening to the sound of black holes colliding. So here is that. You're going to see, on the top and bottom, you're going to see the signals at one of—the Hanford site is the site of the LIGO experiment in Washington. The Livingston one is the site of the one in Louisiana.
- 5.109** They have slightly different properties. They're different instruments. They're built by slightly different teams of people. They have different characteristics. No two instruments, no two telescopes are exactly the same. You're going to see over the course of only a half a second, imagine these things orbiting for millions or billions of years, galaxies a billion-plus light years away, and eventually, one day, a billion, 200 million years ago, they decided to come together.
- 5.110** And this is what it did to space-time, as synthesized and converted into audio frequencies. These are the actual frequencies that they reverberate. It's not like this has changed; this is the way that, if you could, you wouldn't hear anything because there's no air, there's no sound waves in space, but this is converted, nothing else done to it than this. Okay, so here we go. [audio plays] So it's that whoop, whoop. The sound is getting more and more intense, and then the frequency is getting higher and higher, and it's just like a bird. When a bird chirps, that's exactly what it does. They call these signals chirps.
- 5.111** I was in Caltech, one of the main players was Caltech, where my friend Kip Thorne is and where Barry Barish is, two of the three men who won the Nobel Prize. I remember walking by their offices in the early 2000s, a long time before this signal was detected. I

used to just hear this" whoop-whoop." I was like, are they birdwatchers? I thought they were physicists. I got to hear this sound before it was ever known by the public, and it's just such a huge accomplishment. So, this is plotting, yeah, the frequency versus the amplitude over time. I'll play one more time.

- 5.112** Then the other thing to notice is, after it ends, it goes to maximum, and it just dies. Because as I said, you can't have gravitational waves from just a single object by itself. It has a huge mass, it distorts gravity, but nothing else. That's the key thing that proved beyond a shadow of a doubt, these two things came together, collided, amplified, exploded gravitational waves, shot out three solar masses times the speed of light squared worth of energy in shaking up space-time itself, and then nothing else happened.
- 5.113** There's just a black hole sitting there. Again, we don't know where it is. We'll never know where it is, because it doesn't emit gravitational waves anymore. So listen for the "whoop-whoop" and then nothing, silence. [audio plays] So that's how we knew that these black holes were real. We confirmed gravitational waves, we confirmed general relativity. The next step that I would like to see, well, let me ask you, what would you like to see?
- 5.114** We've seen the Event Horizon Telescope. We've seen the black hole in the Milky Way. We've seen the imaging of its stellar orbits around it. What would be a third thing that you would put on your bucket list? What would you like to see for the Milky Way's black hole? I would have said, I would like to hear what the black hole at the center of our galaxy sounds like.
- 5.115** Unfortunately, that would mean it would have to meet its end, collide with another object, perhaps another black hole, and that would be the end of one of our most constant companions, the black hole at the center of the Milky Way. Thanks very much. We'll pick up in the next lecture another and perhaps more interesting aspect of black holes.

Lecture 6: Wormholes & Quantum Mechanics

At a glance — Wormholes and quantum mechanics.

- 6.1** Physicists are incredibly creative, imaginative, and very much childlike in our imaginations. Sometimes that's a good thing, and sometimes it's a bad thing. I always say scientists are like children. We are creative. We like to experiment. We also don't like to share our toys. We fight. We're jealous and petty. We've seen some of that throughout this course as well. But today we're going to look at one of the most exciting, imaginative, and truly speculative and mind-bending concepts that are consequences of the theory of general relativity, and especially its application to black holes. In fact, they have the name "hole" in them. They are wormholes.
- 6.2** And in particular, the concept of a wormhole was really not taken so seriously for a long time, until Kip Thorne, who consulted on the movie *Interstellar* and has won the Nobel Prize as well, he was asked by Carl Sagan, another famous astronomer and popularizer of science, to come up with a device that would allow time travel for his book, called *Contact*, which he wrote with his wife, Ann Druyan, who, as I said, has been on my podcast and is a lovely, artistic, creative individual.
- 6.3** One of the things I do on my podcast is, I always ask my guests what they would put on a time capsule, like the ones in *2001: A Space Odyssey*, these monoliths that are featured throughout that movie, which also has a wormhole and a time-travelling device, but not of the same name. We didn't call them wormholes back then. I ask all my guests this, and some say, "I'd put my kid's pictures on it, or I'd put a hydrogen atom on it," or some can be more nerdy than others.
- 6.4** But I asked Ann Druyan, when she was on, I said, "What would you put on a time capsule, guaranteed to last for a billion years?" She said, "Oh, that's easy. I've already done that." I said, "What?" And she said, when she was first dating Carl Sagan in the 1970s, he was trying to come up with this idea for some sort of an icon, a time capsule, that could be put on the Pioneer spacecraft to go into interstellar space. And later the Voyager golden disc meant this.
- 6.5** So they recorded all sorts of sounds and all sorts of music from around the world. And they put a picture of a man and a woman, as if aliens, when they find it, would be mesmerized by this. And it has all sorts of things on it. I don't know if, I know what this thing is, but I don't know if I found this thing or saw this thing, if I know what it would depict. But this is supposed to depict, on this lower right, the Earth at the center, or the sun at the center of these rays, these diagrams, and these are distances to pulsars, are meant to indicate the directions to different pulsars.
- 6.6** This is meant to be a hydrogen atom. This is meant to be a spectral signature of something like that. There's many other things, and on the back of the Pioneer—here we go—yeah, so this is a binary code that tells time of the scan. So this is how they

could replay it. First of all, we've gone through so many different incarnations of recording music and sounds that I don't know if this would be possible for even a smart teenager, Gen Z or whatever they are nowadays, to decode this thing. They'd be like, "Where's the TikTok? How do you get TikTok on this thing?" Let alone a record player.

- 6.7** But anyway, I asked her what she would put on it. She said, "Oh, I already put something on one. My brain waves were recorded within the first month when I was falling in love with Carl Sagan." So you asked about love and connection and meaning. So she has already done this; she's put her brain waves—this record could last for billions of years, traveling out into interstellar space, courtesy of her husband, Carl Sagan, who had, around the same time, consulted with Kip Thorne about how you could get a plausible mechanism for a human being to travel through space-time.
- 6.8** In fact, to go to very, very distant places that, ordinarily traveling, even at the speed of light, would take enormous amount of time, as we know. How could you shorten that to human life spans, to go distances that would ordinarily take thousands or millions of years? So Kip Thorne came up with this concept called the wormhole, and that's what we're going to explore today. It's featured in many, many movies and even in television shows. And the main concept is illustrated here.
- 6.9** A wormhole is a hypothetical shortcut through space and time, the details of which we'll explore but encounter in more detail in our course on general relativity. Now, it's purely hypothetical. You'll see the barriers to constructing one, even not considering the passage of a human being trying to meet her late father, as takes place in the movie *Contact*; even transporting photons, the easiest, most massless things that you can imagine, are nearly impossible to imagine, let alone transporting a human being. So we'll have to talk about how that would be constructed.
- 6.10** But are they theoretically possible? Yes, they are theoretically possible. Emphasis on the theory. If it would be possible to construct it, what's depicted on the top is a section of the universe, part of the universe, here depicted as two-dimensional, but we know it's really four-dimensional. Entering in at one point, say, on the top of the wormhole shown here, allows you to traverse through some distance, physical distance, possibly arbitrarily far away in the same universe, in the same space.
- 6.11** It's shown curved on top of each other, but you should really think of this C-shaped thing as being all of four-dimensional space-time, and then entering in at one point and then coming out at the other point. That's the question. Could you do such a thing? Could you use it to effectively teleport first information? Let's just start with photons, light, radio waves, gamma rays, whatever. Could you see it entering in one place and then come out in another place? That's the simplest, but not easy, concept of a wormhole.
- 6.12** There are other instantiations of the wormhole that allow you to travel not only in your universe, faster than the speed of light or greater distances than could be traversed at

the speed of light in a finite time, but between other universes. This does come as a consequence if you play the right machinations with Einstein's theory of relativity, of general relativity.

- 6.13** The structure of it generally has two main components, mouths and throats. The mouth is just as you think. You come in to one point, you'd come out in another point, and you'd go through, traverse this throat. Now, that would be possible, assuming you could construct it. Of course, mathematically, you can construct it and think about such a thing that connects distant regions of the universe or even different universes.
- 6.14** Now, if you think about it, it kind of looks like there's two black holes. It looks like there's a black hole on top and a black hole at the bottom, and they're kind of connected. Their event horizons and their eventual singularities are shared in a way between the two different mouths along this throat at some particular distance.
- 6.15** Now, it turns out, once you form a wormhole, it can be considered to be stable, but only for a tiny, tiny brief amount of time. In fact, it may be effectively infinitesimal. If you could do it, the first thing you would want to teleport is, say, a photon, just a particle of information. You'd shine in light of a given frequency or a single photon and ask, does it come out some other place?
- 6.16** Now, it has been modeled in computers, but it can also be modeled in a physical laboratory. Again, you have to do a lot of games with it, but physically it can be modeled, but has to be very, very highly simplified. Actually, Einstein thought about these types of entities. He didn't call them wormholes. They were called Einstein-Rosen bridges. They were possible to create but were effectively, dynamically unstable. As soon as you created one, it'd disappear, collapse.
- 6.17** In the movie Contact, which is based on the book Contact, written by Ann Druyan and Carl Sagan, Jodie Foster, who, again, is playing a real-life radio astronomer named Jill Tarter, who works at the SETI Institute, SETI stands for "search for extraterrestrial intelligence," which is really searching for extraterrestrial technology, passively. Instead of actively going to other galaxies, what Jill Tarter and others are doing at the SETI Institute in Mountain View, California, is, they're listening using radio waves, they're looking at light signals, they're maybe looking for other types of long-range messengers that could convey information about the technological prowess of an alien civilization.
- 6.18** It could be, as I talk about in Astronomy 101, there could be life on other planets, but it could be like the mold on a strawberry that we have here on Earth. That is a huge thing that life exists elsewhere outside of Earth. We have no evidence for life, alien technology or even mold on other planets, even in our own solar system, let alone in other solar systems, let alone in other galaxies.
- 6.19** But if you assume that it exists, there's many forms of life, from single-celled organisms up to conscious human entities or superhuman entities, maybe even purely artificial intelligent entities that traverse the cosmos at the speed of light because they're not

bound by corporeal demands, like having mass that accelerate at infinite energy cost. So, some speculate what is an alien is actually artificial intelligence, and that's easy to transport; at least it can go at the speed of light.

- 6.20** So in the case of this, this book, based on the true life story of Jill Tarter, up until the time when she detects aliens, really does more or less represent a young woman, astronomer, using these powerful telescopes around the world, dedicating her whole life at great risk, because people don't really take the search for aliens— they're polluted and corrupted by science fiction and giant green prosthetic foreheads and all sorts of silly movies and things like that. For a serious scientist like Jill or her colleagues and friends of mine that work in this field, it was for a long time considered complete death to your career to study extraterrestrial life, let alone extraterrestrial intelligence. But nowadays it's become extremely popular.
- 6.21** And even I saw an interview, which I couldn't believe it, a lucky podcaster was fortunate enough to interview former President Barack Obama and asked President Obama, "What did you want to know the first day in office, when you came in January 2009?" He said, "I want to know, where are the aliens?" And the interviewer said, "Well, where are the aliens?" And he said, "Well, they exist, but they weren't what I thought that they were." And then they moved on to, like, "Well, what do you think about the Chicago Cubs?" I'm like, I'm going to kill this guy. You just had the former President admit that aliens exist, and that was your follow-up question, "How are the Cubs doing this year?" Anyway, it was incredibly frustrating.
- 6.22** But to search for these extraterrestrial life forms was an obsession of people like Carl Sagan, like Jill Tarter. Many people do encounter this and want to not only think about the existence of other life forms, but wonder how they could be here. If aliens are here, they must have technology way beyond us. They must have technology, not just be mold on a strawberry. They must have something like we have, the ability to create technology.
- 6.23** Only if they create technology, even in an advanced life form, a chimpanzee right now on Proxima Centauri planet, around Proxima Centauri, we have no idea if there are no chimpanzees there. I can't prove that there are no dolphins swimming in oceans on the surface, because they have no technology, there's no way to do it, unless we go there. So the concept in Contact that was put forth is, let's go to the star system. Of course, now there's a problem.
- 6.24** Going to a star system for a scientist like Jill Tarter or Jodie Foster, played by her, going to another star system in a human lifetime requires some way of teleportation, effectively breaking the light barrier of travel time to a distant civilization. That is accomplished through this device called a wormhole in the movie. Now, it's not really talked about how this happens. They have this technology, it's like a spinning wheel that drops and falls into this tesseract, and then all of a sudden, she's floating around in space.

- 6.25** This is not only a wormhole, which could be microscopic in size; you could think of making a quantum wormhole subatomic, below the size of a proton. That's theoretically possible to think about it, whether or not such thing could ever exist. That would be the simplest thing, to make a tiny wormhole. But this has to be a human-traversable wormhole, meaning that the throat has to be at least as big as the person that's trying to get through it.
- 6.26** So, this kind of a depiction, that's Kip Thorne in the middle there I made, going between these two different regions of space-time, these two holes, black holes on one side, black hole on the other side, and a throat connecting them, and you have to be at least a diameter of a meter or two meters for them to get through— that's an enormous challenge. Now, how this is solved in the movie Contact, you can search for on YouTube, the trailer for it.
- 6.27** But in the scene where she's going through it, this is a necklace she got from her father. And then she realized somehow that the chair was causing some difficulty going through the wormhole. Not exactly sure how she realized that. Then she's floating around in this capsule. It becomes transparent. Now she's coming near this star system, this gas cloud. This is presumably the accretion disk around a black hole. It's very beautiful, very human. She doesn't know what's going to happen.
- 6.28** Then she ends up, she falls asleep, softly landing on the surface of this planet that's near the star Vega. And then she's transported back in time. And there's the black hole that was perhaps the throat of the entry point for the wormhole's throat. And then it's all sort of interesting. And then she comes back and wakes up and is back on Earth, and nobody believes her because they just saw, from their perspective, they saw this ship drop through a tesseract or whatever, and then it fell and just hit this net that caught it. It lasted a fraction of a second.
- 6.29** But for her, her experience lasts 20 minutes or something like that. The time dilation effects of extreme gravity near the black hole, presumably inside of the throat of the wormhole, and then out the other black hole, extend her experience of all time by factors of a thousand. Then on Earth, they kind of think she's crazy until they check and they say, "Oh, there's actually a 17-minute gap in the video recorded inside the—" Okay, so none of this is possible, but it's fun speculation.
- 6.30** And it is grounded in physics that at least one of the progenitors of Kip Thorne would later win a Nobel Prize in relation to his predictions and what black hole collisions would sound like. And I've talked to Kip on my podcast about that very discovery. Now, making a wormhole, going through a wormhole is one thing; that's a big challenge, surviving it, but making it is almost even greater in terms of a challenge.
- 6.31** To make a wormhole, to stabilize it against collapsing in on itself, requires some form of matter and energy that we do not believe exists, or at least have never encountered. It's matter that is called negative energy density. Almost every form of matter, even

antimatter, has positive energy. In other words, if you take an antimatter particle, an antiproton, and collide it with a proton, you have two protons' worth of mass energy. To create them, you need two times that amount of energy, or two positrons and electrons, as we discussed in previous lectures.

- 6.32** But negative energy is something different. So positive energy, if you have, you must say, for example, if you have a gas trapped in a cylinder and you want to compress the gas in the cylinder, you must supply work to that gas. That's ordinary matter. It's supported by thermodynamic energy and pressure that I'll talk about in the next lecture. But in order to have negative energy, what would happen is, as you compressed the piston, it would keep going sucked in. It would be pulling on you to get sucked in.
- 6.33** There's no matter that's really like that. It's almost like a vacuum. A vacuum, if you were in this room and you had room pressure surrounding a piston, and there's a perfect vacuum in there, it will suck it in. Sometimes it is called vacuum energy. But the vacuum that we are talking about in that circumstance is the absence of all matter. This would be a form of matter, like a gas, but acts like a vacuum. It has negative energy. That's what it would mean to have negative energy. We know of no such substance that does that.
- 6.34** To make a vacuum, we remove ordinary matter. To make this type of matter would be the opposite of that. It would be true, honest-to-goodness matter energy, and that it would act as a vacuum pulling in outside forces, doing work on the outside world, and that would keep this thing expanded. So this anti-form of gravity creates anti-gravitational force or expansion. That keeps the wormhole inflated.
- 6.35** In cosmology, in Peterson Academy, I talk about the existence of what we ascribe to the energy of the vacuum, what's called dark energy in cosmology, that seems to cause the universe to accelerate in its expansion. But we don't believe it's a form of matter that we can control or manipulate. It's a form of energy, but not a form of matter. This would be exotic matter that has negative energy density, that if you do work on it, it gets smaller and smaller. The opposite basically happens. It's almost like a rubbery substance that pulls in and has tension everywhere in space.
- 6.36** Now, if you had such an object like that, it would basically act in the opposite way of ordinary matter, say in a black hole, which tends to cause convergence, gravitational lensing, and the bending of light as if there's matter like a lens. Or like we showed yesterday in previous lectures with this water, which is also acting as a lens, distorts. I can see, I'm looking over here, but I can see people over there, because their light is getting refracted and bent to my eyes. They seem like, even though they're physically there, I actually see them over here.
- 6.37** Lenses do that with ordinary matter. If this were filled with anti-gravitating matter, exotic negative energy density matter, then if I were looking in this direction, I would

be seeing effectively, everything would be coming in in this direction, and it would be almost looking like it would be like a divergent lens, which doesn't really exist. So I would be looking in, and everything around me would be—you would see me, my eye would be spread out all around in all directions.

- 6.38** Of course, you've never seen anything like that because we don't have any object that behaves as if it has negative energy density. We've only seen things that have either positive energy density or a perfect vacuum. If this were a perfect vacuum, I'd see straight through. Nothing would get in the way. There'd be no bending of light whatsoever. So this requires anti-bending refracting material.
- 6.39** As you came through and went through the throat, you would see all sorts of distortions. Again, the opposite of what you'd see from a positively gravitating object, the opposite of positive gravitational mass. And so when you went through the throat, you wouldn't see it as a tunnel, as if you would carved out of positive energy space-time. You would see it as a negative or a tunnel that wraps around itself or like a sphere. It's very bizarre to conceive of it. If you looked back and turned around, you'd see the opposite. Again, not a tunnel, not going through the throat. You would see the refraction or bending of all the light around you, focused back in from where you started.
- 6.40** Now, what happens to a clock as you go through a wormhole? If you could make a stable wormhole, which nobody knows how to do, and if you had the material that had this negative energy density, which we don't even know exists, then you can ask the question, what would happen to a clock? Well, what happens to a clock around ordinary matter? We have the time slowing down. So, according to relativity, for general relativity, in a positive mass environment, time slows down. It's kind of like a negative anti-mass time, so things are going to be speeding up. And also, the speed of time flows faster for observers that are not moving than for the moving observer, as we showed.
- 6.41** And we also have a gravitational time dilation that occurred. Remember, that was responsible for the event horizon being the last thing we ever see, from a traverser through the black hole's event horizon. So you'll have less time passing in one of the mouths of the wormhole compared to the other one that you entered into. Therefore, you'll come back desynchronized, and effectively, time would be flowing at a different rate, even in the same galaxy or the same part of the universe or same universe. So effectively, you could go back to an earlier part of time.
- 6.42** Now, in the movie, there's plot defects because she sees her father not when he was a baby or something like that, or not even when she was a baby when he died. In the movie, he dies very young of a heart attack. And so, she's trying to get back in time to go see him. Why does she see him at the age that he died, not when he was a baby? Or why can't she go back in time to kill or to have heart surgery before he even had a

heart attack? So a lot of plot issues or whatever, but let's not ask too many questions. It's really good fun.

- 6.43** So if we wanted to think about how we would synchronize these clocks in order to have this happen, we'd start with a home-built—you're homemade, and you get it from Amazon or you buy it, build it yourself, get with a wormhole. It has two holes on the outset and one at the exit, and then they're connected by a throat, which is the wormhole, the Einstein-Rosen bridge. You go in at relativistic speeds, near the speed of light, compared to the mouth that we labeled A.
- 6.44** You go through at the speed of light; you have a time lag. We leave here 2024, call it 2026. You come out and then you teleport back. You can come out, say, seven years younger if you're going at 99% of the speed of light. This is just one example. So, in this case, if you entered in the year 2023, you could emerge in the year 2024, 2123, 100 years [later], 99 years [later]. Again, this is predicated on you moving near the speed of light, also known to be impossible, require infinite amount of positive energy.
- 6.45** So you take an infinite amount of positive energy nearly, combine it with something that doesn't exist, negative energy density material to build the wormhole, to keep it stabilized. You'd better do it quickly because there's a lot of, as we'll see at the very end of this particular lecture, we'll see how long they actually last for in the most optimistic case. Of course, we already talked about the grandfather paradox where you go back in time, in the movie, Back to the Future, which I don't think involves a wormhole; at least I don't remember it having a wormhole.
- 6.46** But there are examples of this that there are solutions to it where you can't go back and change the future, if future events are fixed. And this is reminiscent of aspects of this Harry Potter book, Prisoner of Azkaban. So there are all sorts of things that could be possible. They're fun to think about. But most importantly, for those of us in California, they make for great Hollywood action scenes.
- 6.47** So, if you basically do anything, can you violate the laws of causality, where cause must precede effect? Possibly. But the problem is that, even if you could make it, these things are dynamically unstable. There are loops, both psychologically, physically, biologically, and mathematically, that could be used to, say, prevent or have a no-go scenario that would forbid us from ever making such a machine. For example, if things can go back in time, you could destroy the experiment that sent the particle that created the negative energy density material. And so, those kinds of issues would have to be worked out.
- 6.48** Typically, physicists work them out by saying, imposing laws like the Prisoner of Azkaban is not a scientific book, but they impose, "You can't change the future." You can't go back in time and push the asteroid out of the way so the dinosaurs survive; because if they survive, mammals like us maybe wouldn't have come to evolve in the

way that we have, and then we wouldn't be here asking about how we could go back in time and create a dinosaur-saving asteroid deflection system.

- 6.49** There are all sorts of things that people think about. But the most important challenge, in my view, is the instability creates a collapse in something much faster than even has a word for it. So we have words for all sorts of things like millisecond, nanosecond. I said that light travels one foot per nanosecond. We don't have a word for how many fractions of what fraction of a second this quantity is. These things seem to collapse even under the most optimistic circumstances.
- 6.50** You buy your negative energy density material on Amazon, you make your wormhole. Even if you could, connecting these two regions, they are separated, they collapse and destroy themselves in this almost $10^{-\text{googol}}$ seconds, so $1/\text{googol}$ seconds. So this isn't even possible to construct how you could teleport something even traveling at the speed of light more than something like a trillionth of a trillionth of a trillionth of a trillionth of a trillionth of the diameter of an electron. So this is just incomprehensible.
- 6.51** These are completely beyond the purview of what scientists could even contemplate studying. And they're super fun to think about. Theoretically, wormholes are a valid solution, but they're different than the valid solution that black holes were known to be from Karl Schwarzschild in 1916. Einstein immediately recognized that this is valid. And we do know of compact, massive objects; we live near one called the sun. It's relatively compact, huge amount of mass.
- 6.52** We could make predictions, test it. Confirmed just three years later, 1919, by Eddington, that matter does tell space-time how to curve, and curved space-time tells matter how to move. Also, it tells light how to move. There's a lot of paradoxes that are present in the case of wormholes. There are a lot of seemingly incontrovertible obstacles to their formation, but a paradox does not make something implausible. A paradox does not make something impossible. Famous ones exist.
- 6.53** The most famous paradox of all, at least in my experience, is called Zeno's paradox. Zeno's paradox, Zeno, I believe, of Alexandria, 2,500 years ago, said that motion is impossible. That sounds weird. Because to move from one side of this room to the other side of this room, first I must go halfway. Then I must go half of the remaining way. Then I must go half of that, which is an eighth, a sixteenth. So motion becomes impossible. I never get from point A to point B. That's Zeno's paradox.
- 6.54** Zeno didn't know about the laws of infinite series, and we now know that that's a convergence series. But the idea was pretty interesting, that he would say that you can't ever go from point A to point B. Now, he didn't mean it; he knew he could get to the local, I don't know what they called it, 7-Eleven, back in [Elea], wherever he was. But paradoxes don't necessarily mean that there's no resolution. The grandfather paradox is another one we just talked about.

- 6.55** But there's a big paradox called the black hole information paradox, which is related to the quantum mechanics of black holes and the thermodynamics of black holes, which suggests that possibly black holes may not only swallow up everything, but they may even destroy information. Now, it doesn't sound that implausible that you could destroy information. If I take this computer and throw it on the ground, I destroy a lot of information. But with godlike technology, we know we could reconstruct it.
- 6.56** I've had many a computer destroyed by many a toddler in my house, or so I thought; I was still able to extract the data from the computer. It looked hopeless. But now—and I just have physicist power, I'm no god— but with godlike power, it's even possible to take a book, and I would never do this because I have an aversion as an author and as a scholar to burning books, and it's detestable to me to think about it, but as a thought experiment, it is possible in theory, for a godlike entity, way beyond the most superintelligent AI you could even imagine, from the ashes and heat, from a book that was burned, god forbid, to reconstruct exactly what was in that book. It is not impossible.
- 6.57** But black holes may be the ultimate cosmic furnaces, destroying information beyond any recognition or reconstruction. Now, to encounter that, we're going to have to explore pillars of physics of the 20th century, 21st century, and maybe 22nd century, when we encounter black hole thermodynamics, black hole quantum mechanics, and the nuclear processes that link them together. And that's our next lecture.
- 6.58** To solve the black hole information paradox, first, I must give you a crash course in thermodynamics, in quantum mechanics, and connect them to general relativity in a way that black holes uniquely are well suited to. We've covered general relativity, and with general relativity, quantum mechanics and nuclear physics made up the three pillars of 20th-century physics. Indeed, we're still learning a lot more about each one of these subjects, but we've covered a great deal in general relativity.
- 6.59** Now we're armed with enough knowledge about black hole properties to start exploring the frontier of what we do not yet know, namely what happens to information that has the misfortune of traveling beyond the event horizon. To do that, I want to equip you and arm you with a basic structure of quantum mechanics. We encountered the titanic intellect of Isaac Newton, who, Einstein claimed, was even greater than he in terms of not only contributions to physics, but to all of human civilization. Einstein, again, was not a humble man, usually. For him to say that about another physicist took a great deal of respect.
- 6.60** Newtonian mechanics is concerned with the clockwork behavior of masses in motion, obeying the three laws of Isaac Newton, the law of inertia, the law that force equals mass times acceleration, and the law that every action has an equal and opposite reaction. Along the way, Newton had to invent the calculus, the theory of tiny infinitesimal changes, of integral and derivative calculus, which is a course taught here at Peterson Academy by another wonderful lecturer.

- 6.61** And applying the laws of calculus and the laws to the laws of mechanics, $F=ma$, inertia, and equal and opposite reactions, Einstein, sorry, Newton, was able to come up with proofs of why the observed behavior of planets and stars acted the way that they did. And in the 1700s, this was considered to be the final word that not only showed that physics is deterministic, but maybe all of humanity is deterministic. Everything that we do just operates as a robotic clock-like mechanism.
- 6.62** That got shattered in the late 1800s, and really, in 1900. It was kind of curious that it happened exactly on the century mark, indicating the 20th century. There are basically three different aspects of science that revolutionized our understanding that Newton wasn't the final word. He's certainly the final word when it comes to normal speeds and normal gravitational fields and normal-sized objects. But when you get to the very massive, the very small, and the very fast, we know that he's wrong. We need special relativity, general relativity, and quantum mechanics.
- 6.63** Two aspects of quantum mechanics that came about in the early 1900s had to do with the properties of photons and what are called black bodies. Black bodies are sources that are heated, of ordinary material, to some temperature that then radiate their heat perfectly in all wavelengths, at all times, at all temperatures. The sun is one of the most perfect black bodies that there is. It emits light due to its temperature.
- 6.64** In my cosmology class, I heat up different objects to different temperatures, some thousands of degrees Kelvin. Depending on their temperature in the Kelvin or absolute scale, they glow with a different color. When you heat something to about 3,000 Kelvin, it glows reddish hot. When you heat something to 5,600 Kelvin, it glows yellow like our sun. When you heat something to 30,000 Kelvin, it glows white-blue, like white dwarves and blue giants that we talked about in previous lectures.
- 6.65** But it wasn't clear what would happen if you kept heating stuff up. Would it be that you had infinite amount of energy? This was called the ultraviolet catastrophe, because it wasn't understood exactly how the black body emission would behave as a function of frequency or wavelength. Higher frequency, shorter wavelength meant that you had larger and larger energy.
- 6.66** There was another effect that was just demonstrated, wherein if you shot photons, particles of light, onto certain metals, that there would be an ejection of electrons with certain energies. Those energies were essentially quantized, so the amount of energy that could be produced is not unlimited or infinitely divisible; that light seemed to come in packets, bundles that we now call quanta. That's the origin of quantum in quantum mechanics.
- 6.67** We've already talked about the anomalies in Newtonian gravity that could not be explained by another planet that was hypothesized to exist, called Vulcan. It had to be that there was a curvature of space and time. If you just curve space, you will not get

the anomalous behavior of Mercury. You have to bend how time behaves, dilate time as well as space.

- 6.68** The quantum revolution was, first and foremost, a statement that energy is not infinitely divisible and continuous. The real numbers that we normally count with, integers plus the rational numbers plus irrational numbers, those are real numbers, those numbers are continuous. You can take a number and get arbitrarily close, add in many decimal places you like. But you can't do that for energy, meaning that energy comes in discrete packets. The carrier of the packet of energy in the quantum world is called the photon.
- 6.69** Einstein himself came up with this concept—again, another tremendous breakthrough, not only describing relativity at this time, special relativity in 1905, but describing the photoelectric effect, discovering that molecules existed, and doing many, many other great contributions. Classically, you could have any energy you want. No, Einstein and Planck, Max Planck said, "No, it can only come in discrete units of chunks called quanta."
- 6.70** In fact, we see that when we look at atoms. We see that there are only certain allowed energies that electrons can move between. The quantum mechanical revolution put to, killed off the notion that energy was purely classical; it could have any value that you desired. No, it could only come in discrete quantum transitions; and in fact, that matter was quantized into electrons, into quarks, into muons, neutrinos, other particles. Those are the things that came about as a result of the quantum revolution.
- 6.71** A further consequence is that particles and waves could be viewed as two complementary ways of looking at the same phenomenon. Light behaves like a wave. It can refract, it can bend just like waves around a jetty on the ocean. It can bend around it, refract. It can reflect off of other surfaces just like a wave can. But it also can behave in certain circumstances like little tiny cannonballs or particles of light. That's the photon description, which is right.
- 6.72** In the famous double-slit experiment, which is the most possible quantum mechanical experiment that one could do, it's a building on what was discovered in 1800 by Thomas Young, which is, if you have light and you pass light through two different slits, these slits are shown over here, vertical lines, basically, that the light will interfere and make alternating dark and light bands. In fact, that's how the interferometer, called the Event Horizon Telescope, works. That's how the LIGO interferometer works as well.
- 6.73** These phenomena for light are familiar with us. We see the rainbow pattern of interference, light and dark bands for light. But then it was found that, well, if it works for light, and everything is quantized, then you should be able to take matter, particles of matter, shoot them into a double slit, and they should interfere with each other; that you could have probability for matter to appear at a location and not appear at another location, and matter from one side of the slit interfere with matter from another.

- 6.74** Exactly that was found in the early 1900s. If you shoot electrons, now you're not shooting light, if you shoot a beam of electrons at a double slit, they will also diffract, and you'll have interference patterns that will look like this, where there'll be different amounts of light and dark spots, showing that light can act like a particle or a wave, and that matter, electrons, can act like a particle or a wave. Certainly in the case when they're orbiting around a nucleus in the Bohr model, they act just like a particle.
- 6.75** When they interfere with each other, that's a wave phenomenon. So at the quantum scale, we really can't distinguish and differentiate between quantum waves and quantum particles. They lose their identity. And this was discovered and predicted. It's known as the de Broglie wavelength hypothesis, for massive particles act like they have a wave, with shorter wavelengths associated with more massive, more fast-moving particles.
- 6.76** The higher the momentum, P , the shorter the wavelength, the higher the frequency of oscillation of the matter wave. Matter can be a wave; matter can be a particle. Light can be a wave; light can be a particle. All that distinguishes that behavior is where you're testing them. What regime are you looking at them at? When you collide objects near the speed of light at the Large Hadron Collider, you're colliding particles. You could think of that as exhibiting particle collisions, not wave-like collisions.
- 6.77** Because of the wave-like properties of matter, it is not possible to say exactly where an object made of matter is located at any time. All that could be said is, there's a probability to find a particle with some given amount of probability in a certain volume at a certain time. That is known as Heisenberg's uncertainty relationship. The more accurately you try to know an object's position, the less you know about how it is moving. The more accurately you know where it is, the less accurately you know how fast it's moving.
- 6.78** I use an analogy like this. If we turned off the lights in here, it was completely dark, and I told you there's a ping-pong ball rolling around on the floor, find the ping-pong ball. So, you could do it randomly. You'd find it eventually. None of us else are here. It's just in a room. It's just going to keep moving around with some velocity. It's just rolling around. Nothing else is going to impact it. Ignore friction for the time being. You can't see it at all. Your eyes are closed. The room is pitch dark. The ping-pong ball is painted perfectly black.
- 6.79** How would you do it? Well, you go around and look for it. I found it! You put your hand on. Oh, I found it. Where is it? Oh, it's right here. Well, where is it exactly? Oh, let me close where it is. Okay, now I stopped it. It stopped moving. I know nothing about its speed anymore. I know exactly where it is. The same comes true for an electron, except for an electron, you can't do what you could do with a ping-pong ball.
- 6.80** A ping-pong ball is a classical massive object. It's macroscopic. You could, if I said, "Now you can use your eyes. Now you can use a high-speed camera," you could do it

instantly. There it is, and that's how fast it's moving. I know exactly how fast it's moving. I didn't stop it. I didn't do anything to it. How did you do it? You looked at it. What did you have to look at it? Photons, light. Light's reflecting from the light off the ping-pong ball to your eye.

- 6.81** You measure a millisecond later, you know its velocity, you know exactly where it is and where it's going and how fast it's going. But an electron, when you look at it, the light has energy that's comparable to the energy of the electron that deflects it, moves it. When you try to look at the electron, you affect the electron. You can't simultaneously know exactly where an electron is and exactly what its momentum or velocity and times its mass is. That's the Heisenberg uncertainty limit.
- 6.82** Another attribute of quantum mechanics is that, unlike classical objects, quantum mechanical objects can exist in what's called a superposition of different quantum states. The famous Schrödinger's cat experiment postulates that if you have a cat inside of a box, you don't know what it is; inside the box, there's a radioactive sample, a uranium nucleus, and it's going to randomly decay into lighter particle lead, for example, and it will emit light, a photon, and it will emit an alpha particle. So it will change a little bit its nuclear structure.
- 6.83** And in so doing, it will emit a photon, which is quantum mechanical. If there's a detector inside this box, which is otherwise dark, the detector releases an amount of cyanide that is enough to kill the cat if it detects a photon. Now, it's not guaranteed. There's not as much radiation in there that's guaranteed to produce a decay. Plus, the half-life of uranium is tens of [millions] of years. So even in a short amount of time, there's almost no guarantee that you'll produce a photon which will cause the cyanide capsule detector to go off to kill the cat.
- 6.84** But quantum mechanically, we have to say that the cat is in this weird superposition of being alive and dead at the same time, and the only way for us to know what state it's in is to do an observation of it, to open up this box. In so doing, we can affect what state the cat is in. The observer is influencing by doing a measurement, just like we do with the ping-pong ball. We measure where the ping-pong ball is, we influence it, we make a destructive or changing the quantum mechanical state.
- 6.85** One of the last aspects of quantum mechanics that's mysterious, highly non-classical, completely unlike anything else that we've discussed in classical physics, is that quantum mechanics allows for particles to become entangled, meaning that their states—and electrons are a lot like black holes; they have mass, spin, and charge. If you create two particles, a positron and an electron, say you create, crash together two high-energy photons, that what will happen is, you can create a positron-electron pair.
- 6.86** And because they have to conserve momentum—their spin has to be conserved, and their charge has to be conserved, one is positive, one is negative—if you measure, say, the charge of one, instantly you know the charge of the other one. So, for example,

let's say we create a positron-electron pair, and they shoot out in two different directions. And one of them goes to the star Vega, [25] light-years away from us. Jodie Foster is still there, hanging out with her dad or doing whatever she's doing.

- 6.87** And then on Earth, eventually, I get around to it, I get tenure, and I decide to say, "What was the spin of this electron over here, compared to the spin of that electron over there, on Proxima Centauri, or sorry, on Vega?" I measure spin up, because the total spin has to be zero. I know instantaneously the spin is down. [25] light-years away, I know what you're measuring. I can't communicate with you that fast. I know nothing else about it. But my measuring this thing being spin down somehow affects your measurement of it being spin up. You have no other choice.
- 6.88** This has been tested everywhere on Earth that you could possibly test it. We've even used distant quasars to test this affects reality, that these photons, electrons, subatomic particles, only at the quantum scale, they seem to be entangled. But that never stopped scientists from looking at ways that you could maybe make this macroscopically; that I could say, well, if I could make something here and entangle it with something far away, effectively, I know exactly what that object is.
- 6.89** So, aren't I teleporting? If I encoded all the thoughts in my mind, for example, couldn't I teleport my thoughts into your brain? And so, scientists work on this—a macroscope—so far, we haven't been able to do this. We've done it for light particles, we've done it for electrons, and nothing else so far.
- 6.90** And then the unification between the quantum mechanical and the astrophysical comes about courtesy of the quantum mechanical process of nuclear fusion that creates the power behind all stars. This takes place as an explanation; quantum mechanics provides an explanation for not the periodic table of the elements, although it does explain that, but the periodic table of the subatomic particles.
- 6.91** It turns out, there's 17 subatomic particles. They're listed here. There are six different types of quarks. There are six different kinds of leptons. The most famous lepton is the electron. There are others we've encountered so far, the muon. Remember, we talked about time dilation due to high velocity propagation. The muon lives a lot longer. If it's traveling 98% of the speed of light, it can live something like seven times longer than it would at rest in a laboratory. The muon is unstable, but it is an elementary particle. It's not made of anything else.
- 6.92** There's a heavier version of the muon, also unstable, called the tau particle. Then there are carriers of force. The only one that I expect you to be familiar with is the photon. These are the things that allow forces to be interchanged between, say, electrons and electromagnetic force. There's hypothesized to be a graviton, but it has not been discovered. Then there's something called the Higgs boson, which was only recently discovered about 14 years ago.

- 6.93** Everything in this room is made of three things, up quarks and down quarks. They make up protons and neutrons and electrons. Every chemical element in this room, in your body, on Earth, in a black hole, that made up a black hole, was only made up of three of the 17 particles. And yet, we get this incredible diversity of the periodic table of the elements; again, they're only made up of neutrons, protons, and electrons.
- 6.94** So, there must be something about the arrangement and the permutation of the different possibilities of combining these subatomic particles together that gives the magic that makes the periodic table so different. There must be some reason why we humans and all the chemistry in life we see around us and in the universe is only made of up quarks, down quarks, and electrons. We have no idea why that's true, but we do believe that these are the only particles and force-mediating carriers that exist.
- 6.95** This comes in to play because there is one figure that I mentioned that's not on this table, that is a force. The four forces of nature, again, are electromagnetism. There's two forms of nuclear force, strong and weak—that's where the Z boson and W boson come into play—and the gluon. Then there is gravity, the fourth force, but there's no gravity particle on there. It's hypothesized to exist but not been detected, because we do not have a quantum theory of gravity.
- 6.96** Classical gravity of Newton was revolutionized by Einstein's classical theory of general relativity. There's no quantum mechanics, there's no Planck's constant, there's no subatomic particles in gravity. But to understand the behavior of these particles and forces that are quantum mechanical under the influence of gravity requires a new breakthrough that we do not yet have, called quantum gravity.
- 6.97** The theory of quantum gravity does not exist. The theory of quantum weak force does exist. The theory of quantum strong force does exist. The theory of quantum electrical force does exist. The theory of quantum gravity does not exist. There are many candidates that are attempting to explain why and how gravity could be quantized. They go by the names of things like string theory. That is an attempt to explain macroscopic gravity by submicroscopic, subelementary particle strings, namely that these particles, the quarks and electrons and so forth, muons, are made of tiny vibrating strings of pure energy.
- 6.98** But that theory is a long way off from being validated. It makes no experimental predictions. It cannot be proven wrong. There's lots of challenges with it; and yet, right now, it's sort of the best that we have. That would be an entire course of its own. But in this case, we do not have a comprehensive theory that explains how gravity behaves on the subatomic scale. Quantum gravity is missing.
- 6.99** So, we have to figure out how that would work and how it could manifest itself. It would presumably be relevant in two scenarios. One, at the beginning of the universe in the Big Bang. Two, at or near the singularity, and at or near the event horizon of a

black hole. And what ends up happening to these carriers of force, like the photon, like the gluon, they behave very differently near the event horizon and near the singularity.

- 6.100** At the singularity, you must have quantum mechanics and gravity married together in some way, or you cannot explain it. Right now, it's just a pure mystery. We have no idea how it behaves. Fortunately, we have a little more information about how the event horizon behaves, quantum mechanically. Ironically, to deal with that, we're going to have to understand the processes that occur astrophysically, and then thermodynamically.
- 6.101** Thermodynamics was a field invented to explain the behavior of steam engines and coal and mechanical work and potential energy and thermal energy. That was in the 1700s and 1800s. Why, when you fire a cannon, does the cannon get hot? What does mechanical friction relate to heat? How much heat can you create? How much heat energy can you extract from falling gravitational masses, interconversion of potential and chemical energies?
- 6.102** We talked about that briefly, but we will explore thermodynamics in the next lecture, and we will explore the properties of quantum mechanics at the boundary of the event horizon. We'll see that what was discovered by Stephen Hawking was the marriage of thermodynamics, the physics of the 1800s, with the physics of the 20th century in quantum mechanics. That'll be our next lecture.

Lecture 7: Entropy & Thermodynamics

At a glance — Entropy and thermodynamics.

- 7.1 We left off in the previous lecture talking about the beautiful implications of the theory of quantum mechanics and how counterintuitive it is. If you thought that was bizarre, well, we go through the wormhole to describe how thermodynamics and quantum mechanics get married together at the surface of a black hole's event horizon. Things are going to get even weirder. But that's where we're heading.
- 7.2 You see, when you combine gravity, thermodynamics, and quantum mechanics, all bets are off. You have to have a self-consistent theory of three different parts that refuse, seemingly, to play well together. It's taken some of the greatest minds in human history, including Stephen Hawking, to confront the many mysteries and paradoxes that seem to arise when you try to force a shotgun marriage between these three disparate but incredibly important pillars of science—thermodynamics, quantum mechanics, and gravity.
- 7.3 Let's go into the impossible. You'll see why I named my podcast after that phrase at the end. Let's go back almost 300 years. The birth of thermodynamics came about with the birth of a new science that was predicated on new technology, very reminiscent of how artificial intelligence has come about to develop new ways of thinking about intelligence, consciousness, brains, etc, based on the increase in technology associated with computing. Without the increased computing power, we wouldn't be talking about artificial intelligence nearly to the extent that we do.
- 7.4 Similarly, in the 18th and 19th centuries, the newfangled science of the day was thermodynamics, literally the movement of heat. The connection between thermodynamic processes and quantum mechanics would not be clear for over 100 years, and it would take another 50 years after that for the scientists that studied gravity to try to reconcile properties of thermodynamics, heat, work, and energy with that of black holes.
- 7.5 We'll get there, but let's go back to the 1800s, the Industrial Revolution, starting in United Kingdom, Scotland, and northern and central England. The steam engine was perhaps one of the first inventions that was useful for converting some form of energy to another form of useful work. Energy can be defined as having two different properties or propensities—potential energy, the ability to store energy for use in some other scenario, and kinetic energy, the energy of motion.
- 7.6 The interconversion between the two became of interest in the 1800s. Because if you could create a machine that would take in energy and also supply energy, then perhaps it would be possible to have an infinite or perpetual motion machine supplying its own energy and existing forever. The realization that that would violate principles of physics

wasn't made clear until much, much later. It was sort of a postulate that you could make such thing as a train that powered itself.

- 7.7** The connection between the technology of the day, the literal hottest field of technology was thermodynamics in the industrial revolution. How efficient could you make a train? How close to the maximum theoretical 100% efficiency of taking a piece of coal, converting it to mechanical energy or heat, could you get? That was the subject of a French scientist named Sadi Carnot. The insight first had to be dependent on inventing the laws of what we call thermodynamics.
- 7.8** Before this time, people thought that heat or energy was a stored property of a material and its temperature depended on how much or how little of a substance called caloric this object possessed. When we think back, it's not so crazy to think about that in those terms, but it's very mechanistic, and it was very much reliant on the attempt to extrapolate the Newtonian physics, the clockwork universe of cause and effect and mechanisms, that would then lead to the deterministic universe of planets and projectiles that we've talked about. It was a natural extension, but now trying to apply it to the microscopic world of heat and the interconversion of energy into heat.
- 7.9** There were laws that became postulated that are summarized as the four laws of thermodynamics, that we'll get to in a second. But the players involved in it are very colorful characters, not the least of which was this gentleman, Ludwig Boltzmann, whose famous constant k we named after him, Boltzmann's constant. Boltzmann was sort of one of the first people to start thinking about the microscopic world in a way that would differ radically from just shrinking down Newtonian physics of particles, projectiles, and planets, down to sub-visible scales.
- 7.10** At this time, it was still 80-100 years before the existence of molecules was even proven by the observations of Brownian motion and the descriptions of their dynamics by Albert Einstein. Albert Einstein was unique in that he pioneered the fields of general relativity, special relativity, quantum mechanics, but also thermodynamics and the properties of molecules. No other scientists really did that, with that degree of expertise spanning so many different fields.
- 7.11** What Boltzmann would be famous for is a concept that a lot of people hear about and maybe say from time to time, but don't really fully understand. It's known as entropy. Entropy essentially boils down to information. How much can you know about a system? Scientists distinguish between two types of systems, microscopic states, microstates that exist within macrostates, so two different configurations.
- 7.12** You can have this room be the environment, the universe, for example. And then you can ask how subsets of that universe behave, and they can behave in radically different ways. You can have entropy, which is a characteristic of the amount of information, which itself is the degree to which you can distinguish information from random noise. So, something with complete random distribution, say we're sending Morse code or

we're sending bits of information, zeros or ones, if that string of bits is completely random, it conveys no information.

- 7.13** And there's an infinite number of ways something can be random. But there's only one way that something can be perfectly ordered. For example, there's one way you could have a perfect order of zeros and ones. It could be all ones or all zeros. If you had a distribution of either all ones or all zeros, you know the existence of the other all-one or all-zero, the complement of it. There's only one way, effectively, everything is the same bit. There's only one state that has all zeros, one state that has all ones.
- 7.14** But there's an infinite number of states with an infinite number of bits that have a random distributed pattern. In fact, you could have many, many numbers where you have billions of ones in a row and you think, "Oh, it's going to be completely uniform distribution of ones," and then all of a sudden it goes to zeros for all of time and all eternity. There's an infinite number of those distributions, so entropy is order. Order and information are intertwined with one another. Entropy measures how many different states are there that you could possibly be in.
- 7.15** So in a room with multiple people, we could arrange only a finite number of permutations of chairs. That's only the possibility of arranging it. We increase the number of people in the room, the number of chairs in the room, it increases exponentially how many different states there are. For example, there's only one state where I just had a delicious cup of coffee with cream in it. There's only one state before I mixed the cream into the coffee, where cream and coffee were 100% separated, zero and one. But once I mixed them together, not only are there an infinite number of those possible states, it's impossible to unscramble them back into pure coffee and pure cream.
- 7.16** Humpty Dumpty's men cannot put Humpty Dumpty back together again. There's an infinite number of ways an egg can be broken, and there's only one way it can be ordered, structured, low entropy. Life goes, and biological processes and physical processes and thermal processes, goes from low entropy to high entropy. In fact, we believe the Big Bang, the origin of our universe, was extremely low entropy, very ordered, and then became extremely disordered, obviously, as things go on.
- 7.17** Now, within systems, we can obviously construct more and more order, but it comes at the cost of more and more energy. And so, at Boltzmann's equation, it says that the entropy of a system depends on a constant, now called Boltzmann's constant, which relates entropy to temperature, and the logarithm, which is a mathematical function, of the number of states that there are.
- 7.18** So imagine if you have a collection of coins. You have 10 coins. They can either be heads or tails. There's one arrangement only where there are 10 heads alone. There's one arrangement, there are 10 tails, and there's many, many arrangements of all these different possibilities of heads and tails. So the number of possible states of the coin is

W. To make it a reasonable number, you take the logarithm of it. The logarithm base 10 of a number is the exponent to which 10 must be raised to, to get that number. So the logarithm of 100 is 2, the logarithm of 1,000 is 3. It's a way of keeping huge numbers like 1,000, a million, a trillion, converting them to things like 3, 6, and 12. So it scales with the number and it makes things a little bit easier to manipulate, but not much.

- 7.19** It also has the property that when you take two systems of a given entropy and you add them together, you effectively can add their entropies together. So you have two cups of coffee and you have two cups of milk. You can start off in the beginning, they both have zero entropy, complete order, coffee in one cup, cream in the other cup. Then you mix them together, they have a very extreme amount of entropy corresponding.
- 7.20** Now, you could theoretically separate it. You could, if you had an infinite amount of time, money, and energy, you could devise a Manhattan project. You could spend all the country's resources in going in, and with a microscope, and selecting every cream molecule and separating it from every coffee molecule—I don't even know how you do it, but you could theoretically do it. But it causes a rapidly increasing amount of energy. To reverse entropy requires immense energy. This is the link between temperature, energy, and entropy.
- 7.21** The formula is that the number of different states and arrangements you can have is this equation, $S = k \log W$, which is on Boltzmann's tombstone, and he actually, tragically, took his own life. Perhaps, some say, he was frustrated by the implications of his theory, but they are far-reaching and powerful and used to this very day. He was almost the right person to create the quantum theory, and it's kind of a deep tragedy that he never lived to see that.
- 7.22** The connection between quantum mechanics and black holes will go through thermodynamics, and that will then lead to concepts of the entropy of a black hole. And what happens if you take something with very, very low entropy, like a book, where the words are very well ordered, even though there's a lot of information, and it has a lot of entropy compared to a single particle by itself, those words come in a very strict order.
- 7.23** There are many, many ways of taking a book and scrambling up the letters, than there are ways of arranging the book in a way that's actually possible to read. There's only one way in which War and Peace can be written that's perfectly understandable in its current form, if there's a near infinite number of permutations of all the letters in it.
- 7.24** So, too, with your DNA. There's only certain ways and patterns at which the four different [nucleotide] base pairs can join together. We'll see, in the final lecture, a beautiful connection between the theory of evolution and the sort of cosmic DNA and the amount of black holes that exist within a universe. Some say there's a connection.

The more fertile a universe, the more black holes it has. And that's called cosmic natural selection, cosmic version of Darwinian evolution. We'll talk about that later.

- 7.25** Here are the laws. There's four laws. It's too bad there's not three, like Newton's laws. But there are four laws, and the first one is the zeroth one, and that is because absolute zero is the minimum temperature scale. And we talk about temperature as being the property that comes to equality between two different objects. We call that thermal equilibrium. We measure temperature of an object. If it's in thermal equilibrium with its surroundings, we can say that they share the same temperature.
- 7.26** You and I are not in thermal equilibrium with this room. If we were, we would be not alive. In fact, if you watch CSI documentaries or movies or whatever they are, when you see forensic scientists, when they come to a crime scene, they often measure the temperature of the body. It's kind of gruesome, not the kind of job I'd like to do. Why do they do that? Well, they know the temperature of the room, they know the temperature that the body died at, and they know how things come to thermal equilibrium over a certain amount of time, given their composition—mostly water, some other proteins and bone and minerals and stuff like that. We know the composition of the human body.
- 7.27** Therefore, if we measure its temperature, we know the cooling curve of that body, and we measure it. If it's the same temperature as the room, we can't tell how long ago it died, the person was killed or died. But if it's anything above that temperature, as it would be, or if it's anything below that temperature, if it's super hot inside the room, then we can measure the time at which the deceased stopped being alive, and that's how they measure in crime scenes. They actually measure temperature, utilizing equilibrium thermodynamics. They use the zeroth law.
- 7.28** The next law, called the first law of thermodynamics, is that energy is conserved. We cannot destroy energy. We can destroy matter. We can destroy hopes and dreams, but we cannot destroy energy. It has to go somewhere. If you take a pendulum, and it's oscillating very fast back and forth, and you stop the pendulum from moving, it will actually be heating up your hand, causing a temperature increase in your hand and the molecules within it, changing their average kinetic energy. You'll change the potential energy to kinetic energy; nothing got destroyed.
- 7.29** We already mentioned that entropy always increases the so-called second law of thermodynamics. Things tend towards randomness. They don't tend, under ordinary circumstances, to go towards order. Now, this can be used and is used by some people to construct what's known as the arrow of time. Because remember, in Newton's laws, the situation where a pendulum is swinging back and forth is symmetric with respect to time. You can't tell, if you're watching a movie of a pendulum, you can't tell if you're watching that movie playing forward as it was recorded or if somebody's playing it backwards. The laws are indistinguishable.

- 7.30** So why is it, when you convert, instead of a pendulum moving or a baseball or billiard balls moving back and forth on a table, you also can't tell—they're just moving around, colliding with each other or planets in their solar—you can't tell if you're watching it in forward or reverse. Purely by looking at it, the laws of Newton are reversible. If molecules are just shrunken-down billiard balls, they should obey the laws of Newton, but they don't. You can distinguish the direction of time.
- 7.31** You don't see eggs coming back together into shells, yolks. And does anyone know what the white of an egg is called? This is one of my favorite things. I tell you something that's really funny. It's called a joke. When you talk about a collection of people, you talk about them as "you folks." The white of an egg is called a— you're supposed to say yolk, but it's albumen. The technical word is albumen. That's about as much biology as I know. You never see them separate; after they separate, you never see them rejoin into order. Coffee never becomes unmixed.
- 7.32** That says that, as time increases, entropy increases. You could say, well, a clock—sometimes I call it an entropy meter. It's measuring entropy. Again, it's hard to define time. Some people say it's just what a clock measures. It's not really that satisfying. But it's a puzzle, because time seems to be reversible when it comes to macroscopic physical processes. So, why isn't it on the microscopic scale? We don't actually know.
- 7.33** Then thirdly, you can't reach absolute zero. These are the rules of thermodynamics. They're taken as postulates. There's no deep proof of how they observe. They're empirically observed facts.
- 7.34** Now, when it comes to black holes, what is the relevance of these? We haven't talked about the fact that black holes have a temperature. Remember, I said they have these features: mass, charge, and spin. But then those mass, charge, and spin can be decomposed into the effects of the energy of the black hole. More mass that went into a black hole, there's more energy that went into a black hole.
- 7.35** So John Archibald Wheeler, the same scientist who said that matter tells space-time how to curve, and space-time's curvature tells matter how to move, the same scientist who came up with the term black hole, also had a thought experiment. He didn't do this experiment. But imagine you have a hot cup of tea or coffee. Therefore, it's a high-end entropy; the molecules are moving extremely rapidly in the hot water.
- 7.36** Now, you pour it into a black hole. Once it goes through the black hole's event horizon, that energy and the associated entropy is now transferred from the universe outside the event horizon. The watchers—now, they have lower entropy. Somehow, they lower their entropy just by pouring their coffee into a black hole, but that then decreases entropy. That violates the second law of thermodynamics. So this is a paradox. Where did the entropy go?
- 7.37** The energy, the mass increased in the black hole. Its size will have changed, its event horizon changed, its ergosphere could have changed. All sorts of things could change.

Now, again, it's a thought experiment. But somehow we reduced the entropy of the universe by letting it get swallowed up by a black hole, like an entropy vacuum cleaner. Could you use this? Could you do something with this to prove that thermodynamics is not violated?

- 7.38** Actually, scientists had a big puzzle about this, and it lasted for several years until an Israeli scientist named Jacob Bekenstein started to think about this problem. Can a black hole have entropy? If so, where does the entropy reside? What determines the property that we call entropy of a black hole? What determines how many states or arrangements, microscopically and macroscopically, are available? How many arrangements of the properties of a black hole can you have, with only these three properties?
- 7.39** He discovered that there was a phenomenon that was completely puzzling and unlike any other physical circumstance that we know about or certainly encounter. And that's that the area of a black hole surface—these are Schwarzschild radii, these are simple Schwarzschild black holes—the area of a black hole, not spinning yet, although you can generalize this if you like, the more mass you put into it, the more matter that falls into it, the higher its energy gets, the higher its entropy gets. When its entropy gets larger, its volume gets larger, but its area gets larger, too.
- 7.40** It turns out, it's the area of the black hole's event horizon. This fictitious layer, this fictitious boundary, it's not a real boundary. Remember, if you're going through it, you just pass right through it. Somebody else sees you stopping and being frozen in space, but you pass right through it, until you get spaghettified. And he had a conjecture that the area of a black hole is where all the entropy lies.
- 7.41** So it's very different than a coffee cup or a human being. The amount of entropy in a coffee cup with cream in it is the amount of volume it has, not the amount of area it has. It's the number of molecules of coffee and the number of molecules of cream that it has. That's what determines its entropy, is its volume.
- 7.42** But black holes are different. It's their area that determines how much entropy they have. And so, what was found was by Stephen Hawking, that the area of a black hole never goes down. If you put matter into it of any kind or energy or information into it, its area increases. And so, that kind of looks like the second law of thermodynamics. You can only increase the property of the black hole by its area increasing. That's like the entropy only gets increased.
- 7.43** So we said, what if there's a proportionality between the black hole's entropy or disorder measure, the number of states, the logarithm of the number of states that it has, and its area? So they found a proportionality between them.
- 7.44** And so, when you throw in Wheeler's tea, from the tea cup, the area increases, the entropy of the black hole increased because its area increased, and the entropy of the outside world decreased by the same amount. This sign that's shown here, dA —this is

the change in area, ΔA — is either greater than or equal to 0, so it can stay the same. When it stays the same, it's equal to 0, the change is equal to 0, so the change in entropy went from taking some entropy from the universe outside the event horizon, and it exactly equals the amount of increased entropy on the black hole's event horizon. That's how the entropy was saved, the second law of thermodynamics was saved.

7.45 But that wasn't enough for Stephen Hawking. Hawking thought that Bekenstein was wrong. He actually thought that he could prove him wrong, that the black holes don't have entropy and they don't have this property where the area increases. And what he did is, he looked at the quantum mechanical behavior of particles and fields like light and matter, as they approach the event horizon of a black hole.

7.46 And what he found was there was something very, very strange about particles. Because you could have a particle-antiparticle pair, which would be created; that's something called virtual particles. That's not really that important. You could think about a photon and another photon coming together with just the right energy to produce an electron and a positron. So photons have no charge, a positron has positive charge, an electron has negative charge. Charge is conserved, energy is conserved, matter is not conserved. So two particles pop out: a positron and an electron.

7.47 But if the positron falls into the black hole, for example, and the electron escapes, it looks like the black hole is actually a white hole, is actually a source of electrons come spraying out of this black hole. Now, don't ask how this happens, but it can happen and does happen all the time, what are called virtual photons, colliding with each other.

7.48 Now, the opposite process also occurs, where two electrons, an electron and a positron, come together just at the edge of the horizon of the black hole. They annihilate each other, disappearing into a photon and an anti-photon, which is another photon, going in different directions. So if one is pointing towards the event horizon, the other one is pointing away from it, you don't see the one that went away from it. You only see the one that's coming towards you, for example.

7.49 Now it looks like the black hole is a source of photons. That's weird. It looks like it shouldn't be that way. They're outside the event horizon, so they can reach us. We can theoretically see this glowing light from the black hole. It turns out, it glows exactly as you might expect a glowing thermal object to emit, as if you had a chunk of iron glowing with some temperature.

7.50 Now, the temperature that it glows at is almost infinitesimal. It's almost as close to absolute zero as you could imagine. It's a very, very weak process. These things will glow with an equivalent temperature, an entropy that allowed this relationship in the lower left. The Hawking entropy, S , is equal to Boltzmann's constant times the area of the black hole, divided by something called the Planck length squared and times the

number 4. So this is the entropy of a black hole, and we can connect this to the temperature of the black hole as well.

- 7.51** So black holes have a temperature, but their temperature is, as I said, astronomically, almost negligibly small. Trillions of trillions of trillions of fractions of a degree above absolute zero. But it is not nothing. And so, theoretically, given enough time, a black hole will keep emitting radiation, glowing and emitting energy, shrinking its horizon, reducing its size, and the black holes will eventually evaporate. And you will no longer have a black hole where you already had a black hole, originally had a black hole. It may take 10 to the trillion trillion years to occur, depending on the mass of the original black hole, but it will eventually evaporate.
- 7.52** Hawking said two things—one, that black holes ain't forever, and also black holes ain't completely black. They do emit light, photons, heat, very, very cold heat, but heat nonetheless. They also evaporate; they disappear, given enough time. There's a connection between the thermodynamics, the entropy, the 18th-century physics and 19th-century physics, and then the quantum mechanics that go into the production of virtual photon-antiphoton pairs. That's quantum mechanics. Black holes provide this unique laboratory to connect together the physics of the 1700s with the physics of the last century.
- 7.53** So there's a connection between the surface gravity of a black hole and its temperature, and the event horizon's area and the entropy it produces. This was discovered by Hawking, perhaps his greatest accomplishment of many accomplishments, but this is certainly the one that was most unexpected. Nobody expected black holes to have heat, to have temperature, to have radiation. It's not the classical radiation of a glowing hot object or a super cold glowing object, but there is this connection between gravity of the black hole and the thermodynamic temperature that they emit.
- 7.54** So the gravity of the black hole acts as a temperature; it's proportional to the gravity. And we will see that the amount of entropy—and what happens to entropy at the surface of a black hole at the event horizon, rather—can be indicative of what will be the ultimate fate of information about the outside universe that happens to fall into the black hole's event horizon. So the final equations where you connect together black holes, gravity, with thermodynamics, is linking together via the first law of thermodynamics, which is the expression of heat as a form of energy that's conserved.
- 7.55** We also have a connection between the amount of charge on a black hole, finally coming into play in this course; the amount of rotational energy, angular momentum or spin, and the area. So this connection between mass, which is proportional to energy, the area of a black hole connected to its entropy, the spin of a black hole and its charge, they all go together in this equation of the black hole law of thermodynamics.

- 7.56 Then you can connect this mass via $E = mc^2$ to total energy of the black hole. There's electrical work, which can be converted to heat. There's electrical work and charge. There's rotational work, which could be theoretically extracted or used to drive the curving of space-time near the ergosphere. Then, of course, there's the area and the event horizon, stored area and proportionate entropy that it stores.
- 7.57 So what was found—and this is observed by LIGO in the experiment we just talked about a lecture back, in the combination and interaction of two black holes with two different event horizons, because they have two different masses, two different spins, two different charges—one starts with an area in entropy A_1 , one starts with an entropy A_2 . They come together to make a new black hole that's not rotating, therefore, won't give off any gravitational waves anymore, but will nevertheless have a much greater surface area, much greater total entropy. And we'll see, we'll have a lower Hawking temperature. So, as entropy never increases, decreases for macroscopic systems on Earth, coffee, eggs, etc., black hole entropy also doesn't decrease.
- 7.58 And finally, we can't actually reach a zero surface gravity state where the black hole completely evaporates to zero temperature in a finite amount of time. That's the last parallel to the third law. Remember, there's four laws. The zeroth law is the statement about equilibrium. First law is conservation of energy. Second law is increase of entropy. Third law is the existence of an absolute zero temperature, below which nothing exists. When we found black holes coming together, we could test this area addition law, and we found that it was borne out by the observation of colliding black holes.
- 7.59 Now, the last thing I want to draw your attention to is that the entropy formula that Hawking came up with is shown here again. It's Boltzmann's constant, which is purely thermodynamic, a classical theory; the area of a black hole, which comes from solving for the event horizon's radius, which depends only on the mass of the black hole for a non-spinning black hole; and then Planck's constant squared involves— sorry, the Planck length involves something called Planck's constant, which is the proportionality between energy and frequency that we said quantized all bits and packets of energy cannot be in packages smaller than Planck's constant times the frequency.
- 7.60 So, since you have a fundamental limit on the amount or quantization of energy, and you can express any bit of information, literally in terms of zeros or ones, all information can either be expressed in terms of zeros or ones in combinations. It could be very long combinations of it. But for example, I just had my genes sampled by one of these ancestry services. The genetic code is a 4-bit code, but I get back this computer list of thousands and thousands of base pairs and whatnot. But it all comes in binary digital code. You can express the quaternary code of G, C, A, and T, of DNA, in terms of a binary code. It's just, that gets much larger.
- 7.61 It's like, the Hawaiian language has as many words as the English language, but they only have 13 letters, I believe, in the Hawaiian language, and we have 26 in the English

language, and there's [33] in the Cyrillic alphabet. But all these things can be interconverted between each other, but they may need fewer letters, when they have more letters of the alphabet, to express the same word. The same thing happens with binary code. We may need a lot of bits, but as long as we can quantize that information into zero or one, then there's a connection between the quantum information and binary and energy.

- 7.62** Energy is quantized, information is quantized. Therefore, energy can't be created or destroyed. Therefore, information can't be created or destroyed. So what happens when you throw information into a black hole? So the problem is not throwing information into a black hole; it's what happens to the information that radiates out from the black hole. I told you that there's an infinite number of ways you can have a random sequence of an infinite number of bits, but there's only one configuration of all zeros, only one configuration of all ones.
- 7.63** Just like, I always say, it's very easy—let me say it like this. It's very hard to make yourself 10,000 times happier. If you had a lot of money and you had a lot of time and great health, you can't really make yourself a thousand times healthier. You can't increase your life to a thousand times, and some people wouldn't want to do that anyway. But you can make your life a thousand times worse in a millisecond. There's almost an infinite number of ways you can ruin your life, ruin your happiness, make your life infinitely worse.
- 7.64** So that's the entropy of happiness. That means there's more states of being unhappy than being happy. If you've ever been happy in life, raise your hand if you've ever been happy. So it's very hard to stay happy. It's like, some people lose weight, you can lose weight, but it's very hard to keep the weight off. I know that from experience.
- 7.65** But the question is, how many different ways can you destroy information when you have a black hole that's slowly, over trillions of years maybe, but nevertheless truly emitting radiation in the form of heat, which is pure random information? What happened to that information? Is it gone from the rest of the universe? Could you somehow reconstruct it? This is a big controversy in physics.
- 7.66** So, does information survive the black hole? Quantum mechanics says no, you cannot destroy—yes, it says yes, you cannot destroy information. But Hawking radiation, thermal heat-based radiation, is indistinguishable from random noise. Even if there was information to begin with, if you put in War and Peace, and out comes pure heat, well, if you also put in 12 Rules for [Life], let's give an homage to Jordan Peterson, you also get out heat. So they're indistinguishable. How could you possibly know the information in one is radically different from the information in the other? Therefore, has information been destroyed?
- 7.67** There are only a few different ways to get out of this paradox. It's still an ongoing question, but the attempts to solve it and to understand what happens to information

is one of the most cutting-edge branches of physics because it involves this discrepancy between the laws of quantum mechanics, the laws of thermodynamics, and, if you will, the laws of what happens at the fundamental level to information and energy.

- 7.68** Since we did talk about genetics in our discussion of binary codes, quaternary codes, information storage, we think all of our DNA is unique. We're all individual, unique arrangements of DNA, and our heritage is all different and unique and wonderful. The question arose that the theory of evolution, of natural selection, evolving modifications, mutations that are then selected for by their fitness to survive, passing on their information to subsequent generations, has been called the most powerful idea, one of the most powerful ideas in human history—Charles Darwin's really world-changing book, *On the Origin of Species*.
- 7.69** And it really came at a time when people didn't really understand how, certainly didn't understand the nature of DNA, of chromosomes, 100 years before that. So, how is it possible that these implications could be understood without a corresponding microscopic and physical understanding of how information, heredity, is passed on?
- 7.70** So, a corresponding ambitious goal by a friend of mine, Lee Smolin, is that, to explain the properties of the survival of our universe and our ability to ask the question of why we exist, has to be explained in a universe that tends towards extreme entropy, as I talk about in my cosmology class in Peterson Academy. So how is it that there's order that gets established in a universe that tends to disorder?
- 7.71** It's not to say that you can't locally reduce entropy. I mean, we have a refrigerator in our house that reduces the energy and reduces the entropy. It can freeze water, making it into a very highly ordered crystal lattice, as opposed to being a highly disordered, very random gas or liquid. We can locally reduce entropy; we just can't globally extract entropy from the entire universe, reducing it overall within the universe as a macroscopic state.
- 7.72** So the question is, could similar processes take place in our universe to take the chaotic, random information and disorder that goes as the universe evolves in time? Can we explain the low-entropy original state of the universe by using elements like DNA, but not of a biological sort, but of a cosmic sort? So the seeds of cosmic structure, of the formation of entire universes, and their fecundity, their ability to support life, their fertility, according to many physicists, including Lee Smolin, is that the answer comes down to black holes.
- 7.73** It's a fascinating, if not highly controversial, subject that has yet to derive physical evidence for it. But it's plausible, it's interesting, and it involves our favorite subject, black holes. Again, some of the major features, although by no means all the features we've talked about in this class, are shown for the black hole, to remind you of where and how far we've come from. We have the singularity, which is the end of space and time, the point at which our laws of physics no longer are compatible between gravity

and quantum mechanics, the physics of the very massive with the physics of the very small.

- 7.74** We do not have a theory of quantum gravity. Right now, it remains elusive. But we know the existence of the event horizon. We have never seen a singularity. We postulate its existence, and we can do sorts of estimates on the mass of the singularity, et cetera, and the properties near it from its effects on the event horizon, the accretion disk, and other things like the ergosphere that we talked about.
- 7.75** We know we've seen the [photon sphere] shadow in the Event Horizon Telescopes. We've seen the dynamics of massive bodies as they orbit within several thousand times of the innermost stable circular orbit. We've seen these powerful relativistic jets that come out. We know that black holes can come about in a variety of ways. There are several different masses of black hole progenitors. These are the ancestors, stars that would later become black holes.
- 7.76** There's three different classes that we talked about in the stellar formation cycle of black holes. One was small, 5 to 100 solar mass stars times our Sun's mass. Intermediate ones, 100 to 100,000. Those become quite a bit larger. They are typically found in clusters of stars that merge together and with multiple black holes coming together over billions of years. Then there are supermassive black holes that are found at the center of our galaxy, millions of solar masses in mass. Our Sagittarius A* black hole, 4 million solar masses. The one in the galaxy M87, something like billions of solar masses.
- 7.77** But then there's another class of black hole that we didn't talk about, that's been discussed in the cosmology lectures that I give here. Those are called primordial black holes. These are black holes born at the beginning of our universe shortly after the Big Bang, just a mere second or two after the origin of the universe. And black holes, because their life span exists much longer than the potential life span of our universe, given its physical constants, and because it's impossible, as we said, to destroy a black hole. To destroy a black hole would mean to destroy information. To destroy information, it would be to destroy energy. Where would it possibly go?
- 7.78** The core crux of Lee Smolin's idea is that primordial black holes, and black holes that exist from these other three candidate progenitor sources, are actually the source not only of future galaxies and stars, but of entire universes. And the idea is sort of as follows. After a black hole is formed from a stellar mass death, from a large solar mass, the core collapse, the formation of the supernova, and then the leftover corpse of the star's core, if it's greater than about three solar masses, we said, now it'll be collapsed into a black hole.
- 7.79** A black hole is a point with no matter but pure curvature. Space-time is essentially curved so sharply that light cannot escape. That's the definition. There may or may not be a singularity associated with them. The singularity, in some cases, you can have

singularities that don't have event horizons. These are interesting things to contemplate. But all we know is that it's possible that you could have a black hole created out of what are called density fluctuations. Again, we talk about this in detail in our course on cosmology here.

- 7.80** So they do not originate from the death of stars. They originate from purely quantum mechanical and non-astrophysical sources after the Big Bang takes place. The smaller ones, the smaller you are, the hotter your Hawking temperature is. The hotter your Hawking temperature is, like a star's luminosity, the quicker you burn out and fade away. Ironically, paradoxically, it's the small primordial black holes that could still exist today. These are things with masses of trillions of kilograms, which sounds like a lot, but it's actually not that much. These are intermediate, way smaller than, say, stars.
- 7.81** For example, it could be that these objects, primordial black holes, are what are responsible for the existence of what's called dark matter in our universe. We know that the ordinary matter that we're made up of, protons, neutrons, my favorite particle, the crouton—no, there's no crouton—but the particles we talked about in the periodic table, of the quarks and leptons, that those particles make up less than about 1% of the mass of our universe. The rest of the mass, something like 50 times more of the mass in the universe, is dark. It gives off no light, it absorbs no light. It seems to only interact gravitationally.
- 7.82** There are many candidates for what it could be. I get into that in my course on cosmology, but primordial black holes are one such example. We know that dark matter must have existed since the Big Bang. While black holes from stars did not exist since the Big Bang, the only black holes that could exist are primordial black holes. Again, we've never seen one; they're quite small. They don't have very powerful ergospheres, [photon spheres], or the large event horizons. They're very difficult, if not impossible to see, by method of Event Horizon Telescopes.
- 7.83** They can be seen, and some have claimed to potentially be seen or called gravitational microlensing. They behave like chunks of matter, but they're not chunks of matter. They're chunks of curvature floating through space, distorting the path of photons as they travel from distant objects to us. So they can act like lenses that warp and distort space-time more distant than they are.
- 7.84** So the theory of cosmic natural selection suggests the following. Universes are not unitary. The word universe means everything rolled into one. That's what uni means. Verse means roll, and uni means one. But actually, we have had problems in the past, assuming uniqueness about our location in the universe or our location even in the solar system. The original thought was that the Earth was the center of the universe. Then it was thought the Earth is just the center of the solar system. Then it was discovered, well, no, the Sun is the center of the solar system. And then it was said, no, the Sun is the center of the universe. Then it was discovered that we live in a galaxy. And then we said maybe the galaxy is the center of the universe.

- 7.85** So, again and again, we've thought that we are unique. So, too, might we be making the same pre-Copernican mistake that the universe is the base reality, when in fact it's the multiverse, a multiplicity, perhaps, of an infinite number of universes within which we inhabit one such small region that just so happens to have just the right properties to form life, to form conscious creatures, to form technology, to form great vast scientific architecture, to then contemplate why it is that we exist. This whole circular process I describe in great detail in cosmology.
- 7.86** But the mechanism is that black holes, primordial black holes in some cases, in some cases stellar black holes, are seeds that encode the DNA and the property of the preceding universe, survive whatever cataclysmic end of the universe awaits our universe, for example, and then live on to spawn a new universe whose ability to reproduce is determined by the number and properties of the black holes within it. So it's exactly like the survival of the fittest.
- 7.87** Those organisms, according to Darwin, that have adapted mutations that are more favorable to do things that other organisms did not adapt to, famous ones being like the eye, and even at the much lower level, when the first cells did not have cell membrane surrounding them, they didn't last so long, they got eaten up by other cells because they weren't as fit. Survival of the fittest implies the mutations lead to natural predisposition to survive, that when you survive, you pass your genetic contributions into the future. So, too, with black holes.
- 7.88** So, the course on Evolutionary Inference, taught by my friends Heather Heying and Bret Weinstein here at Peterson Academy, go into this in great detail; that the goal of evolution is to lodge your genes farther into the future than your competitors. So, too, with the cosmic natural selection— the more black holes there are in the universe, the different types and abundance and distribution of the black hole properties, the more munificent they are, the more likely that universe is to pass on its genetic seeds into the future. Again, controversial, but very, very fascinating. Very much pertinent to our study of black holes.
- 7.89** So, how does it happen? How do we get reproduction, information storage, mutation, selection, and then a cyclical autocatalytic process that sustains itself? Again, controversial, but here's the rough outline. Every black hole can produce a new universe. There are processes by which, because they are emitting particles of light called Hawking radiation, it is possible for those particles of light, under extreme circumstances, to create matter particles.
- 7.90** So, Hawking radiation, theoretically, could be used to create, eventually, for certain masses of black holes, could then create enough actual matter to explain a matter distribution in a universe. It might not survive for more than 10^{96} seconds, but if you have an infinite number of black holes, all with an infinite number of different properties, then some of them will have properties, again, according to this

controversial theory, that would then be more amenable, more prolific at producing more and more black holes in such a universe.

- 7.91** So, the black hole spawns a universe. The more black holes you have, the more opportunities you have to reproduce, to then lodge your design or genetic components, if you will, into the future, and the more and different abundance of different universes that could occur. So the theory that he came up with is more black holes, more success, more selection, more fitness for the future reproduction, population, of the multiverse.
- 7.92** So now this is not accepted by mainstream. There's a book about it, *The Life of the Cosmos*. There are many objections to it. There are even some predictions, which is a good thing, because when you make a prediction, you could be wrong. When you're wrong, you can be proven false, and then you can get another attempt at the same explanation through different means that is more likely to have a chance to be correct.
- 7.93** So Lee Smolin predicted many of these things. There's lots of criticism about it, including the black hole information paradox that we talked about before. Leonard Susskind has also been on my podcast—that you could have a barrier to transmitting information, including the black hole survival from one universe to another. Penrose has shown, though, on the other hand, two things—one, that the black hole singularity has a high level of entropy, unlike the Big Bang in our universe, which is low entropy, highly organized. Now it's highly disorganized as the universe expands and cools off.
- 7.94** But he also showed that the black holes are effectively the only entities that can survive through a collapsing universe. In a separate paper he worked on, if you have our universe, eventually it could collapse for various reasons; we don't think it will, so again, keep paying your taxes. But if it did collapse, or a universe that had a similar number of black holes, but a different mass density, it would collapse. I talk about this in cosmology.
- 7.95** The black holes can survive the ultimate Big Crunch, the opposite of the Big Bang, where the universe's matter collapses in on itself, sometimes to rebound, cycle back, and make a new universe; black holes can survive the Big Crunch, as long as it's not too horrific. And if they do, they persist into the next universe, carrying with them information from the preceding universe, and from the preceding universe to that. So black holes, in certain cases, could be like diamonds, forever.
- 7.96** So a lot of people have a lot of complaints about this theory, how they could work. But there is a beautiful connection between a lot of cosmic problems that are outstanding. I think it deserves attention as a beautiful theory, possibly wrong, probably wrong, to be honest with you. But it does explore certain things, like the abundance of dark matter, the properties of dark matter.
- 7.97** Instead of inventing new types of particles that we've never seen before, postulating that they exist, postulate that things that we know exist, black holes, are responsible

for the mysterious missing matter that we do observe but don't know what it's made of. Then we get rid of the ability to, essentially, have to finely tune the parameters in order for us to exist. There's a narrow range of these parameters like Planck's constant, the gravitational constant, the speed of light. If you change those too much, it's very hard to get things like stars to actually create nuclear fusion, for example.

7.98 And in this scenario, it explains why there's bound to be some regions of the multiverse where there's a universe that has just the right properties and its black holes to self-sustain, supporting natural selection, evolution, fitness, and survival in this autocatalytic process that could continue for trillions of years. Given that the universe that we observe is only 13.8 billion years old, that's many, many bites at the cosmic apple.

7.99 Our last lecture, we will conclude with a summary of the course, but we'll also get into the physics that really kicked off a lot of my recent addiction to black holes, and that's the physics of the black hole Gargantua, in *Interstellar*. Now you are equipped with all the tools needed to go beyond the event horizon, understand the movie, amaze and impress your friends, and that will be our final topic. So, come back in a few minutes. Thank you.

Lecture 8: Journeys Through Space-time

At a glance — Journeys through space-time.

- 8.1 We started this course by describing black holes as the end of gravity. And now we've reached the end of the course. But it's not the end of the story. Because as you'll see, there are so many magical mysteries that we have waiting for us on this tour of the cosmos, and I hope many, many more future lectures that you'll be involved with here at Peterson Academy.
- 8.2 But I'd be remiss and a poor educator if I weren't able to explain to you, so you can explain to your friends and family, what is actually happening in that most famous of all black hole movies, *Interstellar*. Not only does it have an Academy Award-winning soundtrack and tremendous portrayal by Matthew McConaughey and Anne Hathaway and other actors and actresses. But it is a faithful as possible reproduction of the science that goes into these objects as has ever been portrayed.
- 8.3 So much so that four years after the movie came out, Kip Thorne, one of the advisors on Christopher Nolan's movie, *Interstellar*, won the Nobel Prize for his work on black holes. As we've shown, he's played a role in many films, including *Contact*, the wormhole. But he also took the time to write a scientific paper with the visual effects team behind the movie *Interstellar*'s animations and CGI in order to have an accurate representation and to give back to the public the information that he had been working on for his whole career.
- 8.4 We'll talk about that and how that came to be, and I'll share a conversation that I've had with him recently that I know that you'll enjoy. And at the very end, we'll go through a review of the course, including the materials that you can use to continue your education. This is not the end. This is the beginning of our journey, including books by Kip Thorne and others that will take you farther than we've even gone in this course. I hope you'll stay tuned and stay in touch for that.
- 8.5 So Gargantua, the object, the real main character of the movie in many ways. Most people watch it for the love story and the father-daughter story and the apocalyptic end of the world, a tale that's being told that has to be redeemed by Matthew McConaughey. It's quite interesting. I like it, but I like the physics better.
- 8.6 So let's go through the properties of this massive monster called Gargantua and relate it to all the different topics that now you are equipped to understand. Everything that we've covered in this class has culminated in this moment. More than that, what's better than being able to one-up your friends at your next cocktail party by explaining the physics of this movie, which is also so important in our popular culture.
- 8.7 So according to Kip Thorne, the black hole had to have very specific properties. And Kip Thorne had a contract clause that would allow him to demand that they get the physics

right, or else he wouldn't approve of it being released, and Nolan was willing to abide by that. So the parameters that he chose were chosen for a very specific reason. This was a supermassive black hole, 100 million times the mass of our Sun, much more massive than the black hole in the center of our galaxy, which tells us that this galaxy is not—that this black hole is nowhere near Earth. And so to get there would require some other type of tool and technology, which conveniently happened to also be invented or popularized by Kip Thorne, namely the wormhole. So that plays a role in the movie as well.

- 8.8** It had to have a spin that was 14 nines. So 99.9999999999998% of maximum. We call that a near-extremal black hole. If it has 100% of the maximum rotation, it's called the extremal. And so it's far, far more dynamic in its rotation than is the Sagittarius A* black hole at the center of the Milky Way.
- 8.9** So this was chosen, again, for very specific reasons. It had to be highly rotational and it had to be highly massive, as we'll see why. It had to have a metric that was one of the properties that we've associated, the gravitational bending of space and time, that distorts trajectories, that is the metric that's called the Kerr solution. It's in contradistinction to the Schwarzschild solution, which is a non-spinning black hole.
- 8.10** This black hole, we don't know about its charge. Presumably, it has some charge associated with it. It also has these properties in order to be semi-realistic to allow survival of matter not to get torn apart by tidal forces or spaghettification that requires it to have this ergosphere property, but it also has to have an innermost stable circular orbit that could support the motion of the spacecraft, the Endurance, around the black hole without getting torn apart.
- 8.11** In fact, there's the villain in the movie—again, spoiler alert—12 years, you had your chance. Matt Damon plays that when they come to this [Mann's] planet, and different things are happening with the villain, Matt Damon is there, and they have to do some negotiations with him, and you all know what happens.
- 8.12** But that had to have very specific properties, too. Those are dependent upon the mass, charge, and spin of the black hole—all things we've talked about. Couldn't have designed a better course for you to understand a movie than this course at Peterson Academy. First of all, it is a rotating, spinning black hole. Therefore, it exhibits frame dragging. The twisting of space-time along with it, as would a bowling ball spinning in a vat of honey, taking the points within it along for the ride.
- 8.13** Now, it has near-extreme maximum possible spin, and it has such a large mass that those two combine together to exhibit properties that allow for the spacecraft to traverse nearby it and even have a planet nearby it that doesn't get torn apart by tidal forces, as we'll see. Now, inside the ergosphere, space-time is dragged along with it. So space can actually be moving faster than the speed of light at a certain distance close to the ergosphere, within the ergosphere. But it also enables a process that we'll talk

about called the Penrose process, named after Sir Roger Penrose, a friend of mine as well. And this process allows for the return voyage of Matthew McConaughey.

- 8.14** It wouldn't be a Hollywood ending if he doesn't get back home. We all know he gets back home. It's not an easy trip. The process by which that can occur is called the Penrose process, and we'll talk briefly about that. Now, the depiction that's shown in the movie shows an orange glowing black hole, which is similar to the orange color that we've seen in the Event Horizon Telescope images, but that's a coincidence, and we'll explain why in just a bit.
- 8.15** We do see this behavior of this classic shape that I told you is reminiscent of Saturn's rings. But unlike Saturn, the gravity of the black hole is so intense that it can allow us to see the backside of the ring see through the planet itself, in this case, through the accretion disk, through the black hole, rather. You see the warping of space-time allows you to see the accretion disk on the backside and on both directions, on both coming over the top and under the bottom. That's what this black hole looks like.
- 8.16** The ergosphere means it's twisting around and it's rotating, but a black hole without an accretion disk would be much harder to see. The more massive, as we said, the black hole can support a tighter and tighter innermost stable circular orbit.
- 8.17** Now, the team that Nolan worked with was a visual effects team called Double Negative, and that's the name of their company. And they created a simulation code. This is four years before the Nobel Prize that Kip Thorne won, and five years before the detection of the event horizon showing this image that looks like this in terms of its color and exhibits the properties of the photon sphere and parts of the accretion disk. So it's a stunning accomplishment.
- 8.18** Now, I do want to show you—the work that they did together resulted in a paper. The paper came out again in 2014 when the movie came out. This is a physics paper in a physics journal. The physics journal is called Classical and Quantum Gravity. I'm a subscriber. I read it for all the articles every month. I'm sure you will subscribe to it, too. You can get this article for free, unlike many, many of the papers in science nowadays. This is available freely online to anybody.
- 8.19** One of the unique things is that it's typical to only see papers that are written by people at universities like me with a .edu email address. I've almost never seen a paper published by somebody, say, with a Gmail account or a .com account. This is their company, Double Negative, dneg.com. So there are their emails.
- 8.20** Here's the abstract summary of the paper. Interstellar is the first Hollywood movie depicting a black hole as it would actually be seen by somebody nearby. Our team, Double Negative, in collaboration with physicist Kip Thorne, created a code, a massive computer code called Double Negative Gravitational Renderer to solve equations for light beam propagation through curved space-time—it would render for Nolan, it wasn't just enough to have it on a computer, but for IMAX. It's an enormous, enormous

screen if you've ever seen an IMAX. This would hopefully be useful, they say, for the future. It's quite amazing.

- 8.21** Again, a first in really Hollywood and science history for these things to be depicted together in such accurate and precise renderings that trace perhaps a useful path for the future that people could actually use for doing scientific research, but also giving back. Again, scientists work for the public. For us to continue to earn our vast resources, I'm a public university professor, but the point is that we get to do this work because of taxpayers like you and the support of the United States government, for example. To have science give back to science fiction and then science fiction in Hollywood, have this virtuous cycle is quite wonderful, at least from my perspective.
- 8.22** Now, why is the black hole orange in the movie? We saw images from the black hole event horizon that was imaged by the EHT telescope that are orange, too. It has nothing to do with this orange. They wanted an orange color to be reminiscent of the Sun's temperature. As I said, the Sun is a black body. It glows with a temperature of about [5,800 Kelvin, about 5,500 degrees Celsius].
- 8.23** But if it was actually the temperature of a supermassive black hole that we find typically in galaxies—again, our galaxy's black hole is neither supermassive; it's supermassive, but it's not nearly as massive as this object. It's [25] times less massive—but if it was actually more active than a typical galaxy with this number of stars and its black hole equivalent mass, then it would be glowing at millions of degrees Kelvin, and that would obviously be pretty dangerous for the astronauts to go by it.
- 8.24** So they had to make it a little contrived to have a black hole that's cool. It's cool enough to be, say, the temperature of the Sun so the astronauts don't get melted. And the planet with the liquid water on it, we'll see in a second, had these waves and stuff in it. It had liquid water because it's a star—sorry, it has a temperature nearly of our Sun. It's not a star. It's being illuminated by the accretion disk, which is glowing with the temperature of our Sun. So it's made it a Hollywood contrivance, right? So they could live there without being irradiated.
- 8.25** Now, let's look at Miller's planet. So in the movie, there's this ticking beat that goes off every, I think it's every hour and it's every second or so. It's 60,000 times different. Every second on this planet is 60,000 seconds. There's only, as you know, about 86,000 seconds in a day. Basically, every second or so, 1.7 seconds is a day. They have this beat, if you remember in the movie, like a pulsing sound, and that's to represent days on Earth.
- 8.26** But as we know, the astronauts don't know about that. First of all, they're not within the event horizon anyway. They're not within the ergosphere, they're outside of it. That's why they can see the accretion disk in this image here. But the only way to get that is if the black hole is spinning near the maximum velocity that could possibly be spinning, roughly 99.9 with 15 nines and then an 8% of 100% of the maximal spin rate.

- 8.27** The orbit of the planet is effectively at this innermost stable circular orbit. The planet is going in a circular orbit, not an elliptical orbit. It's very different. It's being illuminated by an orange glowing light, but that light is coming from the collision of matter being converted to heat. The frictional process of glowing happens to be at the exact temperature needed to have liquid water on Earth for a planet roughly at that same distance. This was one of the points that Kip Thorne said was a non-negotiable point in his contract.
- 8.28** Now, on that Miller's planet, again, near the innermost stable circular orbit, outside the ergosphere, outside the event horizon for sure, there's a huge amount of gravitational pull from this mass of the Gargantua black hole, pulling on the ocean in this watery world, creating this enormous tidal wave that is just literally occurring twice a day in high tide, and then perpendicular to a twice a day low tide where it's dry.
- 8.29** So there are regions of the planet that are dry, and then every so often, maybe a day or so, the planet is rotating, but the ocean is staying stationary, but the planet's rotating inside this ocean. So different parts of the planet experience high tide, and then the opposite side of that planet is a high tide, and then perpendicular are two low tides. There are low tides and high tides making this enormous wave that comes out twice per day.
- 8.30** Now, how could this possibly be? Again, if you had a large planet, it has a small, close-in radius for spaghettification, as we said before. You wouldn't even feel real discomfort. Here, they're basically walking around in the same gravity, mostly on watery land and stuff like that, watery surface. There are a little pool of water, so they're not getting drowned. Then all of a sudden this huge tidal wave comes in.
- 8.31** Now, fun fact, all of the scenes of them being filmed in the ocean on the ocean surface, they were actually filmed on Earth. That's a fun fact. This would have torn them apart if it were a smaller black hole. Again, counterintuitive. The ergosphere gets smaller, allowing you to get closer to the event horizon and then feel the warmth of the glowing accretion disk.
- 8.32** I mentioned before another Nobel Laureate, Sir Roger Penrose, collaborator with Stephen Hawking. He had proposed separately, not for the movie. He's featured in the movie *The Theory of Everything*, which we'll conclude the final slides of the course with, talking briefly about that movie. But he's featured in that movie, but not in this movie. But his idea is featured in this movie because there has to be some way for you to get—even if you're not in the event horizon, you're still close to the danger zone. You need some mechanism to get you out of the ergosphere and into interstellar space to get back to your wormhole and then get back to Earth. That's the plot. Almost none of this is plausible, but at least there's some hand-waving physical justification for it.
- 8.33** It actually is somewhat related to the Hawking radiation process of virtual photon creation that we talked about before. Namely, that as something is spinning inside of

the ergosphere, if it breaks apart into two or more different pieces of roughly equal mass, then one of them will fall into the black hole through the event horizon and be lost forever, eventually going into the singularity, it won't get ripped apart immediately, eventually it will get ripped apart, and then it will go into the singularity, vanished forever.

- 8.34** Then another half of the spaceship, let's say you broke the spaceship in half, launched half of its mass towards the ergosphere, inside the ergosphere, inside the event horizon, into the singularity, the other one has to, by this process, like equal action caused equal reaction, the other part shoots off and goes on its way and diverges away, saving the day for the spacecraft, the Endurance. So one part gets sacrificed, and that's that robot that looks like a Twix, Twix bars. That's called the TARS. I forget what TARS stands for, but it's a robot. And that is allowing Cooper to get thrown back, basically. It allows Anne Hathaway's character to get thrown back.
- 8.35** So the question is, is this thoroughly plausible or not? Well, it's hard to say, but at least there is a mechanism. Again, this only, to our knowledge, can relate to things near massless, like photons and subatomic particles. It's not really clear if this could happen for a macroscopic process like a spaceship. But let's just say it could happen. Now, your problem is you've got to get into a wormhole and get back to home from 10 billion light-years away.
- 8.36** Remember, this cannot be our Milky Way's supermassive black hole, Sagittarius A*. It's not massive enough. It's super hot. It doesn't have this ergosphere behavior. It has a lot of X-rays, and so this wouldn't happen. But it turns out, thanks to Kip Thorne, again, this process could invoke a wormhole where you have this exotic matter, negative energy density matter, which we don't know exists. We don't even think it's possible that it exists.
- 8.37** Luckily, there happens to be a wormhole near Saturn, and that you can come into one throat and come out by Gargantua, and the converse process can happen as long as you escape via the Penrose process from the ergosphere. So you're putting a lot of things together and everything comes out just right. The Endurance gets out, the infalling module gets attracted into the black hole, never to be seen again. The outgoing one goes into this wormhole's throat, into the entrance, and then comes out, luckily for them, near Saturn, which is just a mere [~75] light-minutes away from Earth, and so therefore, he can make it back home. But by then, his daughter is now 80 or 90, and he's one or two years older than he was when he left.
- 8.38** There's this other plot device that I won't really get into. It really has nothing to do with anything we've talked about. It's more a way of visualizing things like the fourth dimension. Remember that we can't visualize with our massive brains. We see things in two dimensions. We have two retinas that synthesize a stereoscopic projection of what a three-dimensional image would look like, but we don't see physically through objects into three-dimensional space as a four-dimensional object would see.

- 8.39** They use this object called the tesseract, which is a four-dimensional version of a cube, if you start off with a point on a plane, point is a zero-dimensional object, you move it in some direction, that makes a line. If you take that entire line and move it in a perpendicular direction, you get a square. If you take that square and move it in a different perpendicular direction, you get a cube. If you take that cube and move it in a different perpendicular direction, in four dimensions, you get a tesseract.
- 8.40** Now, the shadow of a cube, if we have a cube here in a bright light and I shine a shadow of a cube onto the floor, the shadow is shaped like a—what does the shadow of a cube look like on projection? It looks like a square. If I take a square and shine a flashlight in two dimensions, what does its shadow look like? A straight line. If I take a straight line and ask what its shadow looks like, it looks like a point. Therefore, the shadow of a four-dimensional tesseract—I can't stress how important this is. It's not really important at all. But the shadow of a tesseract is a cube. Yes.
- 8.41** If there were a four-dimensional creature in this room and he happened to have a tesseract or it had a tesseract, and it had a flashlight that could shine in all directions, we would see this cube floating around in space. And they're trying to explain what the four-dimensional thing would look like by a five-dimensional being that projects a four-dimensional image whose shadow is three-dimensional in this library with all these books and strings and basically completely irrelevant and not pertinent to this course, but nonetheless, another way that it's trying to convey a physics idea of dimensionality, of projection of physical space, and how gravity connects everything together.
- 8.42** Again, we have this incredible legacy that science fact became science fiction, and then science fiction was used to create visualizations for scientists to use to continue their exploration in scientific fact. The simulation matched more or less identically to what the shadow of the black hole's photosphere looked like when it was observed five years later. This is a prediction that could have been wrong and was completely validated by observation. It's a stunning achievement. Of course, it introduced many, many people to this topic of time travel, of black holes, of time warps, of wormholes all together in one incredible movie that also has a decent plot and a wonderful soundtrack.
- 8.43** Because as Arthur C. Clarke said—who used a wormhole in his movie, 2001: A Space Odyssey— said, "The only way to discover the limits of the possible is to go beyond them into the impossible." And speaking of into the impossible, then I was delighted that I had on my podcast, Kip Thorne, about a year and a half, two years ago, and I think that's the hope that I've really tried to convey throughout this course that science is done by scientists, and scientists are human beings. Human beings are fascinating, unpredictable, they're nonlinear, but they're capable of great things, especially when they collaborate and they investigate things that seem at first to be beyond comprehension.
- 8.44** Being along as a tour guide along this ride through the universe, I couldn't resist but recap how far we've come on this exploration of these magnificent, mysterious

monsters of the cosmos. So this is a review of it. And really to give you credit for how far you've come in this understanding, starting off with Newton's laws, first postulated in the 1700s, all the way to three-dimensional computer renderings in the most cutting-edge science and science fiction movies that we just encountered with Interstellar.

- 8.45 So everything begins with gravity, and everything ends with gravity when it comes to black holes. And it may be not the end if indeed, Lee Smolin is correct that the death of stars leads to the birth of universes. That's a very, very perhaps wonderful and delightful way to think about it.
- 8.46 Starting with Newton's concepts of universal gravitation, that the same force that pulls the apple from the tree to hit him in the head, supposedly, was pulling on the Moon to cause it to orbit and fall around the Earth. And with a cannon with just the right velocity shooting a cannonball, you could have a cannonball orbit around the Earth forever, if only it had just the right velocity. It turns out to be quite high, as we said. If it got slightly greater velocity than that, it would reach escape velocity never to return to the Earth. The larger the mass of the planet or the star, the larger the escape velocity would become.
- 8.47 And in fact, just a few dozen years after Newton's predictions of the laws of universal gravitation, of mechanics, of calculus, John Michell predicted the existence of a type of star which would exist to burn and make light, but it wouldn't produce any visible light because its gravity would suck those photons back in an analogous procedure of them not reaching escape velocity despite having the propagation speed, the maximum known to exist, namely the speed of light. Now, we know they're not exactly dark stars, the way that he conceived of them, and yet they share some properties, as we've discussed.
- 8.48 When we move from Newtonian mechanics of the human scale of miles per hour, if you will, of hundreds, perhaps thousands of miles per hour and kilograms to size of planets, we found that something was lacking. The faster we traveled and the more massive objects we encountered along our traversing the cosmos, the more weird and unpredictable the universe was to those that only had a classical physical understanding.
- 8.49 It took the genius of Albert Einstein to first come up with the laws that described the properties of velocities when velocities were close to the speed of light, the mechanics of moving bodies. Then he put those moving bodies in gravitational fields 10 years later to come up with the theory of general relativity. Special relativity, what happens to objects as they travel. If you're on a train moving half the speed of light, you throw a baseball at three-quarters the speed of light, how fast does the baseball move? That was completely unknown until Einstein.
- 8.50 Then we said that these had physical observable consequences. It wasn't just something that theoretically could happen. We saw it with honest-to-goodness pieces

of matter, subatomic particles called muons that are elementary, indivisible, would yet survive longer by factors of five times their lifespan. If you could expand your lifespan by five times and still be just the same you, you'd probably take it, too. But you need to accelerate to be close to the speed of light to do so.

8.51 These phenomena do happen in our Earth's upper atmosphere called cosmic ray collisions that produce muons that then decay. We can measure the time dilation effect by the existence of muons that should not be alive by the time they reach the Earth's surface. A small number of them do, exactly consonant with Einstein's time dilation prediction.

8.52 But that wasn't enough. Einstein had to also confront the seemingly irreconcilable difference between motions around massive bodies and motions in free interstellar space, where there were no forces or other fields. How could it be that gravity could manifest itself as a curvature of space? He had to take seriously the idea that acceleration and gravity were one and the same, that there is no way to distinguish between you accelerating with the same force of gravity we feel that's pulling us down, pushing up on us on the Earth's surface, or standing in a rocket being accelerated at that exact same acceleration.

8.53 In doing so, he was able to make further predictions that not only massive objects would be accelerated by gravity, but also massless particles like light, photons, which he also invented the concept of, that they would be deflected. In fact, this was observed by the eclipse of 1919, just four short years after the predictions that he made of general relativity.

8.54 And there's a side note that I didn't mention at the time, but when Einstein predicted in 1915 that gravity caused by masses acts like a lens, like this water that refracts light and allows me to see you guys, even though I'm looking in this direction because of the path that gets bent that steers to my eye and allows me to see around corners as a lens would. He predicted this effect in 1914.

8.55 But it turns out the great Einstein, the man who would later say that if the effect was not observed, then he would feel sorry for God because his equations were right, he actually forgot a little blunder that he made in 1914, which is that he messed up and had a factor of two that was off.

8.56 And it was actually the case that in 1914, there was a similar expedition to the one that would later discover the actual bending of light by the Sun. There was an eclipse of the Sun in 1914, and they sent teams of scientists, astronomers, armed with the same cameras and telescopes that they would later send. They sent them to the place of the eclipse, which was observable only from what we now call Crimea and eastern Ukraine.

8.57 And back then, I don't know what's happening now in that part of the world, but there was a world war going on in 1914 in that part of the world. Of course, now there is another war going on, unfortunately. So the astronomers in 1914 were actually

captured by Russians, and they were held hostage for many months. It was actually a good thing for science. They were eventually released, but it was a good thing because that eclipse, Einstein would have been proven wrong, first of all, because this factor of two.

- 8.58** And the eclipse was much less visible due to both weather phenomena and the position of the Sun and the background stars behind it made it much more apparent that Einstein was correct, finally, in 1919. That eclipse was a much more spectacularly brilliant way to observe this bending of starlight by the Sun. So lucky for Einstein, they didn't do the experiment. There was a war, and there was an impediment that delayed science for making this measurement for five years. Although at the time, I bet he wished he was really mad that they didn't do this measurement, but in the end, he was grateful.
- 8.59** We encountered the concept of the dark star. Again, this is done by people. The people had somewhat semi-tragic lives in some cases, thought of as crazy, madmen, discarded and not really taken seriously, not really gotten the credit that they deserve. There were people that were battling against the most extreme circumstances, literally in the trenches of the war to end all wars and contracting diseases that would later claim his life, Karl Schwarzschild. Yet, even before his death, he was able to submit and get the attention of Albert Einstein to take this seriously, which then allowed Einstein to make predictions about what would be seen, ultimately getting vindicated for his theory. Again, he was this Jewish soldier fighting for what would later lead to the Third Reich. Ironic.
- 8.60** Then there are these brilliant scientists that happened to be born in India that were discarded because primarily of their race and the biases that pervaded in post-Victorian England. He was right, but it would not be verified for about 70 years when he would win a Nobel Prize.
- 8.61** And then the ultimate telescopic glass ceiling that shattered by scientists like Andrea Ghez to predict that you could see the effects of an unseeable star, of unseeable matter, of unseeable curvature, the supermassive black hole in our galaxy using incredible technology called adaptive optics that unblurs and untwinkles stars. Just phenomenal. She was able to do this with her team and colleagues and competitors like Reinhard Genzel.
- 8.62** And then the ultimate instantiation of what general relativity would mean, the ultimate prediction of the existence of objects called black holes by John Archibald Wheeler, who gave us this mantra that space-time tells matter how to move, and matter tells space-time how to curve.
- 8.63** And the beneficiaries of this are all of us, because we can see not only matter going around objects, as Andrea Ghez's group has seen, but we can see the distortion of

starlight and the bending of light from the accretion disk of a black hole projected onto an image made by a radio telescope that literally spans the entire surface of the Earth.

- 8.64** This is the vindication of these ideas, and that the complexity of black holes, that the zoology and taxonomy of black holes, the many different types of black holes, the ancestors of black holes that come in so many shapes, sizes, and varieties, is just incredible. For that reason, it's captivated the minds of the public, wondering what would happen if you were to go to the event horizon, beyond it, through the singularity.
- 8.65** What happens to matter, to light, to forces and fields? How do we ultimately reconcile these two different perspectives when there's only one reality? The observer from great distance seeing the effects of time dilation slowing down, the traveler going blithely, unaware of what's going to happen once he reaches the point of no return. Nothing happens at that moment. But then eventually, the extreme tidal forces lead to the ultimate devastation known as spaghettification.
- 8.66** Then taking this Schwarzschild solution, which is a simplification, but an incredibly valuable and useful one, and then extending it to a property that was known to exist but took 50 years before its implications were explored by Roy Kerr. That was the concept of the spinning black hole. That a rotating black hole would have properties very different from a static or stationary black hole. The concept that the black hole only really exhibits three different characteristics of mass, charge, and spin. We talked primarily about mass and spin in this class. In advanced textbooks and other resources that I'll provide for the course, you will see that there are abilities that scientists have now been able to uncover and how we can understand the charge of a black hole.
- 8.67** And even a concept by my friend, Janna Levin, professor at Barnard College, that you could have a battery made up of a black hole from which you could extract electrical energy, perhaps to power different future technological civilizations in the deep future.
- 8.68** We understood only these mysterious objects called quasars by postulating the existence of a black hole accretion disk that engenders a rotational beaming effect that sometimes will align with our location fortuitously. We'll see this as an incredibly bright, almost dot-like or point-like source with incredible luminosity at tremendous distances. The only way they can be powered is by the engine of creation of this energy, far more powerful than even nuclear fusion, which is the most powerful energy source we believe we've ever encountered here on Earth, and that the accretion disk can actually outdo the nuclear fusion process by factors of 5 to 100. So if we could build, in fact, an ergospheric black hole, perhaps that would be some way of powering not just quasars, but future civilizations.
- 8.69** I have as parting gifts for you all today, the byproduct of the ancestor of our planet and our Sun, in fact. These are meteorites, and I have them for all Peterson Academy students who live in the United States, unfortunately. I will give these to the classroom

audience. I'll have you come up in just a second. And these are meteorites. These are the last bits of ordinary matter that a star will make right before it blows itself apart, spreading this material out into the galaxy. Eventually, some of this material condensed and formed our Sun and our planet. Our planet has an iron core at the center of it. Some of the iron from this progenitor star that blew up, whose name we'll never know, whose properties we'll never understand, is in your blood right now in the iron, in the hemoglobin molecule that allows you to have the energetic processes to sustain generation of sustenance of your life.

- 8.70** We found that stellar mass black holes and stellar origin black holes are very different from other ways of generating black holes, like primordial black holes. But when they occur, they go out with this bang producing the supernova, and then the supernova remnant can either leave behind a neutron star, a giant neutron, which is basically a giant nucleus of an atom, but it's only made of neutrons with a density far, far greater than even these heavy magnetic meteorites that you all get.
- 8.71** When they collapse, they form a neutron star, or if they're more massive than a critical threshold, they'll form a black hole. That's how stars convert into black holes. If you want to know how stars are born, you should take Astronomy 101, Peterson Academy. If you want to know how the universe was born, you should take Cosmology at Peterson Academy. And later, we'll hopefully have offerings on, as I said, relativity and quantum computing. There are many ways you can study not just your own DNA, but your ancestor of the star that made possible life on Earth itself.
- 8.72** We explored using tools and technology of visual observations of the dynamics of the black hole at the center of our galaxy, which may have come about due to the gravitational coalescing of millions of other black holes, or maybe it was swallowing up millions of other stars, or maybe it was swallowing up potatoes. We'll never know. But we do know how massive it is. We do know how it affects time, space, and gravity, thanks to the work of the scientists we've encountered in this class. We've seen that the size of this black hole, this region that we look at, is about the size of our solar system, and yet it's nowhere close to the size of the event horizon, even for a 4 million solar mass black hole like Sagittarius A*.
- 8.73** We've heard the cosmic chirping of black hole collisions. For only when you have binary objects or multiple objects, can you have the emission of radiation that we call gravitational waves. Those gravitational waves come in shapes and sizes uniquely determined by the objects that collided together in a cataclysmic explosion in a galaxy we have no idea where it is, but roughly a billion light years away, and then made this final chirp lasting a fraction of a second, maybe a half of a second, after living for billions of years in orbit around each other, ending in a burst of energy, converting three entire equivalent masses of the Sun into vibrational energy of space-time itself, shaking up space and time.

- 8.74** The amount of energy required, literally astronomical, outshining all the other stars in the entire galaxy from which this binary black hole pair was born. Again, another prediction once thought to be impossible and crazy, even by the great scientist Einstein himself.
- 8.75** The hope is to combine the techniques pioneered by Andrea Ghez with the techniques of LIGO, with the techniques of the Event Horizon Telescope, and do what is called multi-messenger astronomy, combining multiple types of information, optical light, radio waves, and gravitational waves.
- 8.76** There are also plans to do neutrino detection of these processes, and building more and more LIGO-like instruments, perhaps in space, as you'll see, might allow us to detect down to the individual galaxy where these collisions are taking place.
- 8.77** That would be opening up a window into the cosmos exactly like that which was opened up by Galileo, first using a telescope in history to observe the heavens in 1610. Amazing time that we happen to exist in a time we can hear the universe, not just see it. And then, combined with our ability to see and feel and taste these neutrinos, perhaps, it's an unprecedented time for scientific discovery. It allows us to speculate on these even more mysterious, exotic, and possibly outrageous entities like wormholes, tesseracts, time-traveling portals to other dimensions. These bring up the most interesting conundra in all of philosophy, namely paradoxes, causality, locality, and the meaning of the laws of nature.
- 8.78** We saw, too, that connection between black holes, thermodynamics, quantum mechanics, and gravity that I certainly never would have thought to think of that as a connection. These brilliant scientists who came before us have set the path, set the table for us to contemplate the notion and the meaning of things as basic and as taken for granted as information, as entropy, and maybe perhaps of the creation of the universe via a black hole and via its permanent ability to transcend the death throes that would otherwise rip ordinary matter apart.
- 8.79** But black holes are no ordinary things. They are places of pure curvature, places where information may go to die, places where they glow with a heat that will allow them to persist for trillions of years. But glow, they do nonetheless.
- 8.80** And so as we reach the end, the boundary coming so far from Newtonian gravity all the way to relativistic gravity, to black holes, to quantum mechanics, to gravitational waves, and perhaps the unification of gravity with quantum mechanics, which currently is elusive, we may come upon what Stephen Hawking called the mind of God, which is to say a theory of everything.
- 8.81** A single equation, perhaps no longer than the width of a T-shirt that you could put on your shirt, and it would encapsulate all the laws of nature, from the nuclear forces to the electromagnetic force to gravity, subsumed into one master force, a quantized field theory of everything, allowing us to unlock powers that we don't understand currently,

the same way that quantum mechanics enables the transistor, the laser, and the devices we take for granted in electronics, and someday we'll unleash quantum computing, quantum AI, space-borne AI in quantum computing cloud clusters distributed amongst the stars, beaming intelligence throughout the cosmos.

- 8.82** We couldn't comprehend this years ago. All of this made possible by these people, these stories that we have woven together in one coherent whole. For now, when you have the knowledge that you have hopefully been able to glean from this course, you can appreciate the beauty of seeing the scientific discoveries. Because I think a scientist can appreciate beauty just as much as an artist can. Knowing a little bit about the science behind it, to me, makes everything more interesting.
- 8.83** Here's a video from NASA about what it would be like if we took a field trip at the end of this course to a black hole. Come along with me as we finish up and appreciate this fantastic voyage from NASA. There we have it.
- 8.84** Thank you, Professor. I had a question a little bit about information and energy. Then it seems like we're talking about energy from a lot of different perspectives: energy in terms of perhaps joules or energy just in general. How is information an energy? How does that affect our understanding of singularity?
- 8.85** Energy and information are not necessarily the same thing, but it's impossible to have information without energy. We'll explain that in a second. And it may or may not be possible to have information without energy. I said it the other way, energy without information.
- 8.86** So for example, if you have a brand new computer, iPhone, whatever, it comes with some memory chip with it. If you were to take that memory chip, it's blank. So maybe it has nothing in it. But if you want to change the state of that, you have one bit of information on there, you have to do something to the memory to make it change its state. There's no way to do that without applying some amount of energy. When you do that, you can reduce the amount of entropy.
- 8.87** So let's say on your computer, you have just this picture here. It's a lot of information there. It's billions. It's probably 15, 16 megabytes of information. If I want to erase that, let's say it's the only thing on this computer, I want to erase it, I just hit Control, Delete, Command, Delete, whatever, you think it disappears. But actually what's happening is the computer is actually changing the orientation of the electric charge on transistors. In this case, it's electrical transistors called RAM. In the old days, it was a spinning hard drive. It was actually little tiny magnetic fields that were either up for a one or down for a zero. What they would do is they would have a magnet, they'd move a magnet over it, and they'd change everything to a one or everything to a zero. That would erase the information. Obviously, it costs energy to do that. Energy and information are interrelated.

- 8.88** The final question you had is about the singularity. The point of the singularity is that it's almost irrelevant to the concept of information. The information gets created or destroyed at the event horizon. The event horizon is a firewall that literally burns up or destroys all the information if people like Stephen Hawking are right. If other people, like opponents like Leonard Susskind are right, nothing like that happens. But there still would be—the boundary would not be at the singularity; it would be at the event horizon. Okay, next question.
- 8.89** How has understanding relativity changed your perception of the world around you, your existence, and even the biblical concepts of eternity?
- 8.90** I'm more curious about how it might be that you could be affected by scientific understanding of things. I'm curious how you might find it that the understanding of relativity might have an impact on the understanding of scripture.
- 8.91** For example, the Torah, the Old Testament, the Bible, has relatively little to nothing to say about scientific facts. It does have things to say about, apparently, the origin of the universe, that there was no universe, and then there was a universe. That was very controversial, even for Einstein, the greatest mind I think that ever lived. He thought Newton was the greatest mind.
- 8.92** Now, Newton was obsessed with religion. Again, his highest achievement was that he was trying to live as Jesus Christ lived, completely never married, et cetera. He wrote way more about religion than he did about science. Einstein was ambivalent. It wasn't clear if he was a theist or not, but relativity for the person that created it, having this notion that time and space could be manipulated, had basically almost no impact on him spiritually.
- 8.93** I find that hard to believe, especially when you think that for almost all of human history, imagine you have just since the biblical age of maybe 5,000 years ago. I'll go back to the ancient pyramids. Let's say we write down the year when the pyramids of Giza were built 5,000 years ago. Call that year number one. And today is year 5,000 years later, and you were to look at, write down on a ping-pong ball of the year, 1 through 5,000. And you reach into the bag and you pull out a ball and you ask, what did people believe about the age of the universe on that day, at that time? They all said, until the last 100 balls, they would have all said it existed forever.
- 8.94** That was totally in contrast with the biblical idea that the universe came into existence, at least by God, if you're theologically inclined. So that means that people that believed in the Torah and the Bible, they had to live with this crazy— being crazy outsiders for 4,900 years until Hubble and then later Einstein accepted the fact that the universe began in a Big Bang. So the Big Bang incorporates the properties of gravity and therefore, relativity in it. We discuss this in great detail in the cosmology course that I offer.

- 8.95** For me, personally, I find that it's exceptionally beautiful that beauty is a window into the sublime in terms of, does that mean that I will eat a certain food or not go to church or to synagogue on a certain day? It doesn't affect me, honestly, in that way. But I understand people for whom it serves as an inspiration. The sublime beauty is hard to ignore.
- 8.96** Thank you again. You've presented this frontier of multi-messenger astronomy in such a beautiful way. It seems to connect to this relationship that you've drawn out between the scientific and the fictional, the art being created, because there's this pursuit of unblurring our senses, the listening of the LIGO experiment and then the adaptive optics. I'm curious to know, in your opinion, what gets you most excited about tools, whether scientific or creative, fictional, going forward in unblurring our vision of the mystery.
- 8.97** That's a really beautiful way to put it. I didn't tell you what I actually do. I didn't tell you that I am not just an astronomer and a physicist and a cosmologist, but I am an experimental cosmologist, meaning that I build an actual universe—no, I don't do that.
- 8.98** I build telescopes that see light that we can't see called the cosmic microwave background radiation. Some of these telescopes I've shot into space. Some of these telescopes have been in Antarctica at the South Pole. Remember, I told you I've been there twice. Some of these telescopes, my current obsession, is a \$200 million array of instruments at 17,000 feet above sea level in the Atacama Desert of northern Chile, where you have to wear oxygen. There's so little oxygen, you'll die or you'll be severely brain incapacitated—which I can't handle—if you don't wear oxygen.
- 8.99** And to go to these extremes, meaning there has to be some payoff for it. And the payoff is that we are trying to visualize not waves of gravity that come from the black hole collisions. We're trying to visualize, to use your beautiful poetic language, waves of gravity created by the birth of every black hole in every planet and every star, if you will, every chunk of matter in the universe, namely, if the universe began with a Big Bang, it must also have produced a tremendous amount of gravitational waves. We could theoretically detect them if we had an ultra, ultra, super advanced version of LIGO, but it's actually easier to detect them by looking at images called the cosmic microwave background radiation.
- 8.100** I proposed to measure this now, 26 years ago. We still haven't done it. I haven't given up yet. We've gone through successive generations of experiments, more and more sensitive, but also more and more subject to our own biases to want to see the birth of the universe as visualized by these invisible waves of gravity. You're exactly right. That's the thing that captivates me the most. I explore this in my cosmology class. Again, I'm sorry for the advertising, but I'm proud of the class I teach here.
- 8.101** One of the implications of that is if we were to see these waves of gravity, it would mean that the universe, not only a black hole, had a singularity at its beginning, too,

and that it has an event horizon beyond which we cannot see. That theory is called the theory of cosmological inflation. There are just as many scientists who don't believe that's true as believe it's true. And if it did happen in that way, not only would we win Nobel Prizes and so forth, but we would be effectively in the first position in a scientific way to give some credence to the biblical narrative. The biblical narrative seems to suggest that.

- 8.102** Now, I don't approach it from a purely religious perspective, but to say that I don't think about it would be a lie. Because to think about being able to prove or maybe cast doubt upon these ancient models of creation and say, yes, it's consistent with it after, again, 4,900 years of people saying this is rubbish and the universe has existed forever, or it was created and destroyed millions of times, that would be just the most phenomenal opportunity I've ever had.
- 8.103** So that's what I spend my days and nights trying to think about building with my team of students and colleagues, 450 of us around the world, trying for this one goal, knowing at any moment we could be tripped up by a mistake as we were in the past.
- 8.104** That was the origin of my book, which is called Losing the Nobel Prize, not Winning the Nobel Prize, how we made a mistake on the journey of trying to prove something that had these huge, huge stakes, not just the Nobel Prize, but also the existential questions that keep us up at night when we're on the dorm room couch, whether it's at Peterson Academy or whether it's at home. These are the most enigmatic questions I think you can answer. And I just feel this privilege of being able to attempt to answer them, even if I can't answer them, to attempt to answer it is the greatest privilege a scientist can have.
- 8.105** Thank you, guys. It's been a great class.